Foreman notebook, §§15–19

The centre ray read off the list; the five-mark lists along their hinge trees; the masters of $\binom{[5]}{2}$ and two marks doubled, closed; the reading of a ray through the origin; the three-mark centre, set as an examination

22–23 October 2026

This entry follows §14.17 of the running notebook, which was not available in this session; it is written to be appended there. Handbook citations are by printed number and page of the handbook PDF of 20 October 2026; "the note" is the tree-coaction note of 22 October 2026. Every object is a member of the serif family or a construction of the handbook, written with its slots. The receipts are in the script centre_ray.py.

15 The centre ray read off the list

The neighbours of the five-mark centre are the values of the centre ray of the list (Def. 35.9, p. 309), the period $P_{\mathcal{B}}(\sigma\mathbf{1};\sigma,\sigma)$. The interest of the list is one Gramma (Thm. 8.3, p. 54), the period is its residue on the balance wall, and the residue is the Gramma of the letter columns (Thm. 2.9, p. 15). Its share side (Prop. 4.2, p. 25) is the working form of this entry: a fibre integral of Gammas, one per basis for each letter, pinned by equal incidence. The pin is what the list contributes. Closing the fibre at the negative strengths (Thm. 4.5, p. 26) lands on the pairs of k-tuples of bases with equal incidence; the residue at zero shares is the corner of Thm. 38.5(i); the native shares (Thm. 4.3, p. 25) are uniform, the share side of the table's gauge; and the banana is the list whose pin identifies the two letters mark by mark, which turns its share side into the Mellin read of a power of one Gramma.

15.1 The centre period on its share side

Proposition 15.1 (The share side of the centre)**.** [derived]*; receipts* [computed] *Let* $(\mathcal{E},\mathcal{B})$ *be a list,* $n=|\mathcal{E}|\geq 2$*,* $e_0\in\mathcal{E}$*, and* $\operatorname{Re}\sigma>0$*. Then*

$$\Gamma(\sigma)^2\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\int_{\substack{|\mathbf{a}|=\sigma,\ |\mathbf{b}|=\sigma\\ \sum_T a_T\mathbf{1}_T=\sum_T b_T\mathbf{1}_T}}\ \prod_{T\in\mathcal{B}}\Gamma(a_T)\,\Gamma(b_T),$$

the fibre integral of Def. 4.1, p. 25, with one share a_T *for the letter* K *and one share* b_T *for the letter* Ψ *at every basis, pinned by the two letter totals and by the incidence at the marks* $e\neq e_0$*; it has dimension* $2|\mathcal{B}|-n-1$*, and the uniform shares* $a_T=b_T=\sigma/|\mathcal{B}|$ *are an admissible real point.*

Proof. By Thm. 8.3, p. 54, $\Gamma(s)\Gamma(t)\,J_{\mathcal{B}}(\mathbf{m};s,t)=\mathsf{G}_{(\boldsymbol{\mu};s;t);t_{\mathcal{B}}}(\mathbf{1})$ with the unit columns $(\mathbf{e}_e;0;0)$ and the letter columns $(\chi_T/2;1;0)$, $(-\chi_T/2;0;1)$. The linear form $\ell(x;v;w)=\sum_e x_e-\frac{2r-n}{2}(v-w)$ is 1 on the unit columns and 0 on the letter columns, and at the index it is $\kappa_{\mathcal{E}}$ (Thm. 8.2(iii), p. 54): the balance wall is the first wall of this Gramma along ℓ, with the letter columns as facet terms. By Thm. 8.10, p. 56, the period is the residue of $J_{\mathcal{B}}$ there, and by Thm. 2.9, p. 15, at $p=0$ that residue is

the Gramma of the facet terms. Its share side (Prop. 4.2) has one share per letter column; the pins of the two letter scales are $|\mathbf{a}| = s$, $|\mathbf{b}| = t$, and those of the marks at $\boldsymbol{\mu} = 0$ are $\sum_T (a_T - b_T)\chi_T/2 \in \mathbb{C}\mathbf{1}$, which with $|\mathbf{a}| = |\mathbf{b}|$ is $\sum_T (a_T - b_T)\mathbf{1}_T = 0$. At the centre $s = t = \sigma$. The uniform shares satisfy the pins and have positive real parts. □

Receipts. For $\binom{[2]}{1}$ the fibre has dimension one and the integral is four Gammas on a line (Cor. 4.7, p. 26),

$$\frac{1}{2\pi i}\int \Gamma(a)^2\,\Gamma(\sigma-a)^2\,da = \Gamma_{\,1}^{\,4}\begin{bmatrix}\sigma,\ \sigma,\ \sigma,\ \sigma\\ 2\sigma\end{bmatrix}, \qquad \text{so} \qquad P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma) = \Gamma_{\,1}^{\,2}\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix} \quad \text{[derived]}.$$

For $\binom{[3]}{1}$ at $\sigma = 1$ the two-dimensional fibre integral is $2.343907238689463\ldots$, the value of the receipts table below, to fourteen digits [computed].

Proposition 15.2 (The banana's pin)**.** [derived]*; receipt* [computed] *For* $\binom{\mathcal{E}}{1}$ *the pin is* $a_e = b_e$ *at every mark, so each mark carries* $\Gamma(a_e)^2$*, which is the share kernel of the Gramma* $\mathsf{G}_{0;(1,-1)}(1,y) = \int_0^\infty x^{-1}e^{-x-y/x}dx$ *(Prop. 4.2), and*

$$\Gamma(\sigma)^2\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma) = \int_{|\mathbf{a}|=\sigma}\prod_{e\in\mathcal{E}}\Gamma(a_e)^2 = \int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)^n\,dy.$$

At $n = 5$*,* $\sigma = 1$ *the right side is* $39.945588011996245699\ldots$ [computed].

Proof. With $T = \{e\}$ the pins are $a_e - b_e = 0$ for $e \neq e_0$ and the two totals, so $a = b$ and the fibre is the share integral $|\mathbf{a}| = \sigma$ (Def. 4.1). The share side of $\mathsf{G}_{0;(1,-1)}(1,y)$ is $\frac{1}{2\pi i}\int \Gamma(c)^2 y^{-c}dc$, and the read $\int_0^\infty y^{\sigma-1}(\cdot)\,dy$ of a product of n such fibre integrals pins the sum of their shares at σ. □

So the "Bessel moments" are the banana's share side: the pin of a single mark turns the two letters into one Gramma per mark (alias $\mathsf{G}_{0;(1,-1)}(1,y) = 2K_0(2\sqrt{y})$). A general list keeps a share per basis and per letter and couples them only through the incidence; that coupling is the broader arena.

Proposition 15.3 (The corner is the residue at zero shares)**.** [derived] *With* $U_k = \{\pi(k),\ldots,\pi(n)\}$ *the tails of an order* π *and* c *the width (Def. 7.10, p. 49),*

$$\lim_{\sigma\downarrow 0}\sigma^{n-1}P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma) = \sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1},$$

and $\sigma^{n-1}P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)$ *has no term of order* σ*. This is Thm. 38.5(i), p. 330, at the centre question, whose walls are* $\sigma c(S)$ *(Prop. 8.21(ii), p. 59); the term of order* σ *there is a sum of* $\log\theta$ *at greedy bases, zero for a list. On the share side,* $\sigma^{n+1}\Gamma(\sigma)^2P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)$ *tends to the same number: it is the residue of the fibre integral of Proposition 15.1 at the origin, where all* $2|\mathcal{B}|$ *Gammas have their first pole. For* $\binom{\mathcal{E}}{1}$ *the corner is* $n!$*, the* $5!$ *of §14.17; for* $\binom{[5]}{2}$ *the tails of every order have widths* $1,2,2,1$ *and it is* 30*.*

Proposition 15.4 (Closing the centre at the negative strengths)**.** [derived] *For* $k \in \mathbb{N}$*, at* $\sigma = -k$ *the share side closes (Thm. 4.5, p. 26) onto the lattice of pairs* $(\boldsymbol{\alpha},\boldsymbol{\beta}) \in \mathbb{N}^{\mathcal{B}} \times \mathbb{N}^{\mathcal{B}}$ *with* $|\boldsymbol{\alpha}| = |\boldsymbol{\beta}| = k$ *and* $\sum_T \alpha_T\mathbf{1}_T = \sum_T \beta_T\mathbf{1}_T$*, where every Gamma is at a pole,* $a_T = -\alpha_T$*,* $b_T = -\beta_T$*, with the residues*

$\prod_T (-1)^{\alpha_T+\beta_T}/(\alpha_T!\,\beta_T!)$ *(Prop. 1.3(iv), p. 11). The continued period has a pole of order at most* $n-1$ *at* $\sigma=-k$*, and*

$$\lim_{\sigma\to -k}(\sigma+k)^{n-1}\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1}\cdot\sum_{(\boldsymbol{\alpha},\boldsymbol{\beta})}\frac{(k!)^2}{\prod_T \alpha_T!\,\beta_T!}$$
$$=\sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1}\cdot[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k,$$

the sum over that lattice: it counts the pairs of k*-tuples of bases with equal incidence.*

Proof. On the balance wall the period is $\sum_\pi \int_{[0,1]^{n-1}} \prod_{k'=2}^{n} q_{k'}^{\kappa_{U_{k'}}-1} h_\pi(q)\,dq$ in the sector coordinates $\alpha_{\pi(k')}=\rho\,q_2\cdots q_{k'}$ (Thm. 8.7(ii), p. 55, and Thm. 8.10). At the centre $\kappa_{U_{k'}}=\sigma c(U_{k'})$ and $h_\pi=(k_\pi\psi_\pi)^{-\sigma}$, and $k_\pi\psi_\pi=\prod_{k'} q_{k'}^{c(U_{k'})}K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}$ is a polynomial in q with nonnegative coefficients and constant term one (Lemma 8.5, p. 55, weights one). By Lemma 8.6, p. 55, each cube integral continues with at most a simple pole in each $\kappa_{U_{k'}}$ at $-\mathbb{N}$, and at $\sigma=-k$ the $n-1$ poles meet only through the term q^{kc}, $c=(c(U_{k'}))_{k'}$, of $(k_\pi\psi_\pi)^k(k_\pi\psi_\pi)^{-(\sigma+k)}$, with coefficient $\prod_{k'}c(U_{k'})^{-1}[q^{kc}](k_\pi\psi_\pi)^k$. The map $\alpha^v\mapsto q^{(v(U_{k'}))_{k'}}$ is injective on exponents of degree zero, so $[q^{kc}](k_\pi\psi_\pi)^k=[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$. Expanding each letter's k-th power (the share kernel of Cor. 4.4, p. 26, closed at the exponent $-k$) gives the lattice sum. On the share side the factor $(k!)^{-2}$ is the residue of $\Gamma(\sigma)^2$ at $-k$, and the common factor of the lattice points is the residue at zero shares, the corner (Proposition 15.3). □

Corollary 15.5 (One law for the period and its closing)**.** [derived] *If polynomials* $c_0,\dots,c_d$ *satisfy*

$$\sum_{i=0}^{d}c_i(\sigma)\,P_{\mathcal{B}}((\sigma+i)\,\mathbf{1};(\sigma+i),(\sigma+i))=0$$

identically, then for every $k\in\mathbb{N}$

$$\sum_{i=0}^{\min(d,k)}c_i(-k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{k-i}=0.$$

So the least law of the period in its strength is at least as long as the least recurrence of its closings, and the closings obey the reflected law from $k=0$*, with nothing below.*

Proof. Multiply by $(\sigma+k)^{n-1}$ and let $\sigma\to -k$: the terms with $i\le k$ tend to $c_i(-k)$ times the corner times the closing at $k-i$ by Proposition 15.4, and the others are regular there. □

The laws come from the family's own relations. By the letter presentation (Prop. 3.6, p. 20) the centre period is a Breta with one bunch for the shape and one singleton bunch per letter; its letter splits and scale exchanges (Thm. 3.5, p. 19) are the letter and mark relations of the interest (Thm. 8.3(i)–(iii)), and along the centre ray they are the relations of Thm. 35.3, p. 308, whose eliminations are the laws of Cor. 35.6. The converse of Corollary 15.5, that the least recurrence of the closings, read at $k=-\sigma$, is a law of the period, is not derived here; for the banana through seven marks it is not needed, the laws being derived by adding marks (Theorem 15.25). For every list below the minimal solution it gives agrees with an independent quadrature to fourteen digits, and the closings obey it from $k=0$ with nothing below [computed].

Procedure 15.6 (The centre ray from the list)**.** (i) Close the share side at $\sigma = -k$: $x_k = [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$, $k \le k_{\max}$, the letter x_k abbreviating them in this procedure. (ii) Find their least recurrence $\sum_{j=0}^{d} p_j(k)x_{k+j} = 0$ by exact linear algebra and check it on the surplus equations. (iii) Read it at $k = -\sigma$: $c_i(\sigma) = p_{d-i}(-\sigma - d)$, $\sum_i c_i(\sigma)P_{\mathcal{B}}((\sigma+i)\,\mathbf{1}; (\sigma+i), (\sigma+i)) = 0$. (iv) The period is its minimal solution: $P_{\mathcal{B}}((\sigma+1)\,\mathbf{1}; (\sigma+1), (\sigma+1))/P_{\mathcal{B}}(\sigma\,\mathbf{1}; \sigma, \sigma) \to K(\mathbf{1})^{-2}$ (Prop. 35.7, p. 309), the native value of least modulus. (v) Scale it by the residue at zero shares: c_0 has a zero of order $n-1$ at 0, and $[\sigma^{n-1}]c_0 \cdot \sum_\pi \prod_{k=2}^{n} c(U_k)^{-1} + \sum_{i\ge 1} c_i(0)P_{\mathcal{B}}(i\,\mathbf{1}; i, i) = 0$.

Receipts [computed], no quadrature in the procedure:

list	$P_{\mathcal{B}}(\mathbf{1}; 1, 1)$	reading
$\binom{[3]}{1}$	2.34390723868945889060156 2…	$3L(\chi_{-3}, 2)$ (alias)
$\binom{[4]}{1}$	8.41439832211715999779816 7…	$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$
$\binom{[4]}{2}$	2.34390723868945889060156 2…	as $\binom{[3]}{1}$: Cor. 19.17 (p. 143) and duality
$\binom{[5]}{1}$	39.94558801199624569894402…	$M_5(1)$ of §14.17

The four-mark values are those of the note (Ex. 6.3 and 6.2, p. 39). There they are closings of the share side at positive strength: the Gammas of a basis meet in double poles, whose Laurent data are Trailmoments of the counting input and of lattice inputs (Prop. 5.14, p. 33; Lemma 35.20, p. 314).

15.2 The masters are the signs

A native gauge of the centre ray is a point where its direction $(\mathbf{1}; 1, 1)$ is native (Def. 35.4, p. 308): the defect $1 - R_e(\alpha) - R_e^*(\alpha)$ vanishes at every mark, and since $R_e^*(\alpha) = 1 - R_e(\alpha^{-1})$ (Thm. 7.14, p. 50) this reads

$$R_e(\alpha) = R_e(\alpha^{-1}) \qquad (e \in \mathcal{E})\,:$$

inverting the gauge leaves the occupation unchanged. Its native value is $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$.

Proposition 15.7 (The banana has only signs)**.** [derived] *The native gauges of the centre ray of* $\binom{\mathcal{E}}{1}$ *are exactly the signs* $\eta \in \{\pm 1\}^{\mathcal{E}}$, $\eta_{e_0} = 1$, *with* $K(\eta) = \sum_e \eta_e \neq 0$*; their native values are* $K(\eta)^{-2}$ *(Thm. 35.12, p. 310).*

Proof. For $K = \sum_e \alpha_e$ the occupation is $R_e(\alpha) = \alpha_e/K(\alpha)$, so $R_e(\alpha) = R_e(\alpha^{-1})$ for every e says $\alpha_e^2 = K(\alpha)/K(\alpha^{-1})$ for every e. □

So along the banana sequence the native values are $(n-2j)^{-2}$, j the number of minus signs, $\lceil n/2 \rceil$ of them, all simple. At $n = 5$ they are $\frac{1}{25}, \frac{1}{9}, 1$, the symbol of the ray law vanishes there (Cor. 35.6, p. 308), the order is three, and the corner leaves two masters. Every list has the signs with $K(\eta) \neq 0$ among the native gauges of its centre ray; what is special about $\binom{\mathcal{E}}{1}$ is that it has nothing else.

Remark 15.8 (The lowered settings of §14.17). With $g = \mathsf{G}_{0;(1,-1)}(1, \cdot)$ and $h(y) = \sum_m y^m/\Gamma(1+m)^2$, the settings $x^{(j)}(s) = \int_0^\infty y^{s-1} g^{5-j} h^j\, dy$, $j = 0, 1, 2$, satisfy the ray law in the normalization $M(s) = \Gamma(s)^2 P_{\mathcal{B}}(s\,\mathbf{1}; s, s)$, and $x^{(0)} = M$ is Proposition 15.2. Their corners are $120, 0, 0$ [computed], and $x^{(j)}(s+1)/\big(s^2 x^{(j)}(s)\big) \to (5-2j)^{-2}$: they are the sign classes with j minus signs. "The g's outnumber the h's" is $K(\eta) \neq 0$; the tie at four marks is $K(\eta) = 0$, which is not a native gauge.

Proposition 15.9 (The lowered-setting block is the frame)**.** [derived]*; receipt* [computed] *Let* $\Phi(s) = \det\big[x^{(i)}(s+l)\big]_{i,l=0}^{2}$. *By Prop. 35.8, p. 309, applied to the ray law of M (end coefficients* s^6 *and*

-225*)*, $\Phi(s+1)=\frac{s^6}{225}\Phi(s)$ *and* $\Phi(s)=\Phi(1)\,\Gamma(s)^6\,225^{1-s}$*. The amplitudes at the three sign classes,* $x^{(j)}(s)\sim 2\pi^{(5-2j)/2}2^{-j}\rho_j^{5/2}\,\Gamma(2s-\frac{5}{2})\,\rho_j^{-2s}$ *with* $\rho_j=2(5-2j)$*, give*

$$\Phi(1)=\frac{128\sqrt{15}}{16875}\,\pi^3=0.91088056479820303289039718797 5\ldots,$$

which the settings computed directly (quadrature, as a check) reproduce to thirty digits. The corners reduce the third column, so

$$\Phi(1)=\frac{120}{225}\Big(x^{(1)}(1)\,x^{(2)}(2)-x^{(1)}(2)\,x^{(2)}(1)\Big),\qquad x^{(1)}(1)x^{(2)}(2)-x^{(1)}(2)x^{(2)}(1)=\frac{16\sqrt{15}}{1125}\,\Gamma(\tfrac12)^6:$$

the block of §14.17 is the frame of the ray over its corner. The masters do not enter Φ*; their coefficients are minors that the zero corners of the lowered settings kill. So the frame does not shorten the last minor of §14.17.*

15.3 A parallel class closes through the banana

Proposition 15.10. [derived] *Let* $\mathcal{B}$ *be obtained from a list* $\mathcal{H}$ *on the marks* $1,\ldots,m$ *by replacing mark* i *with a parallel class* P_i *of* s_i *marks, so that* $K_{\mathcal{B}}(\alpha)=K_{\mathcal{H}}\big((\alpha(P_i))_i\big)$ *with* $\alpha(P_i)=\sum_{e\in P_i}\alpha_e$*. Then* $[\alpha^v]K_{\mathcal{B}}^k=\big([A^w]K_{\mathcal{H}}^k\big)\prod_i\big(w_i!/\prod_{e\in P_i}v_e!\big)$ *with* $w_i=v(P_i)$*, and the closings of* $\mathcal{B}$ *are*

$$\sum_w\big([A^w]K_{\mathcal{H}}(A)^k\big)^2\prod_i[\alpha^0]\Big(K_{P_i}(\alpha)\Psi_{P_i}(\alpha)/\alpha^{P_i}\Big)^{w_i},$$

with $K_{P_i}=\sum_{e\in P_i}\alpha_e$ *and* Ψ_{P_i} *the letters of the banana* $\binom{P_i}{1}$*: its closings at* $-w_i$*.*

The banana enters the closing only through its own closings, as the parallel node enters the network through its full leans (Cor. 25.9, p. 209).

15.4 The five-mark centres

All connected lists on five marks, up to duality, which leaves the centre unchanged (Prop. 8.14, p. 57); "doubled" and "tripled" are parallel classes. [computed]

list	$\lvert\mathcal{B}\rvert$	$K(\eta)^2$ at the signs (mult.)	roots of the symbol	order	corner	masters
$\binom{[5]}{1}$	5	25 (1), 9 (5), 1 (10)	$25,9,1$	3	120	2
$\binom{[4]}{2}$, a mark doubled	9	81 (1), 9 (6), 1 (9)	$81,9,1$	3	36	2
$\binom{[5]}{2}$	10	100 (1), 4 (15)	$100,4,1,-8$	4	30	3
$\binom{[3]}{2}$, two marks doubled	8	64 (1), 16 (2), 4 (8), 0 (5)	$64,16,4,4,1$	5	44	$3^\dagger$
$\binom{[3]}{2}$, a mark tripled	7	49 (1), 25 (1), 9 (3), 1 (11)	$49,25,9,1,1,1$	6	60	$4^\dagger$

list	$P_{\mathcal{B}}(\mathbf{1};1,1)$
$\binom{[5]}{1}$	39.945588011996245698944028498423995891849782924 77…
$\binom{[4]}{2}$, a mark doubled	5.493901143580844012444257607820691195853638640680 1…
$\binom{[5]}{2}$	4.029714170711049607320695734428406933139867563639 6…
$\binom{[3]}{2}$, two marks doubled	7.865837368364021592421974697446168210982382809458 1…
$\binom{[3]}{2}$, a mark tripled	12.848722141915555957312339931253534636619123934709…

The roots of the symbol are those of the recurrence of the closings, the values $K(\alpha)K(\alpha^{-1})$ at the native gauges; the ray law of the period has their reciprocals. "Masters" is $m-1$ for the least m with a rational relation among $1, P_{\mathcal{B}}(\mathbf{1};1,1),\dots,P_{\mathcal{B}}(m\,\mathbf{1};m,m)$ (PSLQ at 120 digits, coefficients below 10^{15}, values good to 150 digits): over $\mathbb{Q}$, for the symmetric insertion, on the centre ray. In the first three rows the one relation is the corner.

† In the last two rows c_0 and c_1 both carry $(\sigma+1)^2$, so the ray law at $\sigma=-1$ reaches the period at neither -1 nor 0, and it is a second relation, a *second corner*: the boundary term, the constant term at $\sigma=-1$ (Thm. 1.6, p. 12),

$$\operatorname*{CT}_{\sigma=-1}\Big(c_0(\sigma)P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)+c_1(\sigma)P_{\mathcal{B}}((\sigma{+}1)\,\mathbf{1};(\sigma{+}1),(\sigma{+}1))\Big)=-\sum_{i\geq 2}c_i(-1)\,P_{\mathcal{B}}((i{-}1)\,\mathbf{1};(i{-}1),(i{-}1)),$$

is rational [computed]. In lowest terms the two relations read

$$\begin{aligned}&2503-566\,P_{\mathcal{B}}(\mathbf{1};1,1)+88248\,P_{\mathcal{B}}(2\,\mathbf{1};2,2)-908872\,P_{\mathcal{B}}(3\,\mathbf{1};3,3)+1425600\,P_{\mathcal{B}}(4\,\mathbf{1};4,4)=0,\\&13800002-1848045\,P_{\mathcal{B}}(\mathbf{1};1,1)+227950815\,P_{\mathcal{B}}(2\,\mathbf{1};2,2)-3601110627\,P_{\mathcal{B}}(3\,\mathbf{1};3,3)\\&\qquad+11177654265\,P_{\mathcal{B}}(4\,\mathbf{1};4,4)-8565102000\,P_{\mathcal{B}}(5\,\mathbf{1};5,5)=0,\end{aligned}$$

so the masters are $5-2=3$ and $6-2=4$. Write $c_0=(\sigma+1)^2A$ and $c_1=(\sigma+1)^2B$. Proposition 15.4 at $\sigma=-1$ requires $A(-1)\,|\mathcal{B}|+B(-1)=0$, the closing at 1 being $|\mathcal{B}|$; it holds, $B(-1)/A(-1)=-8$ and -7 [computed]. There the recursion of Corollary 15.5 leaves the corner at -1 free, so this is a check of Proposition 15.4 that the recursion does not supply.

$\binom{[5]}{2}$ has native gauges off the signs: on the pattern $\alpha\propto(a,a,b,b,1)$ the native equations have solutions with $K(\alpha)K(\alpha^{-1})=1$ and -8. There is no rational relation among 1, the two masters of $\binom{[5]}{1}$ and the three of $\binom{[5]}{2}$ (coefficients below 10^8, 150 digits), nor with π^2 adjoined.

15.5 The native shares, the roots, and an exact pairing

Every root of the symbol of the closings' recurrence is the reciprocal native value $K(\alpha_c)K(\alpha_c^{-1})$ of a native gauge α_c of the centre ray (Cor. 35.6, p. 308).

Proposition 15.11 (Both reads sit at the native shares)**.** [derived]*; receipts* [computed] *Let the list be connected, and let* $H=2\big(\kappa_{\mathbf{1}}(X_e,X_{e'})\big)_{e,e'\neq e_0}$*, twice the covariances of the indicators under the weight law (Def. 9.2, p. 61), the Hessian of Thm. 35.11, p. 310.*

(i) On the real fibre of Proposition 15.1 the native shares of Thm. 4.3(ii), p. 25, are the uniform shares $a_T=b_T=\sigma/|\mathcal{B}|$*: every basis takes the same share from each letter. This is the table's gauge* $\mathbf{1}$ *read on the share side.*

(ii) The period reads the fibre along its imaginary directions. By Thm. 4.3(iii), which at the period is Thm. 38.5(ii), p. 330,

$$P_{\mathcal{B}}(k\,\mathbf{1};k,k)\sim K(\mathbf{1})^{-2k}\Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}}(\det H)^{-\frac12}.$$

(iii) The closing reads the same fibre on its integer points. Dropping the pin, the multinomial theorem gives $\sum_{|\boldsymbol{\alpha}|=|\boldsymbol{\beta}|=k}(k!)^2/\prod_T\alpha_T!\beta_T!=K(\mathbf{1})^k\Psi(\mathbf{1})^k=|\mathcal{B}|^{2k}$*; the pin keeps the Gaussian of the fibre at the native shares scaled by* k*, and*

$$[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k\sim K(\mathbf{1})^{2k}(2\pi k)^{-\frac{n-1}{2}}(\det H)^{-\frac12}.$$

(iv) So $k^{n-1}\,P_{\mathcal{B}}(k\,\mathbf{1};k,k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k \to (\det H)^{-1}$*: one Gaussian, read once along a contour and once on a lattice, so that the powers of* 2π *cancel and the determinant appears twice.*

Proof. (i) The uniform point satisfies the pins, and $\nabla S = -\log \mathbf{w}$ is constant there, a combination of the two rows of the letter totals; S is strictly concave on the real fibre (Thm. 4.3(ii)). (ii) Thm. 38.5(ii) at the centre: $X(0) = -2\log K(\mathbf{1})$ and $-\nabla_z^2 X(0) = 8\kappa_{\mathbf{1}} = 4H$ on the chart, $y = 2z$; Thm. 35.11 is the same edge read on the generating function of the ray. (iii) The terms $(k!)^2/\prod \alpha_T!\beta_T!$ on the pinned lattice peak at the uniform shares scaled by k; their Gaussian there is the quadratic form of $\sum \log\Gamma$ at the native shares restricted to the fibre, the same form as in (ii), now summed over the integer points of the fibre; $\nabla_z^2 E(0) = 8\kappa_{\mathbf{1}}$ (Thm. 7.14, p. 50) identifies it with H. □

Receipts [computed]: $(\det H)^{-1} = \frac{3125}{16}, \frac{729}{16}, \frac{3125}{81}, \frac{512}{9}, \frac{16807}{192}$ in the order of the table of §15.4. At $k = 70$ the product agrees with these to about 10^{-4}, the error falling like k^{-2}. For $\binom{[2]}{1}$ the product is exact at every k: $k\,\Gamma_{\binom{2}{1}}\!\begin{bmatrix} k,\ k \\ 2k \end{bmatrix}\binom{2k}{k} = 2 = (\det H)^{-1}$ [derived].

Remark 15.12 (The signs are the pin read modulo two). Averaging over the signs detects the incidence modulo two: for every k,

$$2^{-n}\sum_{\eta\in\{\pm1\}^{\mathcal{E}}} K(\eta)^{2k} = \sum_{\substack{|\boldsymbol{\alpha}|=|\boldsymbol{\beta}|=k \\ \sum_T \alpha_T \mathbf{1}_T \equiv \sum_T \beta_T \mathbf{1}_T \ (2)}} \frac{(k!)^2}{\prod_T \alpha_T!\,\beta_T!} \ \geq\ [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k \quad \text{[derived]},$$

since $K(\eta)\Psi(\eta)/\eta^{\mathcal{E}} = K(\eta)^2$. So the sign values $K(\eta)^2$ are the growth rates of the pin's reduction modulo two, with $K(\mathbf{1})^2$ the largest. The full pin keeps them as roots (Thm. 35.12) and adds the native gauges off the signs, below.

Proposition 15.13 (An exact pairing of the period with its closing for the five-mark banana)**.** [derived] *from the ray law; receipt* [computed]*. For* $\binom{[5]}{1}$ *the ray law* $\sum_{i=0}^{3} c_i(\sigma) P_{\mathcal{B}}((\sigma+i)\,\mathbf{1};(\sigma+i),(\sigma+i)) = 0$ *has*

$$c_0 = -\sigma^4, \qquad c_1 = 35\sigma^4 + 70\sigma^3 + 63\sigma^2 + 28\sigma + 5,$$
$$c_2 = -(\sigma+1)^2(259\sigma^2 + 518\sigma + 285), \qquad c_3 = 225(\sigma+1)^2(\sigma+2)^2,$$

and it is self-adjoint under the reflection of the strength: $c_i(-\sigma - i) = c_i(\sigma)$ *for every* i*. So the closings, read as a function of* σ*, solve the adjoint law, and for every* $\sigma \in \{1,2,3,\dots\}$

$$\sum_{i=1}^{3}\sum_{j=1}^{i} c_i(\sigma-j)\ [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{\sigma-j}\ P_{\mathcal{B}}((\sigma-j+i)\,\mathbf{1};(\sigma-j+i),(\sigma-j+i)) = 120,$$

the residue at zero shares, with the closings at negative indices read as 0*.*

Proof. The left side, $B(\sigma)$ in this proof, is the bilinear concomitant of the law and its adjoint: $B(\sigma+1) - B(\sigma)$ is the law applied to the period times the closing at σ, less the period at σ times the adjoint law applied to the closings. The first vanishes by the law. The adjoint law at σ is $\sum_i c_i(\sigma - i)x_{\sigma-i}$, which by $c_i(\sigma - i) = c_i(-\sigma)$ is the recurrence of Corollary 15.5 at $k = \sigma$, zero from $k = 0$ on. At $\sigma = 1$ only $j = 1$ survives, and $B(1) = \sum_i c_i(0) P_{\mathcal{B}}(i\,\mathbf{1};i,i) = 120$ is the corner relation of Procedure 15.6(v). The self-adjointness is an identity of the four polynomials; $B(\sigma) = 120$ holds to seventy digits for $\sigma \leq 55$ [computed]. □

At large σ the closings and the period move by the factors 25 and $\frac{1}{25}$ per step, so $B(\sigma) \to \lambda\chi'(\lambda)\cdot\lim_k k^4\, P_{\mathcal{B}}(k\,\mathbf{1};k,k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$, with $\chi(\lambda)=\sum_i[\sigma^4]c_i\,\lambda^i=-1+35\lambda-259\lambda^2+225\lambda^3$ the symbol and $\lambda=\frac{1}{25}$. With Proposition 15.11:

$$120\cdot\det H=\lambda\chi'(\lambda)\big|_{\lambda=1/25}=\frac{384}{625},\qquad \det H=\frac{16}{3125}\quad\text{[derived]}.$$

The residue at zero shares (the low end of strength, Thm. 38.5(i)) and the Gaussian at the native shares (the high end, Thm. 38.5(ii)) are the two readings of one exact identity between the fibre and its lattice. For the other four five-mark lists the closings solve no adjoint of the ray law in the forms tried: no combination $\sum_{l<d} r_l(\sigma)[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{\sigma+l}$ with polynomial r_l of degree at most 20, nor with the apparent factors of c_0 as denominators up to degree 14, solves it [computed]. In that framework the pairing of the period with its closing is not a single product there; it must carry other solution classes.

Proposition 15.14 (Signs through parallel classes)**.** [derived] *In the setting of Proposition 15.10, $K_{\mathcal{B}}(\eta)=K_{\mathcal{H}}\big(s_1(\eta),\dots,s_m(\eta)\big)$ with $s_i(\eta)=\sum_{e\in P_i}\eta_e\in\{s_i,s_i-2,\dots,-s_i\}$: the sign values of the bananas $\binom{P_i}{1}$ feed the letter of $\mathcal{H}$.*

For the triangle $\mathcal{H}=\binom{[3]}{2}$, $K_{\mathcal{H}}=e_2$, and with one mark tripled, $s=s_1(\eta)\in\{3,1,-1,-3\}$:

$$e_2(s,1,1)=2s+1\in\{7,3,-1,-5\},\qquad e_2(s,1,-1)=-1\quad\text{for every } s,$$

which is the census 49 (1), 25 (1), 9 (3), 1 (11) of §15.4. The second form does not see s.

Proposition 15.15 (Letters that forget a class)**.** [derived] *Write $K_{\mathcal{B}}=A_i\,K_{\mathcal{H}/i}(A)+K_{\mathcal{H}\setminus i}(A)$ with $A_i=\alpha(P_i)$ (Def. 7.3, p. 48), and $A_i'=\sum_{e\in P_i}\alpha_e^{-1}$.*

(i) Tripled triangle. On $\alpha_3=-\alpha_2$ the contraction letter $K_{\mathcal{H}/1}=A_2+A_3$ vanishes, so every copy has occupation zero at α and at α^{-1}, and $K(\alpha)=-\alpha_2^2$, $K(\alpha^{-1})=-\alpha_2^{-2}$ whatever the copies are. The native gauges there are

$$\big\{\alpha_3=-\alpha_2,\ A_1=\alpha_2^2A_1'\big\},$$

a family of dimension two modulo scale, all of native value 1, containing the eight sign classes with $\eta_3=-\eta_2$.

(ii) Two marks doubled. On $P_2=(t,-t)$ the class letter vanishes, $A_2=A_2'=0$, and $K=A_1\alpha_3$ is the letter of $\mathcal{H}\setminus 2$, in which both remaining classes are coloops. The native gauges there are

$$\Big\{P_1=(c,c),\ P_2=(t,-t),\ t^2=\frac{c\,\alpha_3\,(2\alpha_3+c)}{2c+\alpha_3}\Big\},$$

a curve of native value $\frac14$ which on the torus $|\alpha_e|=1$ is a circle through the signs with $(s_1,s_2)=(\pm2,0)$.

Proof. Nativity is $R_e(\alpha)=R_e(\alpha^{-1})$ at every mark. (i) At a copy $x\in P_1$, $R_x(\alpha)=\alpha_x(A_2+A_3)/K(\alpha)$ and $R_x(\alpha^{-1})=\alpha_x^{-1}(A_2'+A_3')/K(\alpha^{-1})$, both zero on $\alpha_3=-\alpha_2$; at α_2 and α_3 the two sides differ by $\mp(A_1/\alpha_2-\alpha_2A_1')$. (ii) At α_3 both occupations are 1 on $A_2=A_2'=0$; at P_1 the equation holds when $\alpha_{1a}^2=\alpha_{1b}^2=A_1/A_1'$, and $\alpha_{1b}=-\alpha_{1a}$ gives $K=0$; at P_2 it holds exactly at the stated t^2. When $|c|=|\alpha_3|=1$, $|2\alpha_3+c|=|2c+\alpha_3|$, so $|t|=1$. □

Remark 15.16 (Local exponents). Put $x_k \approx \lambda^k k^\beta$ into the recurrence of the closings [computed]. At every simple root of every row, $\beta = -2 = -\frac{n-1}{2}$, the exponent of a nondegenerate native gauge (Thm. 35.5(i), p. 308). At the repeated roots the indicial polynomials are

$$(\beta+1)(\beta+2)(\beta+3) \quad \text{(tripled triangle, } \lambda=1\text{)}, \qquad (2\beta+3)(2\beta+5) \quad \text{(two marks doubled, } \lambda=4\text{)}.$$

A family of real dimension d on the torus $|\alpha_e| = 1$, transversally nondegenerate, gives $\beta = -\frac{n-1-d}{2}$. The Hessian of $\log\big(K(\alpha)K(\alpha^{-1})\big)$ there has rank two along the family of (i) and rank three along the circle of (ii) [computed], so these are -1 and $-\frac{3}{2}$. The isolated sign orbit at $\lambda = 1$ (one copy plus, two minus, $\eta_2 = \eta_3$) has full rank and gives -2. The exponents -3 and $-\frac{5}{2}$: [not yet computed]. These families are integral directions in the exceptional set of Thm. 35.19(i), p. 313: the order of the ray law counts native values with these multiplicities, and the families raise the order, and with it the masters, past the number of native values.

Remark 15.17 (Native gauges at complex points off the signs [computed]). $\binom{[5]}{2}$: $K(\alpha)K(\alpha^{-1}) = 1$ at $(\omega^2, \omega^2, \omega, \omega, 1)$, $\omega = e^{2\pi \mathrm{i}/3}$, where $K = K(\alpha^{-1}) = 1$; and -8 at $(1, 1, b, b, 1)$, $b = -(2+\sqrt{3})$, and at its inverse, where $K = -2-2\sqrt{3}$ and $K(\alpha^{-1}) = -2+2\sqrt{3}$. Two marks doubled: 1 at $(\omega, \omega^2, \omega, \omega^2, 1)$. The signs are the points of order two of the torus; points of order three are the next layer that occurs here.

15.6 Closing the banana at strength one

Proposition 15.18 (Four marks close on the half-lattice)**.** [derived] *For* $\binom{[4]}{1}$*, merging the marks in pairs by four Gammas on a line (Cor. 4.7, p. 26),* $\int_{a_1+a_2=c} \Gamma(a_1)^2\Gamma(a_2)^2 = \Gamma^{\,4}_{\,1}\!\left[\begin{matrix} c,\ c,\ c,\ c \\ 2c \end{matrix}\right]$*, leaves a fibre of dimension one, and*

$$\begin{aligned} P_{\mathcal{B}}(\mathbf{1}; 1, 1) &= \frac{1}{2\pi i}\int_{(1/2)} \Gamma^{\,4}_{\,1}\!\left[\begin{matrix} c,\ c,\ c,\ c \\ 2c \end{matrix}\right] \Gamma^{\,4}_{\,1}\!\left[\begin{matrix} 1-c,\ 1-c,\ 1-c,\ 1-c \\ 2-2c \end{matrix}\right] dc \\ &= \big(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}}\big)((0,\infty)) = 7\,\big(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}}\big)((0,\infty)). \end{aligned}$$

Proof. By Lemma 5.16, p. 34, and the shift and reflection of the stacked Gamma (Prop. 1.3(iii),(v), p. 11) the integrand is

$$\Gamma^{\,4}_{\,0}\big[c,\ c,\ 1-c,\ 1-c\big]^2\ \frac{\sin 2\pi c}{\pi(1-2c)} = \frac{2\pi^3 \cos \pi c}{(1-2c)\sin^3 \pi c}.$$

On the lines $\operatorname{Re} c = N + \frac12$ it is $O\big(N^{-1}e^{-2\pi|\operatorname{Im} c|}\big)$, so it closes to the right (Thm. 4.5, p. 26). At $c = m \geq 1$, $2\pi^3 \cos \pi c / \sin^3 \pi c = 2(c-m)^{-3} + O(c-m)$, with no parts of orders one and two, as in Lemma 5.16; the half-integers are zeros. So the residue is $2\,[\varepsilon^2](1-2m-2\varepsilon)^{-1} = -8/(2m-1)^3$, and the closing is $\sum_{m\geq1} 8(2m-1)^{-3} = \sum_{x\in\frac12+\mathbb{Z},\,x>0} x^{-3}$. The last equality is the coset relation at the step 2 (Thm. 5.12(i), p. 32). The half-lattice is the doubled share $2c$ of a merged pair, the lower entry of its stacked Gamma. □

This is the note's value (Ex. 6.2, p. 39), now read as a closing of the share side.

Proposition 15.19 (A mark added is a critical-line kernel)**.** [derived]*; receipts* [computed] *For every* $n \geq 1$*,*

$$P_{\binom{[n+1]}{1}}(\mathbf{1}; 1, 1) = -\sum_{m\geq1} \partial_\sigma P_{\binom{[n]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)\big|_{\sigma=m}.$$

Proof. In the share side of $\binom{[n+1]}{1}$ (Proposition 15.2) merge the first n marks: their fibre at the total d is $\Gamma(d)^2 P_{\binom{[n]}{1}}(d\mathbf{1}; d, d)$, by the same proposition at n marks. The last mark carries $\Gamma(1-d)^2$, and $\Gamma(d)^2\Gamma(1-d)^2 = \Gamma_{\substack{4\\0}}\big[d,\ d,\ 1-d,\ 1-d\big]$, whose right closing reads the derivative of the rest at the positive integers (Lemma 5.16(i)). The rest is holomorphic on $\operatorname{Re} d > 0$, bounded on each vertical line by its value at the real part, and decays as $\operatorname{Re} d \to \infty$, so the closing applies (Thm. 4.5). □

Receipts [computed]. At $n = 2$, with $\partial_\sigma \Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right] = 2\Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right]\big((\log\Gamma)'(\sigma) - (\log\Gamma)'(2\sigma)\big)$ and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ (Prop. 5.14, p. 33),

$$P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = 2\sum_{m\ge 1} \Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}m,\ m\\ 2m\end{smallmatrix}\right]\ (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m-1, 2m-1]) = 2.343907238689458890601562\ldots,$$

the period of two marks times a head of the counting input, to 25 digits. At $n = 3$ the sum is $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and at $n = 4$ it is $39.945588011996245699\ldots$, both to 20 digits.

Remark 15.20 (The five-mark centre is a Trailmoment of a live input). By Proposition 15.19,

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = \big(\mathsf{T}_{0;(1)}[x]\, f\,\mathbb{I}_{\mathbb{Z}}\big)((0,\infty)), \qquad f(m) = -\partial_\sigma P_{\binom{[4]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)\big|_{\sigma=m},$$

a live input (Chapter 6): its density is read from the four-mark centre period, and the four-mark ray law differentiated in σ gives it a recurrence of order two whose inhomogeneity is the values. At $\sigma = 0$ the differentiated law is a relation among the first two slopes, the corner and the values, the term of order σ of Proposition 15.3 being zero. So every $f(m)$ lies in $\mathbb{Q} + \mathbb{Q}(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \mathbb{Q}f(1)$ [derived]. For example $P_{\binom{[4]}{1}}(2\,\mathbf{1}; 2, 2) = (7\zeta(3) - 6)/16$ and $f(2) = -\big(36 + 21\zeta(3) - 2f(1)\big)/32$ (alias $\zeta(3) = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$) [computed]. The first slope is

$$-f(1) = 48\,\mathrm{Li}_4(\tfrac{1}{2}) + 2\log^4 2 - 2\pi^2\log^2 2 - \tfrac{17}{30}\pi^4 = -39.381582123122165624939809345 08\ldots$$

by PSLQ at 60 digits [computed]. In the family these aliases are Trailmoments of index at most four of the counting input at the frequencies 0 and $\frac{1}{2}$ and of the lattice input of density 2^{-x} (Def. 5.2, p. 28). So the four-mark slope sits on the half-lattice of Proposition 15.18, at index four. The masters of the five-mark centre are values of this live Trailmoment, and this is where the closing stops landing on plain lattice inputs.

15.7 Closing the live input of five marks

Throughout this subsection $p(\sigma) = P_{\binom{[4]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ for real $\sigma > 0$ and $f = -p'$; $p\,\mathbb{I}_{\mathbb{Z}}$ and $f\,\mathbb{I}_{\mathbb{Z}}$ are the *values input* and the *slope input* (Def. 5.4, p. 29). The four-mark ray law is

$$\sum_{i=0}^{2} c_i(\sigma)\, p(\sigma + i) = 0, \qquad c_0 = \sigma^3, \quad c_1 = -2(2\sigma+1)(5\sigma^2+5\sigma+2), \quad c_2 = 64(\sigma+1)^3,$$

read off the closings by Procedure 15.6 and derived in Theorem 15.25 below. For a function R on a coset of $\mathbb{Z}$ put $(L^*R)(x) = \sum_i c_i(x-i)\,R(x-i)$, and let L'^* be the same with the derivatives c_i' in place of the c_i.

Proposition 15.21 (The live recurrence)**.** [derived] *The slope input has a recurrence of order two whose inhomogeneity is the values input:*

$$\sum_{i=0}^{2} c_i(\sigma)\, f(\sigma + i) = \sum_{i=0}^{2} c_i'(\sigma)\, p(\sigma + i), \qquad c_0' = 3\sigma^2, \quad c_1' = -6(10\sigma^2+10\sigma+3), \quad c_2' = 192(\sigma+1)^2.$$

Proof. The law holds identically in σ (Theorem 15.25); differentiate it. □

Proposition 15.22 (A mark added, at every strength)**.** [derived]*; receipts* [computed] *For real* $s > 0$ *let* $W_s(d) = \Gamma_2^2\begin{bmatrix} d,\ d \\ 1+d-s,\ 1+d-s \end{bmatrix}$. *Then*

$$\Gamma(s)^2\, P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = -\big(\mathsf{T}_{0;(1)}[x]\,(W_s p)'\,\mathbb{I}_{s+\mathbb{Z}}\big)((s-1,\infty)),$$
$$x^{(1)}(s) = \quad \big(\mathsf{T}_{0;(1)}[x]\,W_s p\,\mathbb{I}_{s+\mathbb{Z}}\big)((s-1,\infty)),$$

with $x^{(1)}$ *the lowered setting with one* h *of Remark 15.8. The period reads the slopes of the word* $W_s p$ *on the coset of its strength, and the lowered setting reads its values. At* $s \in \{1, 2, \dots\}$ *the word* $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$ *is a polynomial with double zeros at* $1, \dots, s-1$*, the cut moves to* 0*, and*

$$\Gamma(s)^2 P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = (\mathsf{T}_{0;(1)}[x]\,(W_s f - W_s' p)\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad x^{(1)}(s) = (\mathsf{T}_{0;(1)}[x]\,W_s p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. Merge the first four marks in the share side of five (Proposition 15.2): $\Gamma(s)^2 P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2 p(d)\,\Gamma(s-d)^2\,dd$ with $0 < c < s$, as in the proof of Proposition 15.19. By the reflection (Prop. 1.3(v), p. 11), $\Gamma(s-d)\Gamma(1-s+d) = \pi/\sin\pi(s-d)$, so

$$\Gamma(d)^2\,\Gamma(s-d)^2 = W_s(d)\,\Gamma_0^4\big[s-d,\ s-d,\ 1-s+d,\ 1-s+d\big]\,,$$

the stacked Gamma of Lemma 5.16, p. 34, at $a = s-d$: double poles at $d \in s+\mathbb{Z}$, with no simple parts. The double zeros of W_s at $d = s-1, s-2, \dots$ remove the poles between the contour and s. The rest is as in Proposition 15.19: $W_s p$ is holomorphic on $\operatorname{Re} d > 0$, of polynomial growth along vertical lines, where the kernel decays like $e^{-2\pi|\operatorname{Im} d|}$, and it decays like $16^{-\operatorname{Re} d}$ to the right; closing to the right (Thm. 4.5, p. 26) reads $-(W_s p)'$ at $d = s, s+1, \dots$. For $x^{(1)}$, expand $h = \sum_k y^k/\Gamma(1+k)^2$ against g^4 (Proposition 15.2 at four marks): $x^{(1)}(s) = \sum_{k\ge 0}\Gamma(s+k)^2 p(s+k)/\Gamma(1+k)^2$, and $\Gamma(s+k)^2/\Gamma(1+k)^2 = W_s(s+k)$. □

Receipts [computed]: at $s = \frac{3}{2}$ both sides of the first identity are 2.27912905076377, and at $s = \frac{9}{4}$ both are 0.129204079141924 (the left side by one-dimensional quadrature of the share side; fifteen digits).

Proposition 15.23 (Summation by parts at the cut: what lands on lattice inputs)**.** [derived] *For every polynomial* R*,*

$$(\mathsf{T}_{0;(1)}[x]\,(L^* R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -24\,R(0),$$
$$(\mathsf{T}_{0;(1)}[x]\,(L^* R)\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,(L'^* R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. By the shifted inputs (Thm. 6.2, p. 42) a shift of the input is a shift of the word, and the cut relation (Prop. 5.3, p. 29) returns the points that cross the cut: for a density φ,

$$\sum_{x\ge 1} R(x)\sum_i c_i(x)\,\varphi(x+i) = (\mathsf{T}_{0;(1)}[x]\,(L^* R)\,\varphi\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - R(0)\sum_{i\ge 1} c_i(0)\,\varphi(i) - R(-1)\,c_2(-1)\,\varphi(1),$$

and $c_2(-1) = c_2'(-1) = 0$. For $\varphi = p$ the left side vanishes by the law. For $\varphi = f$ it is $\sum_x R(x)\sum_i c_i'(x)\,p(x+i)$ by Proposition 15.21, to which the same move applies. So the boundaries are

$$\sum_{i\ge 1} c_i(0)\,p(i) = -4p(1) + 64p(2), \qquad \sum_{i\ge 1}\big(c_i(0) f(i) - c_i'(0) p(i)\big) = -4f(1) + 64f(2) + 18p(1) - 192p(2).$$

The first is $-[\sigma^3]c_0$ times the corner 24 (Procedure 15.6(v)); the second is the term of order σ of $\sigma^3 p(\sigma)$, which vanishes (Proposition 15.3). With the values of Remark 15.20 they are -24 and 0: the lattice Trailmoments in them, $p(1) = 7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$ and $f(2)$, cancel identically. □

So what the closing of the live Trailmoment puts on lattice inputs is the residue at zero shares, and nothing else: neither $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ nor the four-mark slope $f(1)$ survives.

Proposition 15.24 (The fixed block)**.** [derived]*; receipts* [computed] *L^* raises the degree of a polynomial by three, with leading factor $1 - 20 + 64 = 45$, the symbol of the law at $\lambda = 1$, which is not a native value of four marks; so the polynomials are $L^*(\mathbb{Q}[x]) \oplus \operatorname{span}\{1,\, x-1,\, (x-1)^2\}$. By Proposition 15.23, every Trailmoment at index zero of a polynomial word against the values input or the slope input, cut at 0, is a rational number plus a rational combination of the six*

$$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \qquad (j=0,1,2).$$

The share side adds one relation:

$$5\,(\mathsf{T}_{0;(1)}[x]\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 5\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. The splitting: $\deg L^*R = \deg R + 3$, so L^* is injective on polynomials and its image misses exactly three dimensions in each degree. The relation says that each mark takes on average a fifth of the strength. On the fibre of Proposition 15.2 the shares sum to σ, and the integrand and the pins are symmetric in the five marks; so the fibre integral with one share a_e inserted is the same for every e, and the five sum to $\sigma\,\Gamma(\sigma)^2 P_{\binom{[5]}{1}}(\sigma\mathbf{1};\sigma,\sigma)$. Merging the first four marks, the fifth share is $\sigma - d$, and at $\sigma = 1$ the closing of Proposition 15.22 reads its insertion as $-\sum_{x\ge1}\partial_d\big((1-d)p(d)\big)\big|_{d=x} = (\mathsf{T}_{0;(1)}[x]\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{0;(1)}[x]\,(x-1)f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; the right side is $\frac{1}{5}P_{\binom{[5]}{1}}(\mathbf{1};1,1) = \frac{1}{5}(\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. □

Receipts [computed]. At the strengths $2,\dots,5$ the same insertion reduces to rational multiples of this relation. There is no rational relation among 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$ and the five Trailmoments other than $(\mathsf{T}_{0;(1)}[x]\,(x-1)f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ with coefficients below 10^{30} (PSLQ, 360 digits): in that framework, over $\mathbb{Q}$, the block is five-dimensional. Its six values:

j	$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$
0	8.57028044337446126814969582424653728 6…	39.945588011996245698944028498423995 89…
1	0.16106970227667485885131066624851419 8…	0.581162840975212128360890124561738107…
2	0.17187458138353690484768349157529391 6…	0.616835511971624849092924541153028126…

Theorem 15.25 (A mark added carries the ray law)**.** [derived]*; receipts* [computed] *Normalize the ray law of $\binom{[n]}{1}$, $\sum_{i=0}^{d_n} c_i^{(n)}(\sigma)\,P_{\binom{[n]}{1}}((\sigma+i)\mathbf{1};\sigma+i,\sigma+i) = 0$, so that $c_0^{(n)} = (-1)^n\sigma^{n-1}$, as for four marks above and for five marks in Proposition 15.13, and let L_n^* be its adjoint. For $n = 1,\dots,6$,*

$$\sum_{i=0}^{d_{n+1}} c_i^{(n+1)}(s)\binom{x-1}{s-1+i}^2 = \big(L_n^* A_s\big)(x), \qquad A_s(x) = \big((n+1)x - (n+2)s\big)\binom{x}{s}^2,$$

an identity of rational functions of x and s once both sides are divided by $\binom{x}{s}^2$. Read one way it produces the law: the words $\binom{x-1}{s-1+i}^2$ have distinct degrees, so the coefficients of $L_n^*A_s$ in them are the $c_i^{(n+1)}(s)$. Read the other way it proves it: for $2 \le n \le 6$, if the ray law of $\binom{[n]}{1}$ holds identically in σ, so does that of $\binom{[n+1]}{1}$, and the lowered setting $x^{(1)}$ of $n+1$ marks, divided by $\Gamma(s)^2$, obeys it

too. The two-mark period $P_{\binom{[2]}{1}}(\sigma\mathbf{1};\sigma,\sigma)=\Gamma_2\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}$ *obeys* $\sigma P(\sigma)-2(2\sigma+1)P(\sigma+1)=0$ *by the shift of the stacked Gamma (Prop. 1.3(iii), p. 11). So the ray laws of* $\binom{[n]}{1}$, $n\le 7$, *are derived from the stacked Gamma, one mark at a time.*

Proof. Write p_n for the centre period of $\binom{[n]}{1}$, and let $s>0$ be non-integral. Proposition 15.22, with n marks merged in place of four, gives $P_{\binom{[n+1]}{1}}((s+i)\mathbf{1};s+i,s+i)=-\sum_{d\in s+\mathbb{N}}(w_{s,i}\,p_n)'(d)$ with $w_{s,i}=W_{s+i}/\Gamma(s+i)^2$, the terms $d=s,\dots,s+i-1$ being zero by the double zeros of W_{s+i}. On the coset $w_{s,i}(d)$ is $\binom{d-1}{s-1+i}^2$ written with Gammas, so the displayed identity holds there. On $s+\mathbb{N}$ the move of Proposition 15.23 has no boundary for the words A_s and A_s', which vanish at $d=s-1,s-2,\dots$: $\sum_d(L_n^*A)\,\varphi=\sum_d A\,(L_n\varphi)$, and likewise with L_n'. With $(L_n^*A)'=L_n'^*A+L_n^*(A')$ and $L_np_n'=-L_n'p_n$ (the law differentiated),

$$\sum_i c_i^{(n+1)}(s)\,P_{\binom{[n+1]}{1}}((s+i)\mathbf{1};s+i,s+i)=-\sum_{d\in s+\mathbb{N}}\Big((L_n'^*A_s)\,p_n+A_s'\,L_np_n+A_s\,L_np_n'\Big)(d)$$
$$=-\sum_d(L_n'^*A_s)\,p_n+\sum_d A_s\,L_n'p_n=0,$$

and $\sum_i c_i^{(n+1)}(s)\,\Gamma(s+i)^{-2}x^{(1)}(s+i)=\sum_d(L_n^*A_s)\,p_n=\sum_d A_s\,L_np_n=0$. Both sides are holomorphic in s, so the laws hold identically. The displayed identity is a finite computation with rational functions of (x,s). □

Receipts [computed]. The certificate A_s was found by exact linear algebra at each $n\le 6$ and $s\le 5$, the same at every n up to the factor $(n+1)x-(n+2)s$, and the identity was then checked with s indeterminate. At $n=1$ ($p_1=1$, $c^{(1)}=(-1,1)$) it holds as an identity, but the sums of the proof diverge. The five-mark law at $s=\frac12$ and $s=\frac{3}{10}$, from the share side by quadrature: residuals $4.6\cdot10^{-14}$ and $2.3\cdot10^{-14}$ against terms of size 10^2. At $s=0$, with the cut at 0, the same reduction returns the corner relation of $n+1$ marks as $A_0'(0)=n+1$ times the boundary of Proposition 15.23: the corners $(n+1)!=(n+1)\cdot n!$ of Proposition 15.3, here $120=5\cdot24$.

Corollary 15.26 (Two masters over the corner, two lowered numbers)**.** [derived] *For every* $s\in\{1,2,\dots\}$ *there are rationals* a_s,b_s,q_s *with*

$$\Gamma(s)^2P_{\binom{[5]}{1}}(s\mathbf{1};s,s)=q_s+a_s\,P_{\binom{[5]}{1}}(\mathbf{1};1,1)+b_s\,P_{\binom{[5]}{1}}(2\mathbf{1};2,2),$$
$$x^{(1)}(s)=a_s\,x^{(1)}(1)+b_s\,x^{(1)}(2),$$

and in the block

$$P_{\binom{[5]}{1}}(\mathbf{1};1,1)=(\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$P_{\binom{[5]}{1}}(2\mathbf{1};2,2)=(\mathsf{T}_{0;(1)}[x]\,(x-1)^2f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-2\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$x^{(1)}(1)=(\mathsf{T}_{0;(1)}[x]\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\qquad x^{(1)}(2)=(\mathsf{T}_{0;(1)}[x]\,(x-1)^2p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Modulo $L^*(x\,\mathbb{Q}[x])$ *every word* W_s *lies in* $\operatorname{span}\{1,(x-1)^2\}$*: the ray reaches the direction* $x-1$ *of the block only through the value term* $-2(\mathsf{T}_{0;(1)}[x]\,(x-1)p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *of its second master. For example* $(a_3,b_3,q_3)=\big(-\frac{1}{45},\frac{19}{15},\frac{8}{15}\big)$ *and* $(a_4,b_4,q_4)=\big(-\frac{113}{1125},\frac{1907}{375},\frac{944}{375}\big)$.

Proof. The four block identities are Proposition 15.22 at $s=1,2$, with $W_1=1$, $W_2=(x-1)^2$, $W_2'=2(x-1)$. Write $W_s/\Gamma(s)^2=L^*B_s+r_s$ with $\deg r_s\le 2$. Since the display of Theorem 15.25 is

an identity of polynomials and the splitting of Proposition 15.24 is direct, the r_s obey the five-mark recurrence for $s \geq 1$, and at $s = 0$, where $A_0 = 5x$, they obey $\sum_{i\geq 1} c_i^{(5)}(0)\, r_i = 0$. As $r_1 = 1$ and $r_2 = (x-1)^2$, every r_s lies in their span, and the same recursion, from $B_1 = B_2 = 0$, gives $B_s(0) = 0$. Then $x^{(1)}(s) = \Gamma(s)^2 (\mathsf{T}_{0;(1)}[x]\, r_s\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$, and by Proposition 15.23 $P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = (\mathsf{T}_{0;(1)}[x]\, (r_s f - r_s' p)\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) + 24 B_s'(0)$, where $\sum_{i\geq 1} c_i^{(5)}(0) B_i'(0) = A_0'(0) = 5$ carries the corner. □

Proposition 15.27 (The lowered half is a stacked Gamma)**.** [computed]

$$x^{(1)}(1) = (\mathsf{T}_{0;(1)}[x]\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \frac{1}{120}\Gamma {}^{8}_{4}\!\left[\begin{matrix} \frac{1}{15}, & \frac{2}{15}, & \frac{4}{15}, & \frac{8}{15}, & \frac{1}{5}, & \frac{2}{5}, & \frac{3}{5}, & \frac{4}{5} \\ & & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2} & & \end{matrix}\right]$$

to 364 *digits, and*

$$x^{(1)}(2) = \frac{65\, x^{(1)}(1) - 32\, \Gamma(\frac{1}{2})^8 / x^{(1)}(1)}{1125} \quad (360 \textit{ digits}),$$

$$x^{(2)}(1) = \frac{\sqrt{15}}{4\, \Gamma(\frac{1}{2})^2}\, x^{(1)}(1), \qquad x^{(2)}(2) = \frac{\sqrt{15}}{4\, \Gamma(\frac{1}{2})^2} \cdot \frac{65\, x^{(1)}(1) + 32\, \Gamma(\frac{1}{2})^8 / x^{(1)}(1)}{1125} \quad (100 \textit{ digits}).$$

So the block of Proposition 15.9 is $x^{(1)}(1)x^{(2)}(2) - x^{(1)}(2)x^{(2)}(1) = \frac{\sqrt{15}}{4\Gamma(1/2)^2} \cdot \frac{64\,\Gamma(1/2)^8}{1125} = \frac{16\sqrt{15}}{1125}\Gamma(\frac{1}{2})^6$*: the stacked Gamma cancels from the frame, which is why the frame saw only* $\Gamma(\frac{1}{2})$*.*

The values half that the ray reaches is two stacked-Gamma numbers, $x^{(1)}(1)$ and $\Gamma(\frac{1}{2})^8 / x^{(1)}(1)$. The direction $x - 1$ is not among them: $(\mathsf{T}_{0;(1)}[x]\, (x-1)\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$ has no rational relation with 1, $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac{1}{2})^4$, $x^{(1)}(1)$ and $\Gamma(\frac{1}{2})^8 / x^{(1)}(1)$, nor with ten products of $x^{(1)}(1)^{\pm 1}$, powers of $\Gamma(\frac{1}{2})^2$ and $\sqrt{15}$, with coefficients below 10^{12} [computed].

Proposition 15.28 (The block on the four-mark chart and its pencil)**.** [derived]*; receipts* [computed] *On the chart* $y_{e_0} = 0$ *of four marks let* $X = K(\alpha)K(\alpha^{-1})$*, the reciprocal of the stride of the centre ray (Def. 35.9, p. 309), so that* $p(\sigma) = \int X^{-\sigma} dy$*, the value along the ray from the real origin* $(\sigma\mathbf{1}; \sigma, \sigma)$ *(Def. 35.1, p. 307), with* $X \geq 16$ *on the chart (Thm. 35.10(i), p. 309). For a polynomial* R*,*

$$(\mathsf{T}_{0;(1)}[x]\, R\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \rho_R(X)\, dy, \qquad (\mathsf{T}_{0;(1)}[x]\, R\, f\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \log X\, \rho_R(X)\, dy,$$

$$\rho_R(X) = \sum_{m\geq 1} R(m) X^{-m},$$

a rational function of X *whose only pole is* $X = 1$*. In particular*

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = \int \frac{\log X}{X-1}\, dy, \qquad P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2) = \int \frac{(X+1)\log X - 2(X-1)}{(X-1)^3}\, dy,$$

$$x^{(1)}(1) = \int \frac{dy}{X-1}, \qquad x^{(1)}(2) = \int \frac{X+1}{(X-1)^3}\, dy, \qquad (\mathsf{T}_{0;(1)}[x]\, (x-1)\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \frac{dy}{(X-1)^2}.$$

The first is the left closing of Lemma 5.16(ii), p. 34, read on the chart. With the generating function of the four-mark ray from the origin $(\varepsilon\mathbf{1}; \varepsilon, \varepsilon)$*,* $G_\varepsilon(q) = \sum_{k\geq 1} p(k+\varepsilon)\, q^{k-1} = \int X^{-\varepsilon}\, dy/(X-q)$ *for* $|q| < 16$*, carried by the pencil* $K\Psi - q\alpha^{\mathbf{1}}$ *(Thm. 35.10(ii)),*

$$(\mathsf{T}_{0;(1)}[x]\, (x-1)^j\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = (q\partial_q)^j G_0(q)\big|_{q=1},$$

$$(\mathsf{T}_{0;(1)}[x]\, (x-1)^j\, f\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\partial_\varepsilon (q\partial_q)^j G_\varepsilon(q)\big|_{q=1,\ \varepsilon=0}:$$

the block is the jet at $q = 1$ *of the four-mark generating function and of its slope in the origin.*

Proof. Def. 35.1 along the ray from $(\sigma\mathbf{1};\sigma,\sigma)$, the positivity of every term, and $\sum_{m\ge1}R(m)X^{-m}$ summed for $X>1$: $\rho_1=1/(X-1)$, $\rho_{x-1}=1/(X-1)^2$, $\rho_{(x-1)^2}=(X+1)/(X-1)^3$. For the masters use Corollary 15.26; for the pencil, Thm. 35.10(ii) with $k_0=1$ and the origin form $X^{-\varepsilon}dy$. □

Receipts [computed], by three-dimensional quadrature on the chart, to twelve digits:

$$\int\frac{dy}{X}=8.414398322115,\qquad \int\frac{\log X}{X-1}\,dy=39.945588011991,\qquad \int\frac{dy}{X-1}=8.570280443372,$$

against $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the five-mark centre at strength one, and $x^{(1)}(1)$.

Remark 15.29 (What the closing found). (i) The live Trailmoment lands on lattice inputs only through the residue at zero shares (Proposition 15.23).

(ii) The words of the five-mark centre ray reach a plane of the five-dimensional block on each side: on the values side the lowered setting, two stacked-Gamma numbers (Proposition 15.27); on the slopes side the period, whose two masters are

$$P_{\binom{[5]}{1}}(\mathbf{1};1,1)=(\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$P_{\binom{[5]}{1}}(2\mathbf{1};2,2)=(\mathsf{T}_{0;(1)}[x]\,(x-1)^2 f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-2\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)):$$

the slope in the origin of the four-mark generating function at $q=1$, and its second jet less twice the first jet of the generating function itself. They have no rational relation with 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$, $\Gamma(\frac12)^4$, $\Gamma(\frac12)^8$, $(\mathsf{T}_{5;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^4(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the two stacked-Gamma numbers and the direction $x-1$ of the block, with coefficients below 10^{20} (PSLQ, 360 digits) [computed]: in that framework, over $\mathbb{Q}$, they are new numbers.

(iii) The last minor of §14.17 is now written in the masters over the stacked Gamma (Proposition 15.27):

$$\begin{aligned}P_{\binom{[5]}{1}}(\mathbf{1};1,1)\,x^{(1)}(2)-P_{\binom{[5]}{1}}(2\mathbf{1};2,2)\,x^{(1)}(1)&\\ =\frac{65\,x^{(1)}(1)-32\,\Gamma(\frac12)^8/x^{(1)}(1)}{1125}\,P_{\binom{[5]}{1}}(\mathbf{1};1,1)-x^{(1)}(1)\,P_{\binom{[5]}{1}}(2\mathbf{1};2,2)&\\ =4.340002931535662034 07\ldots,&\end{aligned}$$

and it is not a rational combination of 1, $\Gamma(\frac12)^4$, $\Gamma(\frac12)^8$, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and seven products of $x^{(1)}(1)^{\pm1}$ or $x^{(1)}(1)^{\pm2}$ with powers of $\Gamma(\frac12)^4$, with coefficients below 10^{15} [computed]. Its scale is the masters' scale; nothing smaller is missing.

(iv) The rational words and the pole columns of Chapter 6 did not enter: every word of the centre ray is a polynomial, read at index zero. What closed the live Trailmoment is the shifted inputs (Thm. 6.2), the cut relation (Prop. 5.3), summation by parts in its adjoint form (Thm. 6.5, p. 43), and a fixed block at nonpositive index, the counterpart of Thm. 6.9, p. 45.

(v) Aliases, last: $x^{(1)}(1)=8\int_0^\infty tI_0(t)K_0(t)^4\,dt$, whose evaluation by the Gammas at fifteenths is the classical one through the singular modulus of $\mathbb{Q}(\sqrt{-15})$ [recalled]; Theorem 15.25 gives the recurrences of the moments $\int_0^\infty t^{2s-1}K_0(t)^n\,dt$, $n\le7$, by adding marks.

15.8 Open

(i) The certificate for every n. Theorem 15.25 for all n would derive every banana ray law from the stacked Gamma, the law of each list produced by the one before. At each $n\le6$ the same A_s appeared; an operator proof for general n: [not yet computed]. From five to six marks the

symbol of the five-mark law vanishes at $\lambda = 1$ (a native value of five marks), L_5^* raises degrees by less than four, and the block of six marks is built differently: [not yet computed]. A mark added in parallel to a mark of another list: for $\binom{[4]}{2}$ the class decouples and the second row of §15.4 is the square of the three-mark ray (Proposition 16.1); in general the class enters through the law of its parallel cell (§16).

(ii) The direction $x - 1$ of the block, $(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (q\partial_q)G_0(q)|_{q=1} = \int dy/(X-1)^2 = 0.161069702276674858 85\ldots$, which the ray reaches only through the value term of its second master. Expanding h as in Proposition 15.22, it is the lowered setting with h replaced by $y\,\partial_y h$: $\int_0^\infty g^4\,y\,h'(y)\,dy$ [derived] (alias $4\int_0^\infty t^2 I_1(t)K_0(t)^4 dt$, to 25 digits [computed]). Which member of the family it is, and whether a lean of the five-mark list reads it as a period: [not yet computed]

(iii) Three marks: reduce the live Trailmoment of the receipts of Proposition 15.19 at $n = 2$ to the Trailmoment of the odd periodic density modulo 3 at index 2 (alias $3L(\chi_{-3}, 2)$) inside the family; Thm. 5.17 does not reach it, the parity being mismatched. [not yet computed]

(iv) The three masters of $\binom{[5]}{2}$: no identification yet. [not yet computed]

(v) The second corners, from the list. The boundary term at $\sigma = -1$ is a constant term of the continued period (Thm. 1.6), assembled from its Laurent coefficients of orders $-(n-1)$, $-(n-2)$, $-(n-3)$ at $\sigma = 0$ and $\sigma = -1$. The leading ones are the corner and the corner times $|\mathcal{B}|$ (Proposition 15.4); the next vanishes at 0 (Proposition 15.3) and the ray law fixes it at -1. On the share side the third ones are Laurent data of the Gammas at the lattice points of the closing, which are heads of the counting input and Trailmoments of the counting input cut at the data (Prop. 5.13, p. 32; Prop. 5.14, p. 33); the rationality of the two boundary terms says that the Trailmoments cancel and the heads remain. Carrying this out: [not yet computed]. Also: which lists put $(\sigma+1)^2$ into both c_0 and c_1. Here they are the two triangles $\binom{[3]}{2}$ with parallel classes, and not $\binom{[4]}{2}$ with a mark doubled. [not yet computed]

(vi) The last minor of §14.17, $M_5(1)x^{(1)}(2) - M_5(2)x^{(1)}(1) = 4.340002931535 66\ldots$, is the minor of the solutions of native values $\frac{1}{25}$ and $\frac{1}{9}$, so it is the minimal solution of the exterior square of the ray law, of native value $\frac{1}{225}$ [derived]. It is now the masters over the stacked Gamma (Remark 15.29(iii)), so a relation of degree two fixing it would be a relation between the masters and $x^{(1)}(1)$. The exact pairing of Proposition 15.13 is a relation of degree one between solutions, fixed by the residue at zero shares. Whether pairings of the sign classes with closings of the share side twisted by the signs fix relations of degree two such as this minor: [not yet computed]

(vii) The pairing for the other lists: the partner of the period in the adjoint law, when the closings are not. [not yet computed]

(viii) Lean rays. At a lean with uniform masses only $\eta = \mathbf{1}$ stays native for $\binom{\mathcal{E}}{1}$: at a sign the native equation reads $(m-t)K(\eta) = (s-t)\eta_e$ for every e [derived]. The gauges that move off the signs and their values. [not yet computed]

(ix) Families of native gauges. In both cases of Proposition 15.15 the letter collapses at the gauge to the letter of a minor that leaves directions free: a class whose contraction letter vanishes is free, and a class whose own letter vanishes leaves a deletion with coloops, whose relative scales are free. A rule for every list, and the exponents -3 and $-\frac{5}{2}$ of the remark on local exponents. [not yet computed]

(x) A third flavour: $\mathrm{M}(K_4)$ on six marks, an atom, whose full cobasis lean $(0, 2)$ at unit masses is $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (alias $6\zeta(3)$) [recalled]; its centre ray. [not yet computed]

16 The five-mark lists along their hinge trees

A connected list is glued from prime tables and series and parallel nodes at its bunches of width one (Thm. 20.9, p. 166), and its canonical hinge tree (Def. 22.14, p. 189) is read off the Skeleton: the width $c(S) = r(S) + r(\mathcal{E} \setminus S) - r(\mathcal{E})$ of Def. 7.10, p. 49, is a count of the circuit and cocircuit layers, and a bunch of width one is where the list composes itself out of two pieces. The period is read from the leaves up (Thm. 22.15, p. 190). A cell enters only through its hinge wall and its law (Thm. 22.7, p. 185), a series node multiplies the transforms of its cells and a parallel node their dual transforms (Thm. 22.13, p. 188), and a mark left in place is a one-mark cell whose two transforms at the centre are the Gramma $\mathsf{G}_{0;(1,-1)}(1, y)$ (Def. 22.17(1), p. 191). This entry reads the five lists of §15.4 that way. Three are series-parallel, one carries a prime node that acts as a mark, and one is prime.

list	canonical hinge tree [derived]
$\binom{[5]}{1}$	one parallel node
$\binom{[4]}{2}$, a mark doubled	the prime node $\binom{[4]}{2}$, a parallel node of three marks at one marker
$\binom{[5]}{2}$	one prime node
$\binom{[3]}{2}$, two marks doubled	a series node of three marks, parallel nodes of three marks at two markers
$\binom{[3]}{2}$, a mark tripled	a series node of three marks, a parallel node of four marks at one marker

The bunches of width one with two marks on each side are the parallel classes, of width $1 + r(\mathcal{E} \setminus P) - r(\mathcal{E}) = 1$; in $\binom{[4]}{2}$ and $\binom{[5]}{2}$ every set of two marks has width 2, so both are prime.

16.1 A prime node that is a mark

Proposition 16.1 (The doubled mark of $\binom{[4]}{2}$)**.** [derived]*; receipts* [computed] *Replace a mark i of $\binom{[4]}{2}$ by a parallel class of $k \geq 1$ marks. Then*

$$P_{\binom{[4]}{2}[i\to k]}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\; P_{\binom{[k+1]}{1}}(\sigma\mathbf{1};\sigma,\sigma).$$

At $k = 1$ this is $P_{\binom{[4]}{2}} = P_{\binom{[3]}{1}}\,\Gamma_{1}^{2}\begin{bmatrix}\sigma, & \sigma\\ & 2\sigma\end{bmatrix}$, the agreement of the table of §15.1 at every strength; at $k = 2$ it is the second row of §15.4,

$$P_{\binom{[4]}{2},\ \text{a mark doubled}}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)^2.$$

Proof. $\binom{[4]}{2}$ is identically self-dual. The cell of $\binom{[4]}{2}$ hinged at i has the hinge minors $\binom{[4]}{2} \setminus i = \binom{[3]}{2}$ and $\binom{[4]}{2}/i = \binom{[3]}{1}$, dual with the identity and the same weights, so its ratio is identically one, and in every gluing it acts as $\phi(\Delta)$ times one mark, with $\phi(\Delta) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\kappa, 2\sigma - \kappa)$ (Cor. 22.8(i),(ii), p. 186). Its hinge wall is $\kappa = \sigma$, balanced against the parallel cell of the class, whose hinge wall is $k\sigma - (k-1)\sigma = \sigma$ (Thm. 22.13(ii)). By Prop. 8.14, p. 57, $P_{\binom{[3]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)$. What is left is the class glued to one mark, the banana $\binom{[k+1]}{1}$. □

Receipts [computed]: Procedure 15.6 for $\binom{[4]}{2}$ with a mark doubled agrees with $P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)^2$ to 60 digits at $\sigma = 1, \ldots, 4$, and for $\binom{[4]}{2}$ with $P_{\binom{[3]}{1}}\Gamma_{1}^{2}\begin{bmatrix}\sigma, & \sigma\\ & 2\sigma\end{bmatrix}$ at $\sigma = 1, 2, 3$.

Corollary 16.2. [derived] *The ray law of $\binom{[4]}{2}$ with a mark doubled is the symmetric square of the three-mark law: its native values are the products of $\frac{1}{9}$ and 1 in pairs, the roots $81, 9, 1$ of §15.4, and its corner is $6^2 = 36$. Every value lies in $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1) + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)^2$: its two masters are the three-mark centre and its square.*

The same argument at the centre of any identically self-dual list splits off a mark (Cor. 22.8(iii) and Prop. 8.14) [derived]:

$$P_{\binom{[2r]}{r}}(\sigma\mathbf{1};\sigma,\sigma) = \Gamma\,{}_{1}^{2}\!\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}\ P_{\binom{[2r-1]}{r-1}}(\sigma\mathbf{1};\sigma,\sigma).$$

So $\binom{[6]}{3}$ carries exactly the masters of $\binom{[5]}{2}$.

16.2 A series root over banana leaves

Proposition 16.3 (The triangle rows)**.** [derived]*; receipts* [computed] *Let f_{P_k} be the transform at the centre of the parallel cell of a class of k marks; $\int_0^\infty y^{w-1} f_{P_k}(y)\,dy = \Gamma(w)^2 P_{\binom{[k]}{1}}((\sigma-w)\mathbf{1};\sigma-w,\sigma-w)$ for $0<\operatorname{Re} w<\sigma$. Then*

$$\begin{aligned}\Gamma(\sigma)^2 P_{\binom{[3]}{2},\ \text{a mark tripled}}(\sigma\mathbf{1};\sigma,\sigma) &= \int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)^2\,f_{P_3}(y)\,dy\\ &= \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\Gamma(\sigma-d)^2\,P_{\binom{[4]}{2}}(d\mathbf{1};d,d)\,dd,\\ \Gamma(\sigma)^2 P_{\binom{[3]}{2},\ \text{two marks doubled}}(\sigma\mathbf{1};\sigma,\sigma) &= \int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)\,f_{P_2}(y)^2\,dy.\end{aligned}$$

So the tripled triangle is the closing of Proposition 15.22 with the ray of the four-mark banana replaced by the ray of $\binom{[4]}{2}$.

Proof. The triangle is a series root (Def. 22.17(5), p. 191): its cells are the parallel cells of the classes and the marks left in place, the latter with the transform $\mathsf{G}_{0;(1,-1)}(1,y)$. The dual transform of a parallel cell is the product of the k Grammas of its marks (Thm. 22.13(ii)), whose Mellin read is $\Gamma(z)^2 P_{\binom{[k]}{1}}(z\mathbf{1};z,z)$ (Proposition 15.2); its hinge wall is σ, and at $s=t=\kappa=\sigma$ the change of transform reflects $z\mapsto\sigma-z$ (Prop. 22.5(iii),(v), p. 184). The Mellin read of a product pins the sum of the shares, as in the proof of Proposition 15.2, and $P_{\binom{[2]}{1}}P_{\binom{[3]}{1}} = P_{\binom{[4]}{2}}$ is Proposition 16.1 at $k=1$. □

Receipts [computed]: the closing $-\sum_{d\ge1}(W_s\,P_{\binom{[4]}{2}})'(d)/\Gamma(s)^2$ of Proposition 15.22 agrees with Procedure 15.6 for the tripled triangle to 60 digits at $s=1,2,3$; for two marks doubled, opening the list at its unmarked mark (Prop. 19.19, read as in Proposition 16.5 below: the contraction is the four-mark banana, the deletion two parallel pairs) gives 7.8658373683630 by three-dimensional quadrature, against 7.86583736836402159 ….

Proposition 16.4 (Closing the tripled triangle over its input)**.** [derived]*; receipts* [computed] *Let $q(\sigma) = P_{\binom{[4]}{2}}(\sigma\mathbf{1};\sigma,\sigma)$ and $f_q = -q'$.*

(i) q obeys the three-mark law times the shift of the stacked Gamma,

$$\sigma^3 q(\sigma) - 2(2\sigma+1)(10\sigma^2+10\sigma+3)\,q(\sigma+1) + 36(\sigma+1)(2\sigma+1)(2\sigma+3)\,q(\sigma+2) = 0,$$

with corner $12 = 2\cdot 6$ and no term of order σ, and its symbol at $\lambda=1$ is $105\neq0$.

(ii) Summation by parts at the cut 0, as in Proposition 15.23: for every polynomial R,

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\,(L_q^* R)\,q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) &= -12\,R(0),\\ (\mathsf{T}_{0;(1)}[x]\,(L_q^* R)\,f_q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) &= (\mathsf{T}_{0;(1)}[x]\,(L_q'^* R)\,q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + 36\,R(-1)\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).\end{aligned}$$

The last coefficient of the law has only a simple zero at $\sigma=-1$: the shift of the stacked Gamma, $P_{\binom{[2]}{1}}(\sigma+2)/P_{\binom{[2]}{1}}(\sigma+1)=(\sigma+1)/(2(2\sigma+3))$, cancels one of the two factors $\sigma+1$ of the three-mark coefficient $9(\sigma+1)^2$. So the slope boundary there reads $q(1)=P_{\binom{[3]}{1}}(\mathbf{1};1,1)$.

(iii) Every value of the tripled triangle is a rational number, plus a rational multiple of $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, plus a rational combination of the three slope Trailmoments $-(\mathsf{T}_{0;(1)}[x]\,(Rq)'\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $R\in\{1,\,x,\,x^2/2\}$. The words W_s, $s\le 8$, reach all three; the five-mark banana reached two (Corollary 15.26).

(iv) In these coordinates the rational relation of §15.4 for this row, the second corner, is the relation among $1,P(1),\dots,P(5)$, coordinate by coordinate.

Proof. (i) $\sigma\,P_{\binom{[2]}{1}}(\sigma)=2(2\sigma+1)\,P_{\binom{[2]}{1}}(\sigma+1)$ (Prop. 1.3(iii)) multiplies the three-mark law coefficient by coefficient into σ times the display. The corner is the product of the corners $\lim\sigma P_{\binom{[2]}{1}}=2$ and 6; $\sigma\Gamma_2\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right]=2\Gamma(1+\sigma)^2/\Gamma(1+2\sigma)$ and $\sigma^2P_{\binom{[3]}{1}}$ have no terms of order σ (Proposition 15.3). (ii) The move of Proposition 15.23: the value boundary is $-6q(1)+108q(2)=-12$; for the slopes the boundary at the cut 0 is the term of order σ, zero, and the one at -1 is $-R(-1)\,c_2'(-1)\,q(1)$ with $c_2'(-1)=-36$. (iii) As in Proposition 15.24, with $\mathrm{span}\{1,x,x^2\}$ as the complement of $L_q^*(\mathbb{Q}[x])$, and Proposition 15.22 for the values. (iv) Exact arithmetic on the eight coordinates. □

Receipts [computed]. The coordinates reproduce Procedure 15.6 to 60 digits at $s\le 5$. There is no rational relation among 1, the three-mark centre $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, its slope $\partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)|_{\sigma=1}$ and the six Trailmoments

$$(\mathsf{T}_{0;(1)}[x]\,x^j q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\qquad (\mathsf{T}_{0;(1)}[x]\,x^j f_q\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\qquad (j\le 2)$$

with coefficients below 10^{25}, and none among 1, the three-mark centre and the three slope Trailmoments of (iii) below 10^{40} (PSLQ, 300 digits). So the four masters of the tripled triangle are the three-mark centre, entering at the boundary $\sigma=-1$, and the three slope Trailmoments.

16.3 The prime node at a mark

Proposition 16.5 ($\binom{[5]}{2}$ opened at a mark)**.** [derived]*; receipt* [computed] *For a mark e of $\binom{[5]}{2}$ the deletion is $\binom{[4]}{2}$ and the contraction $\binom{[4]}{1}$. Write $X_\backslash=X_{\binom{[4]}{2}}$ and $X_/=X_{\binom{[4]}{1}}$ for their strides on the chart of the other four marks. Then*

$$P_{\binom{[5]}{2}}(\sigma\mathbf{1};\sigma,\sigma)=\int dy\int_0^1 w^{\sigma-1}(1-w)^{\sigma-1}\big(w\,X_\backslash+(1-w)\,X_/\big)^{-\sigma}dw,$$

and on the share side of the shape,

$$\Gamma(\sigma)^2P_{\binom{[5]}{2}}(\sigma\mathbf{1};\sigma,\sigma)=\frac{1}{2\pi i}\int_{(c)}\Gamma(d)^2\Gamma(\sigma-d)^2\,\Phi_\sigma(d)\,dd,\qquad \Phi_\sigma(d)=\int X_/^{-d}X_\backslash^{-(\sigma-d)}\,dy,$$

$0<c<\sigma$, with $\Phi_\sigma(d)=P_F\big(\sigma\mathbf{1};(d,\sigma-d),(d,\sigma-d)\big)$ the centre of the family F of the two children (Def. 8.1, p. 53). At $\sigma=1$,

$$P_{\binom{[5]}{2}}(\mathbf{1};1,1)=\int\frac{\log X_\backslash-\log X_/}{X_\backslash-X_/}\,dy.$$

Proof. The mark is the Breta of its children's even letters (Prop. 19.19, p. 144): at the centre the prefactor and the stacked Gamma are one, and $E_{\backslash e} = \alpha^{\mathcal{E}\backslash e} X_{\backslash}$, $E_{/e} = \alpha^{\mathcal{E}\backslash e} X_{/}$. Splitting the coefficient $wX_{\backslash} + (1-w)X_{/}$ (Cor. 4.4, p. 26) and integrating the shape gives the Gammas, one pair for each child. □

Receipt [computed]: the last integral is 4.0297141707108 by three-dimensional quadrature, against 4.029714170711049607 32 . . . of §15.4.

Proposition 16.6 (The contraction under the deletion)**.** [derived] *On the chart* $X_{\backslash} - X_{/} = (e_2^2 - e_1e_3)/e_4$ *with*

$$e_2^2 - e_1e_3 = \sum_{i<j} \alpha_i^2\alpha_j^2 + \sum_i \sum_{\{j,k\}\not\ni i} \alpha_i^2\alpha_j\alpha_k + 2\,\alpha_1\alpha_2\alpha_3\alpha_4,$$

a polynomial with positive coefficients, so $X_{/} < X_{\backslash}$ *everywhere. The walls of* F *at the strengths* $(d, 1-d)$ *are* 1 *at* $|S|$ *odd and* $2-d$ *at* $|S| = 2$*, so* $\Phi = \Phi_1$ *is holomorphic on* $\operatorname{Re} d < 2$*, and closing to the left (Thm. 4.5, Lemma 5.16),*

$$P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1) = \sum_{m\geq 0} \Phi'(-m), \qquad \Phi(-m) = \int X_{/}^{\,m}\, X_{\backslash}^{-1-m}\, dy :$$

the slopes of the deletion's centre ray with the contraction's even letter inserted m *times.*

Proof. $X_{/} = e_1e_3/e_4$, $X_{\backslash} = e_2^2/e_4$, and the polynomial is read off the monomials of e_2^2 and e_1e_3. The widths are 1 for every S in $\binom{[4]}{1}$ and $1, 2, 1$ at $|S| = 1, 2, 3$ in $\binom{[4]}{2}$. On the line $\operatorname{Re} d = -N - \frac{1}{2}$,

$$|\Phi(d)| \leq \int \left(X_{/}/X_{\backslash}\right)^{N+1/2} X_{\backslash}^{-1}\, dy \to 0,$$

so the left closing applies; at $d = -m$ the kernel $\Gamma(d)^2\Gamma(1-d)^2$ of Lemma 5.16 has the principal part $(d+m)^{-2}$ and no simple part. Equivalently, expand $1/(X_{\backslash} - X_{/}) = \sum_m X_{/}^m X_{\backslash}^{-m-1}$ in the last display of Proposition 16.5. □

The ratio $X_{/}/X_{\backslash}$ is largest at the table's gauge, where it is $\frac{16}{36} = \frac{4}{9}$, so the closing converges like $(\frac{4}{9})^m$ (alias: Newton's inequality $e_1e_3 \leq \frac{4}{9}e_2^2$) [recalled].

Proposition 16.7 (The values at the lattice points are leans of the three-mark banana)**.** [derived]*; receipts* [computed] *For* $m \in \mathbb{N}$*, with* $(e_1e_3)^m = \sum_v c_v \alpha^v$*,*

$$\Phi(-m) = \sum_v c_v\, P_{\binom{[4]}{2}}\big(\mathbf{1} + v; 1 + m, 1 + m\big),$$

and for masses $\boldsymbol{\mu}$ *on the balance wall and every mark* g*,*

$$P_{\binom{[4]}{2}}(\boldsymbol{\mu}; s, s) = \frac{1}{\Gamma(2s)} \int_0^\infty y^{\mu_g - 1} \prod_{e\neq g} \mathsf{G}_{2s-\mu_g-\mu_e;(1,-1)}(1, y)\, dy :$$

$\Phi(-m)$ *is a finite sum of leans of the three-mark banana, Grammas of integer index read three at a time. The first values are*

$$\Phi(0) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad \Phi(-1) = \frac{2 + 2P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)}{9},$$
$$\Phi(-2) = \frac{71 + 56P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)}{810} \quad \text{[computed]}.$$

Proof. On the chart $X_{/}^{m}X_{\backslash}^{-1-m} = \alpha^{\mathbf{1}}(e_1e_3)^m\, e_2^{-2(1+m)}$, and e_2 is both letters of $\binom{[4]}{2}$; each monomial α^v is an insertion moving the masses (Lemma 35.2(iii), p. 308). Then Cor. 22.8(iii) at g, $P_{\binom{[4]}{2}}(\boldsymbol{\mu}; s, s) = \Gamma_{\substack{2\\1}}\begin{bmatrix}\mu_g, & 2s-\mu_g\\ & 2s\end{bmatrix} P_{\binom{[3]}{2}}(\boldsymbol{\mu}|; 2s-\mu_g, \mu_g)$; Prop. 8.14 turns $\binom{[3]}{2}$ into $\binom{[3]}{1}$ at the exponents $(\mu_g, 2s-\mu_g)$; and the share side of the three-mark banana at a lean, $\Gamma(a)\Gamma(b)P_{\binom{[3]}{1}}(\boldsymbol{\mu'}; a, b) = \int_0^\infty y^{a-1}\prod_e \mathsf{G}_{b-\mu'_e;(1,-1)}(1,y)\,dy$ (Prop. 4.2 with the pins $a_e - b_e = \mu'_e - b$, as in Proposition 15.2), whose Gammas cancel the Beta corner. □

Receipts [computed]: $\Phi(-1)$ by three-dimensional quadrature is 0.7430904974869, against $\frac{2}{9}(1 + P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)) = 0.74309049748654642013\ldots$; the leans themselves by one-dimensional quadrature to 40 digits, with PSLQ for the rational coefficients; on the way, $\int_0^\infty \mathsf{G}_{0;(1,-1)}(1,y)\,\mathsf{G}_{1;(1,-1)}(1,y)^2\,dy = \frac{1}{3}$.

Remark 16.8 (What the prime node reads). The prime node reads the three-mark banana at leans. Its values at the lattice points land on $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, and its period at strength one is the sum of the slopes there: the division of Propositions 15.23 and 15.24, values on lattice inputs and slopes carrying the masters, now along the deletion's ray with the contraction's even letter as its stride, of native value $\frac{4}{9}$ at the table's gauge. The three masters of $\binom{[5]}{2}$ are slope Trailmoments of that ray, as the two of $\binom{[5]}{1}$ were of the four-mark ray; the law in m and its closing: [not yet computed].

Remark 16.9 (What the tree decided). A parallel node multiplies Grammas: the banana. An identically self-dual prime node is a mark of ratio one, and the period factors: the doubled mark of $\binom{[4]}{2}$ is the square of the three-mark ray, with no new number. A series root over parallel cells multiplies transforms; its closing has the banana's shape over a product input, and the stacked Gamma in that input's law opens the boundary at $\sigma = -1$, which is where the second corner of the tripled triangle comes from (for two marks doubled: [not yet computed]). A prime node that is not self-dual is opened at a mark into its children, weighed by the Breta of their even letters, and reads leans. The three-mark centre enters the rows whose trees carry a three-mark piece, the class of three or the deletion $\binom{[3]}{2}$ inside $\binom{[4]}{2}$: rows two, three and five. The row with two marks doubled has only pairs and one mark at its leaves; what its masters are: [not yet computed]. Aliases, last: $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = 3L(\chi_{-3}, 2)$, and $\mathsf{G}_{\nu;(1,-1)}(1,y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$, so the leans are moments of products of three Bessel functions K_ν of integer orders.

16.4 Open

(i) The ray of $\binom{[4]}{2}$ with the stride $X_{/}/X_{\backslash}$: the law of $\Phi(-m)$ in m, the summation by parts of its slope input, and with it the three masters of $\binom{[5]}{2}$ (item (iii) of §15.8). [not yet computed]

(ii) Two marks doubled: the closing over its input of two parallel cells, and its second corner as a boundary at $\sigma = -1$. [not yet computed]

(iii) The certificate of Theorem 15.25 for a series root: there the input was one ray, here it is a product, and the laws have orders 6 and 5 (§15.4). [not yet computed]

(iv) $\mathrm{M}(K_4)$ is self-dual only up to a relabelling of its marks, so Cor. 22.8(iii) does not split it (as the handbook notes after Cor. 22.8); opened at a mark its children are $K_4 \setminus e$ and K_4/e, the latter the triangle with two marks doubled of §15.4. Its centre ray read this way. [not yet computed]

17 The masters of $\binom{[5]}{2}$; two marks doubled, closed

Both lists are read at a mark (Prop. 19.19, p. 144): the period is the Breta of the even letters of the two children, its share side is a Mellin–Barnes integral over the split d with the kernel of Lemma 5.16,

p. 34, and the closing (Thm. 4.5, p. 26) reads a values input and a slope input on the lattice of the split. For $\binom{[5]}{2}$ the children are $\binom{[4]}{2}$ and $\binom{[4]}{1}$ (§16.3). For the triangle with two marks doubled the mark is taken in a doubled class; the children are then those of the triangle H with one class doubled, opened at its single mark: the three-mark banana $H/1$ and a pair with a coloop $H \setminus 1$, and the class enters through its Mellin read. In both lists the lattice values of the input are leans of the three-mark banana. This entry finds the law of each input along its lines, the boundaries of its slope input, and the masters. Throughout, $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ is the three-mark centre and $P_{\binom{[4]}{1}}(\mathbf{1}; 1, 1)$ the four-mark centre.

17.1 The input of $\binom{[5]}{2}$ along its lines

For $s \in \{1, 2, \dots\}$ put $\Phi_s(d) = \Phi(d, s-d) = \int X_{/}^{-d} X_{\backslash}^{-(s-d)}\, dy$, the family of Proposition 16.5 on the line s; Φ_1 is the Φ of Proposition 16.6. By Proposition 16.7 every lattice value is a finite sum of leans of the three-mark banana,

$$\Phi_s(-m) = r_s(m) + c_s(m)\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad r_s(m),\, c_s(m) \in \mathbb{Q} \quad (m \in \mathbb{N}),$$

and c_s is the *three-mark coordinate* of the values input.

Proposition 17.1 (The closing at every integer strength)**.** [derived] *For $s \in \{1, 2, \dots\}$ let $V_s(m) = \big((m+1)(m+2)\cdots(m+s-1)\big)^2$. Then*

$$\Gamma(s)^2 P_{\binom{[5]}{2}}(s\mathbf{1}; s, s) = \sum_{m \geq 0} \Big(V_s(m)\, \Phi_s'(-m) - V_s'(m)\, \Phi_s(-m) \Big),$$

and $x_s = \Gamma(s)^{-2} \sum_{m \geq 0} V_s(m)\, \Phi_s(-m)$ is the lowered setting, the values half of the closing.

Proof. Proposition 16.5 and the left closing of Proposition 16.6 at the strength s. By the reflection (Prop. 1.3(v), p. 11), $\Gamma(d)^2\Gamma(s-d)^2 = V_s(-d)\,\Gamma(d)^2\Gamma(1-d)^2$, with double poles at $d = -m$ and no simple parts; the residue at $d = -m$ is $\big(V_s(-d)\Phi_s(d)\big)'$ there. □

At $s = 1$ this is Proposition 16.6.

Proposition 17.2 (The law on a line)**.** [derived] *in Proposition 17.15 On the line $s = 1$,*

$$\sum_{i=0}^{4} c_i(m)\, \Phi_1(-m-i) = 0,$$

$$\begin{aligned}
c_0 &= -8(m+1)^3(m+2)(m+3)(7m+16),\\
c_1 &= (m+2)(m+3)(630m^4 + 4583m^3 + 12351m^2 + 14584m + 6336),\\
c_2 &= -(m+3)(2m+5)(1295m^4 + 11241m^3 + 35869m^2 + 49386m + 24408),\\
c_3 &= 3(2m+5)(2m+7)(385m^4 + 3855m^3 + 13913m^2 + 21153m + 11286),\\
c_4 &= -54(m+2)(m+4)(2m+5)(2m+7)(2m+9)(7m+9),
\end{aligned}$$

and r_1, c_1 obey it separately. Its symbol is

$$216\rho^4 - 330\rho^3 + 185\rho^2 - 45\rho + 4 = (2\rho-1)(9\rho-4)(3\rho-1)(4\rho-1).$$

The lines $s = 2, 3$ have laws of order four and degree eight with the same symbol.

Receipts [computed]. Found by exact linear algebra on the values $\Phi_s(-m)$, $m \le 80$, of Proposition 16.7 (at $s = 1$: 138 equations, 35 unknowns, one solution). At $m = -1$, outside the data, c_0 has a triple zero and the law holds exactly, $\sum_{i\ge1} c_i(-1)\Phi_1(1-i) = 0$ with $(c_1,\dots,c_4)(-1) = (300, -5670, 25920, -34020)$. It was read off first; Proposition 17.15 derives it from the relations along a ray.

Proposition 17.3 (The roots are the native values of the ratio of the strides)**.** [derived] *On the chart $X_/ / X_\backslash = e_1e_3/e_2^2$. At the table's gauge its value is $\frac{4}{9}$, the maximum. On the pattern $(1,1,x,x)$ its native equation is $(1-x)(1+x^2) = 0$, and at $(1,1,i,i)$ the value is $\frac{1}{2}$. Where one mark goes to zero it becomes e_1e_3/e_2^2 of three marks, of maximum $\frac{1}{3}$; where one mark goes to zero and another to infinity it becomes $\alpha_1\alpha_2/(\alpha_1+\alpha_2)^2$, the reciprocal stride of the two-mark banana, of maximum $\frac{1}{4}$. These four values are the roots of the symbol of Proposition 17.2.*

The values input decays like $(\frac{4}{9})^m$; its coordinates r_1 and c_1 decay like $(\frac{1}{2})^m$, and

$$P_{\binom{[3]}{1}}(\mathbf{1};1,1) = -\lim_{m\to\infty}\frac{r_1(m)}{c_1(m)} \quad \text{[computed]}:$$

the three-mark centre is the multiplier that removes the component along the native value $\frac{1}{2}$ of the complex gauge.

Proposition 17.4 (Lines tied by a first-order relation)**.** [derived] *for $s \le 4$ in Proposition 17.15 With $F(m,n) = \Phi(-m,n)$,*

$$\begin{aligned}&4(m+1)(3m+2n+4)\,F(m,n+1) - 12(n+1)(2n+3)\,F(m+1,n+2) - (m+1)(2n-m)\,F(m,n)\\ &+ (6n^2+4mn-3m^2+16n-4m+5)\,F(m+1,n+1) - 6(n+1)(2n+3)\,F(m+2,n+2) = 0.\end{aligned}$$

On the lines $n = m+s$ it ties the line $s+1$ to the line s, so with the law every line is read off the line 1. Receipts [computed]: found at $s \le 4$, the coefficients polynomial in s; the lines 2 to 5 read off the line 1 agree with Proposition 16.7 at every $m \le 80$.

17.2 The slope input and its boundaries

The law holds as a function of the split (Proposition 17.15), so its derivative is the recurrence of the slope input,

$$\sum_{i=0}^{4} c_i(m)\,\Phi_1'(-m-i) = \sum_{i=0}^{4} c_i'(m)\,\Phi_1(-m-i),$$

as in Proposition 15.21. Its four initial slopes $\Phi_1'(0),\dots,\Phi_1'(-3)$ are tied by the boundaries of the law.

Proposition 17.5 (The cut)**.** [derived] *from Proposition 17.2 At $m = -1$, where the four-mark centre $\Phi_1(1) = P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ sits, c_0 has a triple zero; neither $\Phi_1(1)$ nor its slope enters, and*

$$300\,\Phi_1'(0) - 5670\,\Phi_1'(-1) + 25920\,\Phi_1'(-2) - 34020\,\Phi_1'(-3) = -\frac{21380}{63} + \frac{8866}{63}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).$$

The lines 2 and 3 give $\frac{1}{2}$ and $-\frac{5}{9}$ times the same relation.

Proposition 17.6 (The wall at $d=2$)**.** *residue* [derived]*; relation* [computed] Φ_1 *has a simple pole at* $d=2$*, the wall* $2-d$ *of the two-element* S *(Proposition 16.6), with residue* -6*: each of the six two-element* S *collapses (Cor. 8.8, p. 56) to a product of two two-mark centres,* $P_{\binom{[2]}{1}}(\mathbf{1};1,1)^2=1$*; the law at* $m=-2$ *returns the same* -6*. There the law meets the slopes only through*

$$c_2(-2)\,\Phi_1'(0)+c_3(-2)\,\Phi_1'(-1)=96\,\Phi_1'(0)-432\,\Phi_1'(-1),$$

and

$$6\,\Phi_1'(0)-27\,\Phi_1'(-1)=7\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-22$$

(PSLQ, 140 *digits), which is the same as the finite part*

$$\Phi_1(d)=-\frac{6}{d-2}+\frac{21}{2}+\frac{3}{4}\,P_{\binom{[4]}{1}}(\mathbf{1};1,1)+O(d-2).$$

Proposition 17.7 (One free slope, along the three-mark coordinate)**.** [computed] *A third relation,* $1680\,\Phi_1'(0)-24300\,\Phi_1'(-2)=2308\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-7207$ *(PSLQ), with Propositions 17.5 and 17.6, says that* $\Phi_1'(-m)-\Phi_1'(0)\,c_1(m)\in\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *for* $m\le 3$*. By the slope recurrence and Proposition 17.4 it holds for every* m *and every line:*

$$\Phi_s'(-m)=\rho_s(m)+\lambda_s(m)\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)+c_s(m)\,\Phi_1'(0),\qquad \rho_s(m),\,\lambda_s(m)\in\mathbb{Q}.$$

The one free direction of the slope input is the three-mark coordinate of the values input. Minimality fixes it: the slope input has no component along the native value $\frac{1}{2}$*.*

Receipts [computed]. With $\Phi_1'(0)$ from the masters below, $\Phi_1'(-m)/\Phi_1(-m)=0.81342$ at $m=600$, tending to $\log\frac{9}{4}=0.81093$, where a component along $\frac{1}{2}$ would grow like $(\frac{9}{8})^m$. The free slope is

$$\Phi_1'(0)=\int\log\frac{X_\backslash}{X_/}\,\frac{dy}{X_\backslash}=2.7943139847582708461546135530266971416 39\ldots,$$

$$\Phi_1'(0)=2\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-\partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\big|_{\sigma=1}-\int\log X_/\,\frac{dy}{X_\backslash},$$

$$\int\log X_/\,\frac{dy}{X_\backslash}=9.96865275615229094531816226 62\ldots,$$

by three-dimensional quadrature 2.7943139847583; and by the same quadrature

$$\Phi_1'(-1)=0.8280938236176\quad\text{against}\quad 0.82809382361938566084\ldots.$$

There is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, its slope, $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$, the four-mark slope and $\Phi_1'(0)$ with coefficients below 10^{15}.

17.3 The masters of $\binom{[5]}{2}$

Theorem 17.8 (The masters of $\binom{[5]}{2}$)**.** *(i)* [derived] *from Propositions 17.1 and 17.7; (ii)–(iv)* [computed]

(i) Put

$$A_s=\frac{\sum_m\big(V_s\rho_s-V_s'r_s\big)(m)}{\Gamma(s)^2},\quad B_s=\frac{\sum_m\big(V_s\lambda_s-V_s'c_s\big)(m)}{\Gamma(s)^2},$$

$$C_s=\frac{\sum_m V_s(m)c_s(m)}{\Gamma(s)^2},\quad R_s=\frac{\sum_m V_s(m)r_s(m)}{\Gamma(s)^2},$$

each convergent, the coordinates decaying like $(\frac{1}{2})^m$*. Then*

$$P_{\binom{[5]}{2}}(s\mathbf{1};s,s) = A_s + B_s\,P_{\binom{[3]}{1}}(\mathbf{1};1,1) + C_s\,\Phi_1'(0), \qquad x_s = R_s + C_s\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)\,:$$

the same C_s *weighs the three-mark centre in the lowered setting and the free slope in the period.*

(ii) $C_1 = \dfrac{\sqrt{3}}{80}\Gamma_{\!4}^{\,4}\!\left[\begin{smallmatrix}\frac{1}{20},&\frac{3}{20},&\frac{7}{20},&\frac{9}{20}\\ \frac{1}{2},&\frac{1}{2},&\frac{1}{2},&\frac{1}{2}\end{smallmatrix}\right]$ *(200 digits).*

(iii) The ray law of §15.4 and its corner are the boundary at the cut. As functions of the four initial slopes,

$$\sum_{i=1}^{4} c_i^{\binom{[5]}{2}}(0)\,P(i) - 1800 = -\tfrac{55}{27}\,\beta, \qquad \sum_{i=0}^{4} c_i^{\binom{[5]}{2}}(1)\,P(1+i) = -\tfrac{898}{189}\,\beta,$$

with β *the left side minus the right side of Proposition 17.5,* $P(i) = P_{\binom{[5]}{2}}(i\mathbf{1};i,i)$*, and* $1800 = 60\cdot 30$ *the corner times* $-[\sigma^4]c_0$*. The lowered setting obeys the ray law with no corner.*

(iv) A second corner lands on the three-mark centre:

$$20\,P(1) - 1672\,P(2) + 5120\,P(3) = 384 - 135\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).$$

It is the ray law at $\sigma=-1$*:* $(c_2,c_3,c_4)(-1) = 40\cdot(20,-1672,5120)$ *and* $\mathrm{CT}_{\sigma=-1}\big(c_0(\sigma)P(\sigma) + c_1(\sigma)P(\sigma+1)\big) = 5400\,P_{\binom{[3]}{1}}(\mathbf{1};1,1) - 15360$*. In the coordinates of (i) it reads* $5C_1 - 418C_2 + 1280C_3 = 0$*, so the free slope drops out.*

Over $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *the masters of* $\binom{[5]}{2}$ *are* $P(1)$ *and* $P(2)$*; the third master of §15.4 is the three-mark centre.*

Proof. (i) Insert Proposition 17.7 in Proposition 17.1; every coordinate is a combination of solutions of the law with the native values $\frac{1}{2}, \frac{4}{9}, \frac{1}{3}, \frac{1}{4}$, so each sum converges. (ii)–(iv) are identities among the sums of (i), checked to 140 to 200 digits. □

	$s=1$	$s=2$
A_s	1.26175500810518477241902922714639 42…	−0.31713709103564459977401900329380 55…
B_s	−0.40624669401528537366902394210824 01…	−0.05302558816228198122536942578617 45…
C_s	1.33133346849845008118245921460185 74…	0.16079473803473841611487043550547 92…
R_s	0.35274808101892850647342823634187 56…	−0.26684642159953763102470310656076 39…
$P(s)$	4.02971417071104960732069573442840 69…	0.00788683420102877864559915176853 24…
x_s	3.47327023494199033975035335634162 78…	0.11004152882326099964359712711845 30…

Receipts [computed]. The four initial slopes solved from $P(1), P(2), P(3)$ and Proposition 17.5 give $P(4)$ and $P(5)$ to 154 digits against Procedure 15.6, and the relations of Propositions 17.6 and 17.7 then come out of PSLQ with coefficients below 10^5. Apart from (iv) there is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Phi_1'(0)$, $P(1)$, $P(2)$, x_1, x_2 with coefficients below 10^8; R_1, C_2, R_2, C_3 are not in $\mathbb{Q}\,aC_1 + \mathbb{Q}\,b\,\Gamma(\frac{1}{2})^2 + \mathbb{Q}\,c\,\Gamma(\frac{1}{2})^4/C_1 + \mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ for any $a,b,c \in \{1,\sqrt{3},\sqrt{5},\sqrt{15}\}$, coefficients below 10^{12}.

17.4 Two marks doubled, closed

Let H be the triangle with the class 2 doubled. Opened at its mark 1 (Prop. 19.19), its children on the chart of $\{2a, 2b, 3\}$ are the three-mark banana $H/1 = \binom{[3]}{1}$, of stride $X_{H/1} = (B+\alpha_3)(B'+\alpha_3^{-1})$, and $H \setminus 1$, the pair $\{2a, 2b\}$ with 3 a coloop, of stride $X_{H\setminus 1} = BB'$, where $B = \alpha_{2a} + \alpha_{2b}$ and $B' = \alpha_{2a}^{-1} + \alpha_{2b}^{-1}$. As $X_{H/1} = BB' + B\alpha_3^{-1} + B'\alpha_3 + 1$, the deletion lies under the contraction. Put $\Psi(a,b) = \int X_{H/1}^{-a} X_{H\setminus 1}^{-b}\, dy$ and $\Psi_s(d) = \Psi(d, s-d)$.

Proposition 17.9 (Opened at a mark of a doubled class)**.** [derived] *For the triangle R with two marks doubled, $0 < c < \sigma$,*

$$\Gamma(\sigma)^2 P_R(\sigma\mathbf{1}; \sigma, \sigma) = \frac{1}{2\pi i} \int_{(c)} \Gamma(d)^2 \Gamma(\sigma - d)^2 \, P_{\binom{[2]}{1}}(d\mathbf{1}; d, d)\, \Psi_\sigma(d)\, dd,$$

and for $s \in \{1, 2, \dots\}$, with $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$ as in Proposition 15.22,

$$\Gamma(s)^2 P_R(s\mathbf{1}; s, s) = \sum_{x \geq s} \big(W_s\, f_s - W_s'\, q_s\big)(x), \qquad q_s(x) = P_{\binom{[2]}{1}}(x\mathbf{1}; x, x)\, \Psi_s(x), \quad f_s = -q_s'.$$

At the lattice points, with $k = x - s$,

$$q_s(x) = \frac{1}{\Gamma(2x)} \sum_{u=-k}^{k} \binom{2k}{k+u} \int_0^\infty y^{x-1-|u|}\, \mathsf{G}_{|u|;(1,-1)}(1,y)^2\, \mathsf{G}_{0;(1,-1)}(1,y)\, dy \;\in\; \mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1).$$

Proof. R is H with its mark 1 replaced by a class of two. In the opening of H at 1 the mark contributes $\Gamma(d)^2$; the class contributes the Mellin read of the product of its two Grammas, $\Gamma(d)^2 P_{\binom{[2]}{1}}(d\mathbf{1}; d, d)$ (Thm. 22.13(ii), p. 188, as in the proof of Proposition 16.3). Ψ_σ is holomorphic on $\operatorname{Re} d > 0$, its one wall being the coloop, and $X_{H\setminus 1}/X_{H/1} < 1$ while $P_{\binom{[2]}{1}}$ decays like 4^{-d}: close to the right as in Proposition 15.22. At $d = x$ the insertion $(BB')^k = \sum_u \binom{2k}{k+u}(\alpha_{2a}/\alpha_{2b})^u$ moves the masses (Lemma 35.2(iii), p. 308), and the share side of the three-mark banana at a lean (proof of Proposition 16.7) has the indices $(-u, u, 0)$: $\mathsf{G}_{-u}\mathsf{G}_u\mathsf{G}_0 = y^{-|u|}\mathsf{G}_{|u|}^2\mathsf{G}_0$. □

Proposition 17.10 (The input's law)**.** [derived] *in Proposition 17.15 On the line 1,*

$$\begin{aligned} x^2(2x-1)^2(60x^2+137x+65)\, q_1(x) - (2x+1)(1680x^5+4316x^4+2782x^3+231x^2-223x-56)\, q_1(x+1) \\ + (2x+1)(2x+3)(2940x^4+9113x^3+7350x^2-89x-1080)\, q_1(x+2) \\ - 18(x+2)(2x+1)(2x+3)(2x+5)(60x^2+17x-12)\, q_1(x+3) = 0, \end{aligned}$$

with the symbol $36\rho^3 - 49\rho^2 + 14\rho - 1 = (\rho-1)(4\rho-1)(9\rho-1)$, and the lines are tied by

$$\begin{aligned} -4(2x-2s-1)^2\, q_{s+1}(x) + 2(2x+1)(16x-1)\, q_{s+1}(x+1) \\ + \big(4x^2 - (8s+1)x + 2s(2s+1)\big)\, q_s(x) - 2(2x+1)(4x-1)\, q_s(x+1) = 0. \end{aligned}$$

The roots are the native value $\frac{1}{4}$ of the pair's period times the native values $4, 1, \frac{4}{9}$ of $X_{H\setminus 1}/X_{H/1}$: 4 at the sign $(1,1,-1)$ of the three-mark banana, where $BB' = 4$ and $X_{H/1} = 1$; 1 where $\alpha_3 \to 0$ and the pair splits; $\frac{4}{9}$ at the table's gauge [derived].

Receipts [computed]: found on $x \leq 91$ (176 equations, 28 unknowns) and at $s \leq 4$; the lines 2 to 5 read off the line 1 agree with Proposition 17.9. The native value 1 sits on the symbol, the case of §15.8(i): the coordinates of q_1 decay only like powers, $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -\lim r/c$ again, and the closing cannot be summed coordinate by coordinate. It is summed whole, with the minimality below.

Proposition 17.11 (Zero shares)**.** *Laurent data and boundary* [derived]*; second relation* [computed] *At* $d = 0$*,*

$$P_{\binom{[2]}{1}}(d\mathbf{1}; d, d) = \frac{2}{d} - \frac{\Gamma(\frac{1}{2})^4}{3}\, d + O(d^3), \qquad \Psi_1(d) = \frac{2}{d} + 2 + O(d).$$

The pole of Ψ_1 *is the coloop's wall: each end of* α_3 *gives* $\int X_{H\backslash 1}^{-1} dy = P_{\binom{[2]}{1}}(\mathbf{1}; 1, 1) = 1$*, and the finite part is* $\int \log X_{H\backslash 1}\, X_{H\backslash 1}^{-1} dy = 2$ *on the chart of the pair. So* $q_1(d) = 4/d^2 + 4/d + O(1)$ *where* c_0 *has a double zero: the law at* $x = 0$ *returns the* 4*, and its derivative is the boundary*

$$56\, f_1(1) - 3240\, f_1(2) + 6480\, f_1(3) = -\frac{1726}{3} - \frac{16}{15}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1).$$

The next relation carries $\Gamma(\frac{1}{2})^4$*, the class's term of order* d*:*

$$3\, f_1(1) - 27\, f_1(2) = 11 + 9\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - 2\,\Gamma(\tfrac{1}{2})^4 \qquad (\textit{PSLQ},\ 130\ \textit{digits}).$$

Hence $f_1(x) - f_1(1)\, c_1(x) \in \mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4$ *for every* x *and, by the lines, on every line: as for* $\binom{[5]}{2}$ *the one free slope lies along the three-mark coordinate. Minimality fixes it: no component along the native value* 1*.*

The free slope is

$$f_1(1) = 2\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - \partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)\big|_{\sigma=1} - \int \log X_{H\backslash 1}\, \frac{dy}{X_{H/1}},$$

$$f_1(1) = 7.458823870410082651014428317985859793 05\ldots,$$

$$\int \log X_{H\backslash 1}\, \frac{dy}{X_{H/1}} = 5.304142870500479140458347501226070 8\ldots,$$

by two-dimensional quadrature 5.3041428705004785 [computed].

Theorem 17.12 (The masters of two marks doubled)**.** [computed]

(i) $P_R(\mathbf{1}; 1, 1) - 8\, P_R(2\mathbf{1}; 2, 2) = 6\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - 2 - \frac{4}{9}\,\Gamma(\frac{1}{2})^4$*.*

(ii) $521\, P_R(\mathbf{1}; 1, 1) - 12288\, P_R(2\mathbf{1}; 2, 2) + 28672\, P_R(3\mathbf{1}; 3, 3) = 648 + 1350\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$*.*

(iii) Through the closing, the corner relation of §15.4 is the minimality alone, with the value $4432269600 = -[\sigma^4]c_0 \cdot 44$*; the second corner is* $-\frac{1741824}{13}$ *times the boundary of Proposition 17.11 plus the minimality, with the value* $-80640 \cdot 2503$*. The second corner is the boundary of the input's law at zero shares, where the class and the coloop put the double pole* $4/d^2 + 4/d$*.*

(iv) The lowered settings $x_s = \Gamma(s)^{-2} \sum_x W_s(x) q_s(x)$ *obey the ray law with no corner; at* $\sigma = -1$ *they have the rational second corner* $80640 \cdot 432$*, and* $239 - 148\, x_1 + 5244\, x_2 - 14336\, x_3 = 0$*.*

With the corner and the second corner of §15.4, every $P_R(s\mathbf{1}; s, s)$ *lies in* $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4 + \mathbb{Q}\, P_R(\mathbf{1}; 1, 1)$*: one master over* $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4$*, and the* $3^\dagger$ *of §15.4 are* $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$*,* $\Gamma(\frac{1}{2})^4$ *and* $P_R(\mathbf{1}; 1, 1)$*.*

Receipts [computed]. The three initial slopes solved from the minimality at $x = 640$ and $P_R(1), P_R(2)$ give $P_R(3), P_R(4), P_R(5)$ to 155 digits against Procedure 15.6; the slope at $x = 548$ is then $1.7 \cdot 10^{-336}$, of the size 4^{-x}. The values:

$$P_R(\mathbf{1}; 1, 1) = 7.865837368364021592421974697446 1682\ldots,$$

$$x_1 = 2.509422383492932056627701120329 7088\ldots.$$

There is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Gamma(\frac{1}{2})^4$, $f_1(1)$ and $P_R(\mathbf{1};1,1)$, nor with $\partial_\sigma P_{\binom{[3]}{1}}$ and $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ adjoined, below 10^{10}.

Remark 17.13 (What the closing found). (i) In both lists the lattice values are three-mark leans, and the slope input has one free direction, the three-mark coordinate of the values; the other initial slopes are fixed by the boundaries of the law, at the cut and the wall for $\binom{[5]}{2}$, at zero shares for the triangle, and the free one by minimality along the leading native value: $\frac{1}{2}$ at the complex gauge $(1,1,i,i)$ for $\binom{[5]}{2}$, 1 at the sign of the three-mark banana for the triangle. The words of the masters, V_s with $s \le 3$ and W_s with $s \le 2$, have degree below what the adjoint of the law adds (six for $\binom{[5]}{2}$; five for the triangle, whose symbol vanishes at $\rho = 1$), so summation by parts in the sense of Proposition 15.23 has nothing to reduce; the content is in the boundaries.

(ii) The corners of the ray laws are those boundaries. For $\binom{[5]}{2}$ the corner and the law itself are the boundary at the cut; for the triangle the corner is the minimality and the second corner is the boundary at zero shares.

(iii) Over $\mathbb{Q}$ the masters are $\{P_{\binom{[3]}{1}}(\mathbf{1};1,1), P(1), P(2)\}$ for $\binom{[5]}{2}$ and $\{P_{\binom{[3]}{1}}(\mathbf{1};1,1), \Gamma(\frac{1}{2})^4, P_R(1)\}$ for the triangle with two marks doubled. The three-mark centre is a master of both; in the triangle it enters through the contraction $H/1$.

(iv) The weight C_1 of the three-mark centre in the lowered setting of $\binom{[5]}{2}$ is a stacked Gamma at twentieths, and the same number weighs the free slope in the period.

(v) Aliases, last: $P_{\binom{[3]}{1}}(\mathbf{1};1,1) = 3L(\chi_{-3},2)$, $P_{\binom{[4]}{1}}(\mathbf{1};1,1) = 7\zeta(3)$, $\Gamma(\frac{1}{2})^4 = \pi^2$; C_1 is the Chowla–Selberg period of $\mathbb{Q}(\sqrt{-5})$; $X_\backslash - X_/ = (e_2^2 - e_1e_3)/e_4$, so $x_1 = \int dy/(X_\backslash - X_/)$ is a period of the quartic surface $e_2^2 = e_1e_3$; $\mathsf{G}_{\nu;(1,-1)}(1,y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$ [recalled].

Remark 17.14 (The leaf and the closing). In each of the four rays met so far, the three-mark ray, the four-mark ray, the input of $\binom{[5]}{2}$ along its lines and the input of the triangle with two marks doubled, the lattice values are $r + c\,\ell$ and the lattice slopes are $\rho + \lambda\,\ell + c\,\ell'$: r, c, ρ, λ rational, the same c in both, and ℓ the centre of the smallest piece, $\ell = P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ for the three-mark ray and the two inputs and $\ell = P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ for the four-mark ray. For the two banana rays this is [derived]: the cut relation and its derivative (the corner and its vanishing term of order σ, Proposition 15.3) have the same homogeneous part, so the one new slope $\ell' = \partial_\sigma P(\sigma\mathbf{1};\sigma,\sigma)|_{\sigma=1}$ follows c through the recursion (exactly, for every lattice point checked). For the two inputs it is Propositions 17.7 and 17.11.

In every case the coordinates r and c carry the leading native value of the law, and that native value sits at a gauge off the positive chart: the sign $(1,1,-1)$ of three marks (value 1), the sign $(1,1,1,-1)$ of four marks (value $\frac{1}{4}$), the complex gauge $(1,1,i,i)$ of $X_/X_\backslash$ (value $\frac{1}{2}$), the sign of the three-mark banana inside the triangle (value 1). The leaf's centre is the multiplier that removes it,

$$\ell = -\lim \frac{r}{c} \qquad \text{[computed]},$$

and the true values decay like the extreme the positive chart reaches: $\frac{1}{9}$ and $\frac{1}{16}$ at the gauges of the bananas, $\frac{4}{9}$ at the table's gauge for $\binom{[5]}{2}$, and $\frac{1}{4}$ for the triangle, from the stratum where $\alpha_3 \to 0$ and the pair splits.

So the closing pairs two kinds of number: Trailmoments of rational coordinate sequences against the closing's words, and the leaf's centre with its one slope. For $\binom{[5]}{2}$ this is Theorem 17.8(i). The stacked Gammas sit differently in the two prime lists. For the banana the whole lowered setting $x^{(1)}(1) = R + C\,P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ is the stacked Gamma at fifteenths (Proposition 15.27), while the weight $C = 1.07051077985786214296521092129478 08\ldots$ of the four-mark centre is not: there is no relation with that stacked Gamma, with $\Gamma(\frac{1}{2})^8$ over it, with 1 and $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$, times $1, \sqrt{3}, \sqrt{5}, \sqrt{15}$,

below 10^{12}. For $\binom{[5]}{2}$ the weight C_1 of the three-mark centre is the stacked Gamma at twentieths and x_1 is not [computed].

17.5 The laws from the relations along a ray

The laws of §§17.1–17.4 were read off exact lattice values. They follow from the two relations along a ray (Thm. 35.3, p. 308), integration by parts in every mark and the shift by the stride. The relation space has finite codimension (Thm. 35.19, p. 313), so every insertion has a law that follows from it (Cor. 35.6, p. 308). Finding it is linear algebra, and what the linear algebra returns is an insertion that certifies the law.

Proposition 17.15 (Certificates)**.** [derived]*; the certificates* [computed] *On the line s of $\binom{[5]}{2}$ put $\langle g\rangle_m = \int g\, t^m\, X_{\backslash}^{-s}\, dy$ with the stride $t = X_{/}/X_{\backslash} = e_1e_3/e_2^2$, so that $\langle t^i\rangle_m = \Phi_s(-m-i)$ and $\langle t^i X_{\backslash}^{-1}\rangle_m = \Phi_{s+1}(-m-i)$. For each law and each line relation $\sum_j p_j(m)\langle g_j\rangle_m = 0$ used in §§17.1–17.3 (the laws at $s \le 3$; the first-order relations and the explicit relations of the lines at $s \le 4$) there is an insertion*

$$H = \frac{P}{e_2^{\,d}},$$

P symmetric in three marks, of degree $2d$, with every exponent at most d and coefficients in $\mathbb{Q}[m]$, with H_e its image under the exchange of the first mark with e, such that

$$\sum_j p_j(m)\, g_j = \sum_{e\in\mathcal{E}} \Big(\alpha_e\partial_{\alpha_e} H_e + H_e\big(m\,\alpha_e\partial_{\alpha_e}\log t - s\,\alpha_e\partial_{\alpha_e}\log X_{\backslash}\big)\Big).$$

For the triangle with two marks doubled the same holds on the chart $\alpha_{2a}=1$ of $\{2a,2b,3\}$ for the laws of Ψ_s (the law of Proposition 17.10 divided by the shift of $P_{\binom{[2]}{1}}$), with the stride $X_{H\backslash 1}/X_{H/1}$, the origin form $X_{H\backslash 1}^{-s}dy$, and $H_e = P_e/(L_1^{d_1}L_2^{d_2}L_4^{d_4})$, where L_1, L_2 are the letters of the three-mark banana, $L_4 = \alpha_{2a}+\alpha_{2b}$, and P_e lies inside the Newton polygon of the denominator.

	law	*first-order relation*	*explicit relation*
$\binom{[5]}{2}$	$d=7$, *degree* 5 ($s=1$); $d=9$, *degree* 7 ($s=2,3$)	$d=3$, *degree* 5	$d=5$, *degree* 5
triangle	$(d_1,d_2,d_4)=(3,3,2)$, *degree* 6	$(2,2,1)$, *degree* 5	$(2,2,1)$, *degree* 5

The degrees are in the line coordinate.

Proof. The identity integrates to the law by Thm. 35.3, once $H_e\, t^m X_{\backslash}^{-s}$ tends to zero at the boundary of the chart. The hypothesis of Thm. 35.3 fails here. Along the line the contraction's exponent goes down as the deletion's goes up, and the direction has $\kappa_{1,S} = (d+d^*)_{\binom{[4]}{1}}(S) - (d+d^*)_{\binom{[4]}{2}}(S)$, which is 0 for $|S|$ odd and 1 for $|S|=2$; on the odd walls the stride does not decay but tends to the ratio of the face. The origin form carries the decay instead. Along a direction w with $w_1\ge w_2\ge w_3\ge w_4$,

$$\operatorname{ord}_w t = w_3 - w_2 \le 0, \qquad \operatorname{ord}_w X_{\backslash}^{-s} = s(w_3+w_4-w_1-w_2) < 0,$$
$$\operatorname{ord}_w H_e \le d(w_1+w_2) - d(w_1+w_2) = 0,$$

the last because no exponent of P exceeds d. So the boundary terms vanish for $m \ge 0$. Both sides of the law are holomorphic in m where the values converge ($\operatorname{Re} m > -2$ on the line 1), so the law holds identically in the split. For the triangle, $\operatorname{ord}_w H_e \le 0$ by the Newton polygon, and $(X_{H\backslash 1}/X_{H/1})^x X_{H\backslash 1}^{-s}$ decays in every direction for $x>0$. □

Receipts [computed]. Each certificate comes from exact linear algebra in under a second; the symmetric ansatz brings the law of the line 1 of $\binom{[5]}{2}$ down to 444 unknowns, and every equation is checked exactly. The certificates of the two line-1 laws were checked again by direct evaluation at random rational points, with the line coordinate negative and non-integral. The two banana rays, whose strides decay in every direction so that Thm. 35.3 applies as stated, have certificates of the same kind, with the denominators $e_1^2 e_2$ (three marks) and $e_1^2 e_3^2$ (four marks).

Corollary 17.16 (What is now derived)**.** [derived] *The laws of Propositions 17.2 and 17.10, the line relations of Propositions 17.4 and 17.10 for $s \le 4$, the slope recurrences (their derivatives in the split), and the boundaries at the cut (Proposition 17.5) and at zero shares (Proposition 17.11, with its Laurent data) are derived. Still computed: the relations among the initial slopes found by PSLQ (Propositions 17.6 and 17.7, and the relation carrying $\Gamma(\frac{1}{2})^4$ in Proposition 17.11), the identities of the closings in Theorems 17.8(iii) and 17.12(iii), and the relations among the masters.*

Remark 17.17 (The centre read off the law)*.* With the laws derived, the exchange rate of Remark 17.14 is an exact identity at every lattice point, not only in the limit. On the three-mark centre ray $p(\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ the three-mark law (its certificate is among the receipts of Proposition 17.15) and the first two values,

$$\sigma^2 p(\sigma) - (10\sigma^2 + 10\sigma + 3)\, p(\sigma+1) + 9(\sigma+1)^2\, p(\sigma+2) = 0,$$
$$p(1) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad p(2) = -\tfrac{2}{3} + \tfrac{1}{3}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1),$$

give $p(q) = r(q) + c(q)\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ with r, c rational, and $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -r(q)/c(q) + p(q)/c(q)$ for every q. The ratios $-r/c$ are

$$0, \quad 2, \quad \tfrac{23}{10}, \quad \tfrac{145}{62}, \quad \tfrac{1331}{568}, \quad \tfrac{242353}{103400}, \quad \tfrac{5495507}{2344600}, \quad \tfrac{46422941}{19805800}, \quad \ldots,$$

all below $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, the error shrinking by 0.1112 per step at $q = 30$, towards $\frac{1}{9}$, the native value of the gauge over that of the sign $(1, 1, -1)$ [computed]: about a digit of the three-mark centre per step of exact arithmetic. On the line 1 of $\binom{[5]}{2}$, $\Phi_1(-m) = \int t^m X_{\setminus}^{-1}\, dy > 0$ and $\Phi_1(0) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, so $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -r_1(m)/c_1(m) + \Phi_1(-m)/c_1(m)$ [derived]; the exact $c_1(m)$ is positive for $m \le 80$, so these ratios are lower bounds as well, with error $6.7 \cdot 10^{-4}$ at $m = 80$, shrinking per step towards $\frac{8}{9} = \frac{4}{9}/\frac{1}{2}$ [computed]. With $V_1 = 1$, Theorem 17.8(ii) is a series of the rationals that the law of Proposition 17.2 produces:

$$\sum_{m \ge 0} c_1(m) = 1 + \frac{2}{9} + \frac{28}{405} + \frac{208}{8505} + \frac{304}{32805} + \frac{16672}{4546773} + \cdots = \frac{\sqrt{3}}{80} \Gamma_{\,4}^{\,4}\left[\begin{matrix} \frac{1}{20}, & \frac{3}{20}, & \frac{7}{20}, & \frac{9}{20} \\ \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2} \end{matrix}\right].$$

17.6 Corrections to §§15.4 and 16

(i) The masters column of §15.4 counts over $\mathbb{Q}$. Over $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ the list $\binom{[5]}{2}$ has two masters, and over $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\, \Gamma(\frac{1}{2})^4$ the triangle with two marks doubled has one (Theorems 17.8 and 17.12).

(ii) In the remark "What the tree decided" of §16.3 the three-mark centre enters every row of §15.4 but the banana, row four included, through the contraction of H at its single mark.

(iii) In the remark "What the prime node reads" of §16.3, two of the masters of $\binom{[5]}{2}$ are slope Trailmoments of the deletion's ray; the third is $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$.

17.7 Open

(i) Done for the lines used here (Proposition 17.15). One certificate with s a variable, as Theorem 15.25 has for the banana. [not yet computed]

(ii) The finite part $\frac{21}{2}+\frac{3}{4}P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ at the wall $d=2$ from the collapse of Cor. 8.8 as a regularized lean of the four-mark banana; the third relation of Proposition 17.7 from the next order. [not yet computed]

(iii) The relation of Proposition 17.11 that carries $\Gamma(\frac{1}{2})^4$, from the next Laurent order at zero shares. [not yet computed]

(iv) Theorem 17.12(i) and (ii) as boundaries: (i) ties the lines 1 and 2 with the coefficient $8=|\mathcal{B}|$. Which insertion reads it (a mean share as in Proposition 15.24?). [not yet computed]

(v) The other weights A_s, B_s, R_s, and C_s for $s\ge 2$, of $\binom{[5]}{2}$, as periods of the surface $e_2^2=e_1e_3$ at its $\mathbb{Q}(\sqrt{-5})$ point. [not yet computed]

(vi) $\mathrm{M}(K_4)$, opened at a mark into $K_4\setminus e$ and K_4/e, the latter the triangle with two marks doubled closed here. [not yet computed]

18 The reading of a ray through the origin

Remark 17.17 read the three-mark centre off its law, and Remark 17.14 found the same exchange rate on four rays. This entry states that reading for every ray through the origin of every family, as derived statements, and runs it where §§15–17 had not looked: off the centre, at a lean of three marks, and on the one connected list on four marks that had not been read. The receipts are in the script ray_reading_s18.py (python-flint, exact arithmetic and ball arithmetic, no quadrature in the procedure).

Throughout, a *ray through the origin* is a ray of Def. 35.1, p. 307, with origin 0 and an integral direction $(\mathbf{m}_1;\mathbf{s}_1,\mathbf{t}_1)$ on the balance wall, $s_{1j},t_{1j}\in\mathbb{N}$, all of whose proper walls are positive, $\kappa_{1,S}>0$ for every nonempty proper S. Its values and its stride are

$$P(k)=P_F(k\mathbf{m}_1;k\mathbf{s}_1,k\mathbf{t}_1)=\int_{y_{e_0}=0} f^k\,dy\quad(k\ge 1),\qquad f=\alpha^{\mathbf{m}_1}\prod_j K_j^{-s_{1j}}\Psi_j^{-t_{1j}},$$

with $\alpha^\star$ the native gauge of the direction on the orthant, where f takes its largest value $f(\alpha^\star)$ there (Thm. 35.10(i), p. 309). The centre ray of Def. 35.9, p. 309, is the case $\mathbf{m}_1=\sigma\mathbf{1}$, $\mathbf{s}_1=\mathbf{t}_1=\boldsymbol{\sigma}$.

18.1 One pencil, two chains

Proposition 18.1 (The closings of a direction)**.** [derived] *Put $x_j=[\alpha^0](1/f)^j$ for $j\in\mathbb{N}$, the constant terms of the powers of the Laurent polynomial $1/f=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$, and*

$$\mathrm{Cor}=\sum_{\pi}\prod_{i=2}^{n}\kappa_{1,U_i}^{-1},$$

the sum over the orders π of $\mathcal{E}$, with tails $U_i=\{\pi(i),\dots,\pi(n)\}$. Then P continues meromorphically in k, with its poles at negative rational k, and

$$\lim_{k\to -j}(k+j)^{n-1}P(k)=\mathrm{Cor}\cdot x_j\qquad(j\in\mathbb{N}).$$

At the centre this is Proposition 15.4: there $1/f=\prod_j(K_j\Psi_j/\alpha^{\mathcal{E}})^{\sigma_j}$ and $\kappa_{1,S}=\sum_j\sigma_j c_j(S)$.

Proof. In the sector coordinates of Thm. 8.7(ii), p. 55, $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(i)} = \rho\, q_2 \cdots q_i$, one has $K_j = \rho^{r_j} \prod_i q_i^{d_j(U_i)} k_{j,\pi}$, $\Psi_j = \rho^{n-r_j} \prod_i q_i^{d_j^*(U_i)} \psi_{j,\pi}$ and $\alpha^{\mathbf{m}_1} = \rho^{m_1,\varepsilon} \prod_i q_i^{m_1(U_i)}$, so by the balance of the direction

$$f = \prod_{i=2}^{n} q_i^{\kappa_{1,U_i}}\, \varphi_\pi(q), \qquad \varphi_\pi = \prod_j k_{j,\pi}^{-s_{1j}} \psi_{j,\pi}^{-t_{1j}},$$

and by Thm. 8.10, p. 56, $P(k) = \sum_\pi \int_{[0,1]^{n-1}} \prod_i q_i^{k\kappa_{1,U_i}-1} \varphi_\pi(q)^k\, dq$. The sector letters have non-negative coefficients and positive constant terms (Lemma 8.5, p. 55), so $(1/\varphi_\pi)^j = \sum_v c_{\pi,v} q^v$ is a polynomial. At $k = -j + \varepsilon$ write $\varphi_\pi^k = (1/\varphi_\pi)^j \varphi_\pi^\varepsilon$. By Lemma 8.6, p. 55, the term q^v has a pole in ε of order at most the number of i with $v_i = j\kappa_{1,U_i}$, so the pole of order $n-1$ comes from $v = j\kappa_1$ alone, with the leading coefficient $c_{\pi,j\kappa_1} \prod_i \kappa_{1,U_i}^{-1}$. Finally $c_{\pi,j\kappa_1} = [q^0]\big(\prod_i q_i^{-\kappa_{1,U_i}}/\varphi_\pi\big)^j$ is the constant term of $(1/f)^j$ in the sector coordinates, and $\alpha^v \mapsto \prod_i q_i^{v(U_i)}$ is injective on exponents of degree zero, so it is x_j for every π. □

Corollary 18.2 (The law and its closings)**.** [derived] *If* $\sum_{i=0}^{h} c_i(k) P(k+i) = 0$ *identically, then* $\sum_{i=0}^{\min(h,j)} c_i(-j)\, x_{j-i} = 0$ *for every* $j \in \mathbb{N}$*, and* c_0 *vanishes to order at least* $n-1$ *at* 0*, with*

$$[k^{n-1}]c_0 \cdot \mathrm{Cor} + \sum_{i \geq 1} c_i(0)\, P(i) = 0.$$

Proof. As Corollary 15.5, with Proposition 18.1 in place of Proposition 15.4; the second statement is the case $j = 0$, $x_0 = 1$. □

So a law of the values is the recurrence of the integers x_j read backwards, and reading it is exact linear algebra on integers. The converse, that the least recurrence of the x_j, reflected, is a law of P, is what a certificate of Thm. 35.3, p. 308, settles, as in Proposition 17.15.

Remark 18.3 (The torus and the chart). [derived] The two sequences are one integrand on two chains. For $|q| < f(\alpha^\star)^{-1}$, $\sum_{k\geq 1} P(k) q^{k-1} = \int_{y_{e_0}=0} f/(1-qf)\, dy$ (Thm. 35.10(ii) with $k_0 = 1$). On the torus $T = \{|\alpha_e| = 1\}$, where $|1/f| \leq (1/f)(\mathbf{1})$ because $1/f$ has nonnegative coefficients, expanding in $1/(qf)$ gives for $|q| > f(\mathbf{1})^{-1}$

$$\sum_{j \geq 0} x_j\, q^{-j-1} = -\frac{1}{(2\pi i)^{n-1}} \oint_T \frac{f}{1-qf} \prod_{e \neq e_0} \frac{d\alpha_e}{\alpha_e}.$$

The pencil $\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\, \alpha^{\mathbf{m}_1}$ of Thm. 35.10 carries both: the positive chart gives its solution at $q = 0$, whose coefficients are the periods, and the torus its solution at $q = \infty$, whose coefficients are integers for a list. Corollary 18.2 is the one equation, inhomogeneous by its boundary terms, read at its two ends, and Cor is the ratio of the two normalizations.

18.2 The slit

Proposition 18.4 (The ray lives on the slit plane)**.** [derived] $G(q) = \int_{y_{e_0}=0} f/(1-qf)\, dy$ *is holomorphic on* $\mathbb{C} \setminus [f(\alpha^\star)^{-1}, \infty)$*, and it continues* $\sum_{k\geq 1} P(k) q^{k-1}$.

Proof. On the chart $0 < f \leq f(\alpha^\star)$ (Thm. 35.10(i)). For q off the slit the segment $q \cdot (0, f(\alpha^\star)]$ misses 1, so $|1 - qf|$ is bounded below by the distance from 1 to that segment, locally uniformly in q, and $\int f\, dy = P(1) < \infty$; dominated convergence gives holomorphy. On $|q| < f(\alpha^\star)^{-1}$ the integral is the series. □

Theorem 18.5 (Positive selection)**.** *(i)–(iii)* [derived]*; the count in (iii)* [computed] *on every ray below. Let* $\sum_{i=0}^{h} c_i(k)P(k+i) = 0$ *be a law of a ray through the origin,* $c_0 c_h \neq 0$*, let* D *be the largest degree of the* c_i*, and* $\chi(\lambda) = \sum_i [k^D]c_i\,\lambda^i$ *its symbol, with roots* λ_c*; it vanishes at the native value of every nondegenerate native gauge (Cor. 35.6, p. 308), and a root need not be a native value.*

(i) For every solution $(x_k)_{k\geq 1}$ *of the law,* $X(q) = \sum_k x_k q^{k-1}$ *satisfies a linear differential equation in* q *with polynomial coefficients, inhomogeneous by a polynomial that depends linearly on* $x_1, \ldots, x_h$*; in* $\mathbb{C}^\times$ *its singular points are the* $q_c = 1/\lambda_c$*. So* X *continues along every path in* $\mathbb{C}^\times$ *that avoids them.*

(ii) The values are the solution whose X *is holomorphic on* $\mathbb{C} \setminus [f(\alpha^\star)^{-1}, \infty)$ *(Proposition 18.4): holomorphic at every* q_c *off the slit, that is at every root* $\lambda_c \notin (0, f(\alpha^\star)]$*.*

(iii) At a singular point q_c*, holomorphy of* X *is* ν_c *linear conditions on* $(x_1, \ldots, x_h)$*,* ν_c *the number of independent local solutions of the equation that are not holomorphic there, at most the multiplicity of* λ_c *as a root of* χ*. At a simple root that is the native value of a nondegenerate native gauge it is one: the branch of Thm. 35.11, p. 310, read at that gauge in place of* $\alpha^\star$*, of exponent* $\frac{n-3}{2}$*, with a logarithm for odd* n*. With the corner of Corollary 18.2, if* $1 + \sum_{q_c \text{ off the slit}} \nu_c = h$ *and the conditions are independent, the law, the corner and positivity determine the values.*

(iv) Then, for a basis $x^{(0)}, \ldots, x^{(h-1)}$ *of rational solutions fixed by the corner,* $P = x^{(0)} + \sum_{i\geq 1} M_i x^{(i)}$*, and the* M_i *are the limits as* $K \to \infty$ *of the solutions of* $x^{(0)}_{K+l} + \sum_i M_i x^{(i)}_{K+l} = 0$*,* $0 \leq l < h-1$*, with error* $O\big((f(\alpha^\star)/\min_{c\neq\star}|\lambda_c|)^K\big)$ *up to powers of* K*.*

Proof. (i) Multiply the law by q^{k-1} and sum over $k \geq 1$: k acts as $q\partial_q + 1$, a shift by i as q^{-i} after removing the first i terms, which leaves a polynomial. The coefficient of the top power of $q\partial_q$ is $\sum_i [k^D]c_i\,q^{-i}$, which vanishes in $\mathbb{C}^\times$ exactly where $\chi(1/q) = 0$. (ii) Proposition 18.4. (iii) The inhomogeneity is holomorphic at q_c, so the singular part of X there lies in the span of the local homogeneous solutions that are not holomorphic, with coefficients linear in $(x_1, \ldots, x_h)$. At a nondegenerate native gauge the value over a steepest-descent disc (Thm. 35.5, p. 308) gives one branch of exponent $\frac{n-1}{2} - 1$; Thm. 35.11 is this computation at $\alpha^\star$, and it is local. (iv) The coefficients of X grow at the rate $|\lambda_c|$ of the singular point nearest 0 at which X is singular. The rational solutions are singular at the q_c off the slit and P is not, so the combination annihilating their $h-1$ largest rates is P, and its own size $f(\alpha^\star)^K$ against the smallest of the others is the error. □

Procedure 18.6 (The reading of a ray)**.** (1) From the family and the direction write $1/f$ and the walls $\kappa_{1,S}$, and Cor (Proposition 18.1). (2) The closings $x_j = [\alpha^0](1/f)^j$, exactly. (3) Their least recurrence by exact linear algebra, reflected (Corollary 18.2): the law; its certificate by Thm. 35.3 as in Proposition 17.15. (4) The native values, the roots of the symbol, and the native gauges, the critical points of $\log f$: the signs and what a lean moves them to, and families. (5) The seeds: the corner, and a second corner wherever c_0 and c_1 share a zero (§15.4). (6) Positivity, Theorem 18.5(iii)–(iv): the masters as limits of rational solutions, at the rate $f(\alpha^\star)/\min_{c\neq\star}|\lambda_c|$. (7) Integer relations among the masters and against the members already in use; what is left over $\mathbb{Q}$ are the masters of the ray.

Remark 18.7 (What sits on the slit)*.* Positivity says nothing at a native value in $(0, f(\alpha^\star))$. For a ray strictly inside its walls f has one critical point on the orthant (Thm. 8.18(ii)–(iii), p. 58), so such a value would sit at a native gauge off the orthant; on every ray strictly inside its walls read in §§15–18 there is none. On a ray along walls the slit carries the faces: on the line 1 of $\binom{[5]}{2}$ (§17.1) the native values $\frac{1}{3}$ and $\frac{1}{4}$ of Proposition 17.3 lie below $\frac{4}{9}$, they come from strata at the boundary of the chart, and the constants there are the faces' own, the three-mark centre among them (Theorem 17.8). That is Thm. 35.15, p. 312, read on the slit.

18.3 Off the centre: a lean of three marks

The centre is the fixed point of Cremona (Thm. 8.13, p. 57), where both letters take one exponent; a lean gives them two. For $\binom{\mathcal{E}}{1}$, $K = \sum_e \alpha_e$ and $\Psi = \alpha^{\mathcal{E}} \sum_e \alpha_e^{-1}$, and at uniform masses $m\mathbf{1}$ the balance is $nm = s + (n-1)t$.

Proposition 18.8 (The lean is the index of the Gramma)**.** [derived]*; receipt* [computed] *For masses* $\mathbf{m}$ *on the balance wall and* $\operatorname{Re} s, \operatorname{Re} t > 0$*,*

$$\Gamma(s)\Gamma(t)\, P_{\binom{\varepsilon}{1}}(\mathbf{m}; s, t) = \int_0^\infty y^{t-1} \prod_{e \in \mathcal{E}} \mathsf{G}_{m_e - t;(1,-1)}(1, y)\, dy, \qquad \sum_e (m_e - t) = s - t.$$

At the centre this is Proposition 15.2; the lean and the offsets move the index slots of the marks' Grammas. In the direction $(2\mathbf{1}; 4, 1)$ *of three marks,* $P_{\binom{[3]}{1}}(2\mathbf{1}; 4, 1) = \frac{1}{6}\int_0^\infty \mathsf{G}_{1;(1,-1)}(1, y)^3\, dy = 0.030080945282262304399884274898349847\ldots$ *in ball arithmetic.*

Proof. On the share side (Proposition 15.1) the mark e carries the share a_e of the basis $\{e\}$ in K and the share b_e of the cobasis $\mathcal{E} \setminus e$ in Ψ, and the pin at e reads $a_e + \sum_{e' \neq e} b_{e'} = m_e$, that is $a_e = b_e + m_e - t$. So each mark carries $\Gamma(b_e + m_e - t)\Gamma(b_e)$, the share kernel of $\mathsf{G}_{m_e - t;(1,-1)}(1, y)$ (Prop. 4.2, p. 25), and the Mellin read of the product pins $\sum_e b_e = t$, as in Proposition 15.2. □

Proposition 18.9 (The native gauges of a lean of $\binom{\mathcal{E}}{1}$)**.** [derived] *In the direction* $(m\mathbf{1}; s, t)$*,* $s \neq t$*, put* $u_e = \alpha_e / K(\alpha)$*,* $v_e = \alpha_e^{-1} / K(\alpha^{-1})$ *and* $\lambda = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$*. At a native gauge* $su_e - tv_e = \frac{s-t}{n}$ *and* $u_e v_e = \lambda$ *for every* e*, so every* u_e *is a root of* $su^2 - \frac{s-t}{n}u - t\lambda = 0$*; with* j *marks at one root* u_- *and* $n - j$ *at the other* u_+*,*

$$(n - 2j)\, u_+ = 1 - j\, \frac{s - t}{ns}, \qquad f = \Big(\prod_e u_e\Big)^{\frac{s-t}{n}} \lambda^t.$$

At $s = t$ *the two roots are* $\pm u_+$ *and this is Proposition 15.7. For three marks in the direction* $(2\mathbf{1}; 4, 1)$*:* $j = 0$ *is the table's gauge,* $f = \frac{1}{243}$*;* $j = 1$ *is* $\alpha \propto (3, 3, -2)$ *and its images,* $f = -\frac{27}{64}$*, the sign* $(1, 1, -1)$ *moved along the real line.*

Proof. Nativity is $m = sR_e(\alpha) + tR_e^*(\alpha)$ with $R_e = u_e$ and $R_e^* = 1 - v_e$ (Thm. 7.14, p. 50), and $u_e v_e = 1/\big(K(\alpha)K(\alpha^{-1})\big)$ for every e. The u_e sum to 1, which fixes u_+. For the value, $\alpha^{\mathbf{1}} = K(\alpha)^n \prod_e u_e$ and $K(\alpha^{-1}) = 1/(K(\alpha)\lambda)$, and the powers of $K(\alpha)$ cancel by the balance. □

Receipts [computed]. The walls of $(2\mathbf{1}; 4, 1)$ are 2 on marks and 3 on pairs, so $\mathrm{Cor} = 6/(2 \cdot 3) = 1$. The closings are $x_j = (4j)!\, j! \sum_{b_1+b_2+b_3=j} \prod_f \big(b_f!\,(b_f + j)!\big)^{-1} = 1, 36, 4620, 762300, 140216076, \ldots$; sixty of them have a least recurrence of order 2 and degree 6, and reflected it is the law

$$\begin{gathered} -3k^2(3k+1)(3k+2)(265k^2 + 636k + 383)\, P(k) \\ +(1721705k^6 + 7575502k^5 + 13047276k^4 + 11068046k^3 + 4766659k^2 + 965100k + 75600)\, P(k+1) \\ +972(k+1)^2(4k+5)(4k+7)(265k^2 + 106k + 12)\, P(k+2) = 0, \end{gathered}$$

$P(k) = P_{\binom{[3]}{1}}(2k\mathbf{1}; 4k, k)$, with the symbol $265(243\lambda - 1)(64\lambda + 27)$: the two native values of Proposition 18.9, which were read off the list before the closings were computed. The corner gives $12600\, P(1) + 68040\, P(2) = 383$; positivity gives

$$P_{\binom{[3]}{1}}(2\mathbf{1}; 4, 1) = 0.030080945282262304399884274898349846733673851 1\ldots,$$

about two digits per step (the rate $\frac{64}{6561}$), against 0.95 for the centre, and Proposition 18.8 confirms it. Over $\mathbb{Q}$,

$$54\,P_{\binom{[3]}{1}}(\mathbf{21};4,1) = 11 - 4\,P_{\binom{[3]}{1}}(\mathbf{1};1,1) \qquad (380 \text{ digits}).$$

The Cremona partner $(\mathbf{31};1,4)$ has the same closings, by $\alpha \mapsto \alpha^{-1}$, and the same law. So the lean is a new coordinate system on the content of the centre: a different law, different native gauges, a different corner, the same master.

The rational coordinates of this ray do not give a gap (Thm. 34.3, p. 300): on $(1, P_{\binom{[3]}{1}}(\mathbf{1};1,1))$ their denominators beyond d_k^2 grow like $e^{8.7k}$, with primes of the apparent factor $265k^2 + \cdots$ among them, against the decay $243^{-k} = e^{-5.49k}$ [computed]; at the centre the excess is $3^{1.88k}$, against 9^{-k}.

18.4 The fourth list on four marks

On four marks the connected lists are $\binom{[4]}{1}$, $\binom{[4]}{2}$, $\binom{[4]}{3}$ and the triangle H with one mark doubled; the first two were read in §15.1, and the third is the dual of the first (Prop. 8.14, p. 57). H is the list of §17.4, where it was a step toward the triangle with two marks doubled; its own centre was not read there. With the doubled class $\{a,b\}$ and the marks $2,3$ its bases are the pairs other than $\{a,b\}$, and with $A = \alpha_a + \alpha_b$, $B = \alpha_2 + \alpha_3$

$$K = AB + \alpha_2\alpha_3, \qquad \Psi = AB + \alpha_a\alpha_b :$$

the letters exchange under $\{a,b\} \leftrightarrow \{2,3\}$, so H is self-dual up to that relabeling. The widths are 1 on marks, on $\{a,b\}$, on $\{2,3\}$ and on triples, and 2 on the four other pairs (Def. 7.10, p. 49), so at the centre $\mathrm{Cor} = 8 \cdot 1 + 16 \cdot \frac{1}{2} = 16$.

Proposition 18.10 (The native gauges of the centre of H)**.** [derived]*; that there are no others* [computed] *The table's gauge, of native value* $\frac{1}{25}$*; the sign* $(-1,-1,1,1)$*, of native value* $\frac{1}{9}$*; and, of native value* 1*, the three curves*

$$\{A = 0,\ \alpha_a^2 = \alpha_2\alpha_3\}, \qquad \{B = 0,\ \alpha_2^2 = \alpha_a\alpha_b\}, \qquad \{A = B = 0\},$$

the second the image of the first under the self-duality; on them the class letters vanish and K*,* Ψ *collapse to monomials, the case of Proposition 15.15(ii) with one class. Every sign of value* 1 *lies on them.*

Proof. At the centre nativity is $R_e(\alpha) = R_e(\alpha^{-1})$ (§15.2). On $A = 0$ also $A(\alpha^{-1}) = A/(\alpha_a\alpha_b) = 0$, so $K(\alpha) = \alpha_2\alpha_3$, $K(\alpha^{-1}) = 1/(\alpha_2\alpha_3)$ and $R_2 = R_3 = 1$ at α and at α^{-1}, while $R_a(\alpha) = \alpha_a B/(\alpha_2\alpha_3)$ and $R_a(\alpha^{-1}) = B/\alpha_a$ agree exactly when $B = 0$ or $\alpha_a^2 = \alpha_2\alpha_3$, and likewise at b; there $K(\alpha)K(\alpha^{-1}) = 1$. The second curve follows by the self-duality. At a sign $K(\eta) = (\eta_a+\eta_b)(\eta_2+\eta_3)+\eta_2\eta_3 \in \{\pm5, \pm3, \pm1\}$ (Proposition 15.14), and $K(\eta) = \pm1$ forces $\eta_a = -\eta_b$ or $\eta_2 = -\eta_3$, which puts the sign on one of the curves. Newton's method from several thousand starts in $(\mathbb{C}^\times)^3$ finds no critical point of $\log f$ off these. □

Receipts [computed]. By Proposition 15.10 the closings are

$$x_k = \sum_{i+j+l=k} \Big(\frac{k!}{i!\,j!\,l!}\Big)^2 \binom{2i+2j}{i+j} = 1, 5, 53, 761, 12661, 229705, 4410665, \ldots,$$

and the least recurrence of 160 of them has order 4 and degree 7 (156 equations, 40 unknowns, one solution). Reflected, the law of the centre of H, $P(k) = P_H(k\mathbf{1}; k, k)$, is

$$\begin{gathered}
k^3(k+1)(144k^3+864k^2+1693k+1075)\,P(k)\\
-2(k+1)(2592k^6+19440k^5+56322k^4+80296k^3+60004k^2+23017k+3580)\,P(k+1)\\
+2(21168k^7+211680k^6+884919k^5+2003456k^4\\
+2646984k^3+2031207k^2+830703k+137260)\,P(k+2)\\
-2(k+2)^2(34848k^5+261360k^4+741842k^3+984642k^2+598948k+127215)\,P(k+3)\\
+225(k+2)^2(k+3)^2(144k^3+432k^2+397k+102)\,P(k+4)=0,
\end{gathered}$$

with the symbol $144(25\lambda-1)(9\lambda-1)(\lambda-1)^2$: the curves double the root 1, an integral direction in the exceptional set of Thm. 35.19(i), p. 313. (A search by total size first returns a recurrence of order 5 and degree 5: this law composed with a factor of order one, whose symbol has the extra root $\frac{131003}{4771109}$.) The corner relation is $17200 - 7160P(1) + 274520P(2) - 1017720P(3) + 826200P(4) = 0$, and c_0, c_1 share the zero $k=-1$; the second corner (§15.4) is rational,

$$57\,P(1) - 807\,P(2) + 1050\,P(3) = 196 \qquad (530 \text{ digits}),$$

the constant term at $k=-1$ being 1176. Positivity (Theorem 18.5: three conditions, one at $\frac{1}{9}$ and two at the double root) gives

$$\begin{aligned}
P_H(\mathbf{1}; 1, 1) &= 4.0962369983378828500555532283512686376346708 96\ldots,\\
P_H(2\mathbf{1}; 2, 2) &= 0.0477164986915131102974120808906636666177626293 6\ldots.
\end{aligned}$$

Proposition 18.11 (The centre of H on its share side)**.** [derived]*; receipt* [computed] *H is a series root over the parallel cell of $\{a, b\}$ and the two marks left in place (Def. 22.17(5), p. 191), so, as in Proposition 16.3,*

$$\Gamma(\sigma)^2 P_H(\sigma\mathbf{1}; \sigma, \sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\,\Gamma(\sigma-d)^2\,P_{\binom{[2]}{1}}(d\mathbf{1}; d, d)^2\,dd, \qquad P_{\binom{[2]}{1}}(d\mathbf{1}; d, d) = \Gamma_{\substack{2\\1}}\begin{bmatrix} d,\ d\\ 2d\end{bmatrix},$$

$0 < c < \sigma$*. At $\sigma = 1$, on $c = \frac{1}{2}$, the integral is* $4.0962369983378828500555532283512686376346709\ldots$ *in ball arithmetic: the value of the positive selection, to* 44 *digits.*

Proof. A series node multiplies the transforms of its cells (Thm. 22.13, p. 188): the two marks give $\mathsf{G}_{0;(1,-1)}(1, y)^2$, whose Mellin read is $\Gamma(z)^2 P_{\binom{[2]}{1}}(z\mathbf{1}; z, z)$ (Proposition 15.2), and the parallel cell gives the transform whose Mellin read is $\Gamma(w)^2 P_{\binom{[2]}{1}}((\sigma-w)\mathbf{1}; \sigma-w, \sigma-w)$ (Proposition 16.3); the Mellin read of the product pins the sum of the shares, and $d = \sigma - w$. □

The integrand is $\Gamma(d)^6\Gamma(\sigma-d)^2/\Gamma(2d)^2$. Its poles at $-\mathbb{N}$ have order four and residues that grow like 16^m, and to the right the double poles of $\Gamma(\sigma-d)^2$ meet a factor that grows faster than every power: neither closing of Thm. 4.5, p. 26, converges. The centre of H is a fibre integral of stacked Gammas that does not land on a lattice, as the five-mark centre does not (Remark 15.20).

The lean. The direction $(2\mathbf{1}; 3, 1)$ has the walls 2 on marks, 3 on $\{a, b\}$, 1 on $\{2, 3\}$, 4 on the other pairs and 2 on triples, so $\mathrm{Cor} = \frac{1}{3} + 1 + 1 = \frac{7}{3}$. On the slice $\alpha_a = \alpha_b = x$, $\alpha_2 = \alpha_3 = 1$ the stride is $x^3/\big((4x+1)^3(x+4)\big)$, whose critical points are the roots of $2x^2 - x - 6 = (x-2)(2x+3)$ [derived]: the native gauge of the direction on the orthant is $(2, 2, 1, 1)$, where $K = 9$, $\Psi = 12$ and $f = \frac{4}{2187}$, and the sign has moved to $(-\frac{3}{2}, -\frac{3}{2}, 1, 1)$, $f = \frac{27}{2500}$; the curves move to curves of value $-\frac{16}{27}$ [computed]. The closings $x_j = [\alpha^{2j\mathbf{1}}]K^{3j}\Psi^j = 1, 48, 10240, 3131184, \ldots$ have a least recurrence of order 4 and degree

25 (196 equations, 104 unknowns, one solution), with the symbol a multiple of $(2187\lambda-4)(2500\lambda-27)(27\lambda+16)^2$; positivity converges at the rate $\frac{10000}{59049}$, about 0.77 digits per step, and the lean's own second corner is rational: $413667900\,P'(1)+6131684550\,P'(2)+12031031250\,P'(3)=24789199$, $P'(k)=P_H(2k\mathbf{1};3k,k)$.

Theorem 18.12 (One content, two rays)**.** [computed]

$$28\,P_H(2\mathbf{1};3,1)=14-3\,P_H(\mathbf{1};1,1)-P_H(2\mathbf{1};2,2),$$
$$5670000\,P_H(4\mathbf{1};6,2)=126002\,P_H(\mathbf{1};1,1)+143398\,P_H(2\mathbf{1};2,2)-522781,$$
$$95681250000\,P_H(6\mathbf{1};9,3)=3048368669-731189098\,P_H(\mathbf{1};1,1)-1115791502\,P_H(2\mathbf{1};2,2),$$

to 530 *digits. The centre ray and the lean ray of H, with two laws, two corners and two positive selections, land on the one space $\mathbb{Q}+\mathbb{Q}\,P_H(\mathbf{1};1,1)+\mathbb{Q}\,P_H(2\mathbf{1};2,2)$.*

Remark 18.13 (What the masters of H are not). [computed] There is no rational relation among 1, $P_H(\mathbf{1};1,1)$, $P_H(2\mathbf{1};2,2)$ and the members in use, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Gamma(\frac12)^4$, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^6/\sqrt3$, $\sqrt3\,\Gamma(\frac12)^2$, $\sqrt3\,\Gamma(\frac12)^2\log3$, $\Gamma(\frac12)^4\log2$, $\Gamma(\frac12)^4\log3$, $\log2$, $\log3$, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)\log2$, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)\log3$, $\Gamma(\frac13)^6/\Gamma(\frac12)^4$ and $\Gamma(\frac12)^8/\Gamma(\frac13)^6$, with coefficients below 10^{24} (420 digits); nor with $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, the two masters of $\binom{[5]}{1}$, the stacked Gamma X at fifteenths of Proposition 15.27, $\Gamma(\frac12)^8/X$, $\Gamma(\frac12)^4$, $\sqrt3\,\Gamma(\frac12)^2$, $\sqrt{15}$, and $\sqrt{15}$ times X, $\Gamma(\frac12)^8/X$ and $\Gamma(\frac12)^4$, below 10^{28} (380 digits). In that framework, over $\mathbb{Q}$, the two masters of H are new numbers. H and $\binom{[5]}{1}$ have the same native values $\frac{1}{25},\frac19,1$ at their centres and different masters: the native values do not decide the numbers.

What the tree expects [not yet computed]. Opened at the mark 2 (Prop. 19.19, p. 144), H has the children $\binom{[3]}{1}$ and the pair $\{a,b\}$ with the coloop 3, the strides $X_{H/1}$ and $X_{H\setminus1}$ of §17.4, and its input is Ψ_σ of §17.4 without the factor $P_{\binom{[2]}{1}}$. Its lattice values are three-mark leans, in $\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, and its slopes should have one free direction (Proposition 17.11). As for $\binom{[5]}{2}$ (Theorem 17.8), the masters over the field of the weights of that closing should be $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ and one free slope, the weights carrying what Remark 18.13 sees as new.

18.5 Two observations

(i) *The exact pairing belongs to the banana.* Propositions 15.13 and 15.11 give

$$\mathrm{Cor}\cdot\det H=-\frac{\lambda^\star\chi'(\lambda^\star)}{[k^{n-1}]c_0}=\prod_{c\neq\star}\Big(1-\frac{\lambda^\star}{\lambda_c}\Big)$$

whenever the closings solve the adjoint of the law and c_0 has degree $n-1$; here H in $\det H$ is the Hessian of Proposition 15.11. The laws of $\binom{[n]}{1}$ for $n=2,3,4,5$ (Theorem 15.25, §15.7, Proposition 15.13) are self-adjoint up to sign, $c_i(-k-i)=\pm c_i(k)$ [computed], and the identity is

$$n!\cdot\frac{2^{n-1}}{n^n}=\prod_{j=1}^{\lceil n/2\rceil-1}\Big(1-\frac{(n-2j)^2}{n^2}\Big)=\prod_{j=1}^{\lceil n/2\rceil-1}\frac{4j(n-j)}{n^2}\quad\text{[derived]},$$

which holds for every n: the low end of strength, the corner (Thm. 38.5(i), p. 330), and the high end, the Gaussian at the native gauge (Thm. 38.5(ii)), tied through the native values. On H, whose law is not self-adjoint, and on $\binom{[5]}{2}$ (§15.5) the closings solve no adjoint and the identity fails.

(ii) *The content belongs to the list.* On $\binom{[3]}{1}$ and on H, two rays each, the masters of the ray are those of the list. The letter and mark relations (Thm. 8.3, p. 54) keep integer data on the balance

wall with rational coefficients, so every integer period of a list lies in the span they leave, and each ray samples that span. A ray with the symmetries of the list may sample only their invariant part, which offsets would break. Hypothesis: every ray through the origin strictly inside its walls, without the list's symmetries, spans the whole content. [not yet computed]

18.6 Open

(i) Certificates of Thm. 35.3 for the three laws of §§18.3–18.4, and the converse of Corollary 18.2 in general. [not yet computed]

(ii) The slit: for rays strictly inside their walls, no native value in $(0, f(\alpha^\star))$; for rays along walls, exactly the faces' native values there, carrying the faces' masters. [not yet computed]

(iii) The content of a list: an offset ray, and a ray breaking the symmetries $a \leftrightarrow b$, $2 \leftrightarrow 3$ of H. [not yet computed]

(iv) The masters of H through its hinge at a mark, as in §§17.1–17.3, with the weights. [not yet computed]

(v) H against $\binom{[5]}{1}$: the local exponents at $\frac{1}{9}$ and 1, the frames (Prop. 35.8, p. 309), and whether one law is a transform of the other. [not yet computed]

(vi) The period matrix of the centre ray of H: the solutions selected at its other native gauges, and whether they are stacked Gammas, as the lowered settings of $\binom{[5]}{1}$ are (Proposition 15.27). [not yet computed]

(vii) A carrier built for the purpose: a list on six marks that is not a graph, with native gauges off the signs and off the torus, read by certificates rather than closings. [not yet computed]

Aliases, last: the x_j are the classical periods of the Laurent polynomial $1/f$, and the law is the Picard–Fuchs operator of its pencil read on coefficients; Theorem 18.5(iv) is the mechanism of Apéry limits, the selection read off the slit (Pincherle's minimal solution, Perron's theorem); H is the cycle matroid of a triangle with one edge doubled; $\Gamma(\frac{1}{2})^6/\sqrt{3}$ stands for $L(\chi_{-3}, 3) = \frac{4\pi^3}{81\sqrt{3}}$; $\mathsf{G}_{\nu;(1,-1)}(1, y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$, so Proposition 18.8 reads the lean as the order of the Bessel functions.

19 The three-mark centre, set as an examination

Remark 17.17 read the three-mark centre off its law as a list of ratios, and Procedure 18.6 states the reading of every ray through the origin. This entry runs the reading once, on that one ray, written as an examination question with its answer key and its marking scheme. A key has to say what a complete solution shows at every step and what each omission costs. So every hypothesis that §§15–18 checked once in passing is checked here, at the step that uses it, and the places where a solution can look right and be wrong are named, with the credit they earn. The ratios of Remark 17.17 become a certified bracket. The receipts are in the script exam_s19.py (python-flint: exact arithmetic for the solution, ball arithmetic only for the independent check of §19.5).

19.1 The question

Question (40 points). Let $F = \binom{[3]}{1}$ be the list on the marks $\{1,2,3\}$ whose bases are the singletons, with unit weights, so that its letters are

$$K = \alpha_1 + \alpha_2 + \alpha_3, \qquad \Psi = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3,$$

and let $\ell = P_F(\mathbf{1}; 1, 1)$ be the period of its centre. By the reading of its centre ray (Procedure 18.6), that is, the law of the ray certified by the relations along a ray (Thm. 35.3, p. 308), its corner, and positive selection (Theorem 18.5), find a rational number q with

$$0 < \ell - q < 10^{-6},$$

and prove both inequalities.

You may cite the handbook and §§15–18 of this notebook. When you cite a statement, check its hypotheses in the case at hand. Exact computation (integers, rationals, polynomials, linear algebra over $\mathbb{Q}$) may be done by machine; report it so that it can be repeated. Floating-point arithmetic, numerical integration, integer-relation searches and comparison with a known decimal value of ℓ may be used as checks. They earn no credit.

19.2 The answer and the scheme

Answer. $q = a_9 = \dfrac{2447622221}{1044248800} = 2.3439071426\ldots$, with

$$\frac{2447622221}{1044248800} < \ell < \frac{6414407843}{2736630400}, \qquad b_9 - a_9 = \frac{6561}{59511099776} < 1.11 \cdot 10^{-7}.$$

Any rational q with a complete proof of $0 < \ell - q < 10^{-6}$ earns full credit, whatever bounds the proof uses (other consequences of positivity, such as the determinants of Thm. 35.10(iii), are admissible), and the points of the scheme then go to the corresponding steps of that proof; §19.4, items 1, 5 and 6, lists the alternatives a grader will meet most often. What the key proves is this.

Theorem 19.1 (The three-mark centre, bracketed)**.** [derived]*; the table* [computed] *exactly. Let c and r be the solutions of*

$$k^2 x(k) - (10k^2 + 10k + 3)\, x(k+1) + 9(k+1)^2\, x(k+2) = 0 \qquad (k \geq 1)$$

with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. For every integer $K \geq 2$, $c(K) > 0$ and $9c(K) > c(K-1)$, and

$$a_K = -\frac{r(K)}{c(K)} \;<\; P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) \;<\; b_K = \frac{r(K-1) - 9\, r(K)}{9\, c(K) - c(K-1)},$$

$$b_K - a_K = \frac{2 \cdot 9^{2-K}}{3(K-1)^2\, c(K)\big(9c(K) - c(K-1)\big)}.$$

Step	What is shown	Points
1	The ray: the chart and its measure, the stride, the values, the walls; $\ell = P(1)$	3
2	The stride: $f \le \frac{1}{9}$; $\frac{1}{9}e^{-\varphi} \le f \le e^{-\varphi}$; $\int e^{-k\varphi}dy = 6/k^2$; what follows at every real $k > 0$	5
3	The closings, their recurrence, the reflected candidate law	3
4	A certificate for Thm. 35.3 and its exact verification	5
5	The law at every real $k > 0$: the integration by parts, with the bounds of Step 2	5
6	The native values; what positivity can select	2
7	The corner: $k^2P(k) \to 6$, so $9P(2) = 3\ell - 6$	5
8	The rational solutions, their signs, the exact table	4
9	The bracket $a_K < \ell < b_K$, the choice of K, the answer	8
		40

19.3 The key

Throughout, e_1, e_2, e_3 are the elementary symmetric polynomials of $\alpha_1, \alpha_2, \alpha_3$, so that $K = e_1$ and $\Psi = e_2$, and $D_e = \alpha_e\, \partial/\partial\alpha_e$.

Step 1 (3 points). The ray. Take the chart $\alpha_3 = 1$ of the mark $e_0 = 3$, with $\alpha_1 = e^{y_1}$ and $\alpha_2 = e^{y_2}$, so that

$$dy = dy_1\, dy_2 = \frac{d\alpha_1\, d\alpha_2}{\alpha_1\alpha_2}.$$

The centre ray at the strength $\sigma = 1$ (Def. 35.9, p. 309) has origin 0 and direction $(\mathbf{1}; 1, 1)$. Its stride and its values are

$$f = \frac{\alpha^{\mathbf{1}}}{K\Psi} = \frac{e_3}{e_1e_2}, \qquad P(k) = P_F(k\mathbf{1}; k, k) = \int_{\mathbb{R}^2} f^k\, dy \qquad (k = 1, 2, \dots),$$

so that

$$\ell = P(1) = \int_0^\infty\int_0^\infty \frac{d\alpha_1\, d\alpha_2}{(1 + \alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 + \alpha_1\alpha_2)}.$$

For real $k > 0$ the same integral defines $P(k)$; Step 2 shows that it converges. The walls of the direction are $\kappa_{1,S} = |S| - d(S) - d^*(S)$ (Def. 8.1, p. 53), the width $c(S)$ of Def. 7.10, p. 49. Every nonempty S has rank $r(S) = 1$, so for S nonempty and proper $d(S) = |S| - r^*(S) = r - r(\mathcal{E} \setminus S) = 0$ and $d^*(S) = |S| - r(S) = |S| - 1$, and

$$\kappa_{1,S} = 1 \qquad (\varnothing \ne S \subsetneq \mathcal{E}).$$

All proper walls are positive, the hypothesis of Thm. 35.3 and of Thm. 35.10, p. 309.

Marking. 1 point for the stride with the chart and its measure; 1 for $\ell = P(1)$ with the definition of the values cited; 1 for the walls. The measure $d\alpha_1 d\alpha_2$ in place of dy makes the integral diverge (the integrand is $\sim 1/(\alpha_1 + \alpha_2)$ at infinity): -1.

Step 2 (5 points). The stride. (a) *The gauge.* The identity

$$e_1e_2 - 9e_3 = \alpha_1(\alpha_2 - \alpha_3)^2 + \alpha_2(\alpha_3 - \alpha_1)^2 + \alpha_3(\alpha_1 - \alpha_2)^2$$

holds because both sides equal $\sum_{i\ne j} \alpha_i^2\alpha_j - 6e_3$, the left since $e_1e_2 = \sum_{i\ne j} \alpha_i^2\alpha_j + 3e_3$. So $0 < f \le \frac{1}{9}$ on the orthant, with equality exactly on the diagonal $\alpha_1 = \alpha_2 = \alpha_3$, which on the chart is the single point $y = 0$. This is Thm. 35.10(i) here, with $\alpha^\star = \mathbf{1}$ and $f(\alpha^\star) = \frac{1}{9}$.

(b) *The decay.* Order the coordinates, $\alpha_{(1)} \ge \alpha_{(2)} \ge \alpha_{(3)}$. Then $\alpha_{(1)} \le e_1 \le 3\alpha_{(1)}$, $\alpha_{(1)}\alpha_{(2)} \le e_2 \le 3\alpha_{(1)}\alpha_{(2)}$ and $e_3 = \alpha_{(1)}\alpha_{(2)}\alpha_{(3)}$, so

$$\frac{1}{9}\,\frac{\alpha_{(3)}}{\alpha_{(1)}} \le f \le \frac{\alpha_{(3)}}{\alpha_{(1)}}.$$

On the chart $\alpha_{(3)}/\alpha_{(1)} = e^{-\varphi(y)}$ with

$$\varphi(y) = \max(y_1, y_2, 0) - \min(y_1, y_2, 0),$$

and since 0 and y_i are both among the three entries, $\varphi(y) \ge |y_i|$: $\varphi(y) \ge \|y\|_\infty$.

(c) *The volume.* For real $k > 0$,

$$\int_{\mathbb{R}^2} e^{-k\varphi(y)}\,dy = \frac{6}{k^2}.$$

The lines $y_1 = y_2$, $y_1 = 0$, $y_2 = 0$ cut the plane into six cones, one for each order of the entries $(y_1, y_2, 0)$, and on each cone φ is the largest entry less the smallest:

$$\int_{y_1 \ge y_2 \ge 0} e^{-ky_1} dy = \int_0^\infty y_1 e^{-ky_1} dy_1 = \frac{1}{k^2}, \qquad \int_{y_1 \ge 0 \ge y_2} e^{-k(y_1 - y_2)} dy = \frac{1}{k}\cdot\frac{1}{k},$$
$$\int_{0 \ge y_1 \ge y_2} e^{ky_2} dy = \int_0^\infty u e^{-ku} du = \frac{1}{k^2},$$

and the three cones with y_1 and y_2 exchanged give the same.

(d) *Consequences*, for real $k > 0$:

$$\frac{6 \cdot 9^{-k}}{k^2} \le P(k) \le \frac{6}{k^2}, \qquad P(k+1) < \frac{1}{9}\,P(k),$$

the second from (a), strict because $f < \frac{1}{9}$ off the single point $y = 0$. And P is continuous on $(0, \infty)$: on $[k_1, \infty)$ the integrands f^k are dominated by f^{k_1}, because $0 < f < 1$, and f^{k_1} is integrable.

Marking. (a) 2 points; the identity must be shown or verified by expansion. The two inequalities of the arithmetic and geometric means, $e_1 \ge 3e_3^{1/3}$ and $e_2 \ge 3e_3^{2/3}$, are an accepted proof of $f \le \frac{1}{9}$. The strict inequalities of Step 9 need only $f < \frac{1}{9}$ on a set of positive measure; without any strictness the bracket is closed at the top, which suffices for the answer. (b) 1 point for the upper side, the decay; citing Thm. 35.10(i) for $f \le C'e^{-c'\|y\|}$ on the chart earns it, and gives convergence at every real $k > 0$. 1 point for the lower side of (b) and the volume (c) together; they serve only the corner, and a solution that obtains the corner by a correct citation of Proposition 15.3 or 18.1 in Step 7 earns this point there. (d) 1 point: convergence, positivity, continuity and $P(k+1) < \frac{1}{9}P(k)$. Step 2(d) credits the proof of the last inequality and Step 9(b) its use.

Step 3 (3 points). The candidate law. Here $1/f = e_1e_2/e_3 = (\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, and the closings of the direction (Proposition 18.1) are

$$x_j = [\alpha^0]\Big(\frac{1}{f}\Big)^j = \sum_{a_1+a_2+a_3=j} \Big(\frac{j!}{a_1!\,a_2!\,a_3!}\Big)^2 = 1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ 35169,\ 272835,\ 2157759,\ \ldots,$$

since the constant term pairs the monomial α^a of $(\sum_e \alpha_e)^j$ with the monomial α^{-a} of $(\sum_e \alpha_e^{-1})^j$, and both have the coefficient $j!/(a_1!a_2!a_3!)$. Exact linear algebra with the ansatz $\sum_{i=0}^{2} p_i(j)\,x_{j+i} = 0$, $\deg p_i \le 2$, on $x_0, \ldots, x_{15}$ (14 equations, 9 unknowns) has a one-dimensional space of solutions,

$$(j+2)^2 x_{j+2} - (10j^2 + 30j + 23)\,x_{j+1} + 9(j+1)^2 x_j = 0.$$

Reflected as in Corollary 18.2, $c_i(k) = p_{2-i}(-k-2)$:

$$k^2P(k) - (10k^2+10k+3)\,P(k+1) + 9(k+1)^2\,P(k+2) = 0. \tag{L}$$

At this point (L) is a candidate. Corollary 18.2 runs the other way, from a law of P to the recurrence of the closings, and sixteen closings do not prove their own recurrence.

Marking. 1 point each for the closings, the recurrence (with the size of the system, or any other exact derivation) and the reflection. Any route to (L) earns the 3 points: the closings are how the law is found, not part of its proof. A solution that takes (L) as proved here keeps these 3 points and earns nothing in Steps 4 and 5.

Step 4 (5 points). The certificate. The origin is 0, so the defects of the origin vanish, and the defect of the direction is (Lemma 35.2(ii), p. 308)

$$c_e = D_e \log f = 1 - R_e - R_e^*, \qquad R_e = \frac{\alpha_e}{e_1}, \qquad R_e^* = \frac{\alpha_e(e_1-\alpha_e)}{e_2},$$

the occupation of e under the basis law and under the cobasis law. Since $\sum_e R_e = 1$ and $\sum_e R_e^* = 2$, $\sum_e c_e = 0$. A *certificate* of (L) is a choice of insertions g_1, g_2, g_3 (Def. 35.1: regular functions on the letter complement; here rational functions of degree 0 whose denominators are products of e_1 and e_2, with coefficients polynomial in k) such that

$$k^2 - (10k^2+10k+3)\,f + 9(k+1)^2 f^2 = \sum_{e=1}^{3} \big(D_e g_e + k\,c_e\,g_e\big). \tag{C}$$

Ansatz: $g_e = N_e/(e_1^2 e_2)$, with N_e homogeneous of degree 4 with coefficients in $\mathbb{Q}+\mathbb{Q}k$, g_e the image of g_1 under the exchange of the marks 1 and e, and N_1 symmetric in α_2, α_3. Multiplied by $e_1^3e_2^2$, (C) becomes a polynomial identity, of degree 7 in α and 2 in k, through

$$e_1^3e_2^2\big(D_e g_e + k\,c_e g_e\big) = e_1e_2\,D_eN_e - N_e\big(2\alpha_e e_2 + \alpha_e(e_1-\alpha_e)e_1\big) + k\,N_e\big(e_1e_2 - \alpha_e e_2 - \alpha_e(e_1-\alpha_e)e_1\big),$$

and it is linear in the 18 coefficients of N_1: 99 equations of rank 10. One solution is

$$g_1 = -\frac{\alpha_1^2\big(k\,\alpha_1(\alpha_2+\alpha_3) + (2+3k)(\alpha_2^2+\alpha_3^2) - (1-k)\,\alpha_2\alpha_3\big)}{e_1^2e_2}, \tag{G}$$

and within the ansatz the others differ from it by one insertion h added at every mark, h with a symmetric numerator (an 8-dimensional space). Such an h changes nothing, because $\sum_e D_e h = 0$ (Euler's identity in degree 0) and $\sum_e c_e h = 0$.

Verification. Write $g_e = g_e^{(0)} + k\,g_e^{(1)}$. The three powers of k in (C) read

$$\begin{aligned} k^2: &\quad \textstyle\sum_e c_e g_e^{(1)} = 1 - 10f + 9f^2 = (1-f)(1-9f),\\ k^1: &\quad \textstyle\sum_e \big(D_e g_e^{(1)} + c_e g_e^{(0)}\big) = -10f + 18f^2,\\ k^0: &\quad \textstyle\sum_e D_e g_e^{(0)} = -3f + 9f^2. \end{aligned}$$

Times $e_1^3e_2^2$ each sum over the marks is a symmetric polynomial of degree 7, and in the basis of products of e_1, e_2, e_3 exact expansion gives

$$\sum_e e_1^3e_2^2\,c_e g_e^{(1)} = e_1^3e_2^2 - 10\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$

$$\sum_e e_1^3e_2^2\,D_e g_e^{(1)} = -2\,e_1e_2^3 - 3\,e_1^2e_2e_3 + 9\,e_1e_3^2, \qquad \sum_e e_1^3e_2^2\,c_e g_e^{(0)} = 2\,e_1e_2^3 - 7\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$

$$\sum_e e_1^3e_2^2\,D_e g_e^{(0)} = -3\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$

which, with $f = e_3/(e_1e_2)$, are the three right sides times $e_1^3e_2^2$. So (C) holds identically in α and k. Proposition 17.15 reported a certificate of this law with the same denominator.

Marking. 1 point for the defects c_e, in any equivalent form. 2 points for a certificate. Any certificate is accepted; it need not be symmetric, and any denominator is accepted for which Step 5 can be carried out. 2 points for the verification: by exact expansion, by the symmetric-function form above, or by exact evaluation of the cleared identity on a grid large enough to decide it. After clearing, both sides are polynomials of degree at most 7 in each α_i and 2 in k, homogeneous in α, so agreement at $\alpha_3 = 1$ on 8 values of α_1, 8 of α_2 and 3 of k (all 192 points) proves it. Agreement at a few random points is a test, not a proof. To test a certificate other than (G), a grader evaluates both sides of (C) at $\alpha = (1, 2, 4)$, $k = \frac{1}{3}$, where the left side is $-\frac{2809}{7203}$ (there $f = \frac{4}{49}$, and (G) gives $g_1, g_2, g_3 = -\frac{85}{1029}, -\frac{310}{1029}, -\frac{424}{1029}$); a certificate written on the chart, with $g_3 = 0$ and D_1, D_2 read as $\partial_{y_1}, \partial_{y_2}$, is tested at the same point $(\alpha_1, \alpha_2) = (\frac{1}{4}, \frac{1}{2})$. A certificate with floating-point coefficients, or verified only in floating point, earns 0 of the 4: the identity has to hold exactly. How the certificate was found does not matter: rational coefficients found by a numerical search and then verified exactly earn full credit.

Step 5 (5 points). The law at every real $k > 0$. Thm. 35.3 states the relations for all *large* k. The law is needed at $k = 1, \dots, K-2$ (Step 8) and as $k \to 0^+$ (Step 7), so the integration by parts is carried out here at every real $k > 0$, with the bounds of Step 2. Fix a real $k > 0$.

(a) $D_e(f^k) = k\, c_e f^k$, so the right side of (C), times f^k, is $\sum_e D_e(g_e f^k)$.

(b) On the chart $D_1 = \partial_{y_1}$ and $D_2 = \partial_{y_2}$, and for every function h of degree 0, $D_3 h = -D_1 h - D_2 h$ (Euler's identity). So

$$\sum_{e=1}^{3} D_e(g_e f^k) = \partial_{y_1} H_1 + \partial_{y_2} H_2, \qquad H_i = (g_i - g_3) f^k.$$

(c) *Bounds.* Every monomial of every N_e occurs in the expansion of $e_1^2 e_2$: for N_1, $\alpha_1^3\alpha_2 = \alpha_1^2 \cdot \alpha_1\alpha_2$, $\alpha_1^2\alpha_2^2 = \alpha_1\alpha_2 \cdot \alpha_1\alpha_2$ and $\alpha_1^2\alpha_2\alpha_3 = \alpha_1^2 \cdot \alpha_2\alpha_3$, with the images under $2 \leftrightarrow 3$, and the other N_e are images of N_1. The coefficients of $e_1^2 e_2$ are positive, so $\alpha^v \le e_1^2 e_2$ on the orthant for each such monomial, and $|g_e| \le \|N_e\|_1$, the sum of the absolute values of the coefficients. For every mark i,

$$D_i g_e = \frac{D_i N_e}{e_1^2 e_2} - g_e\left(2R_i + R_i^*\right),$$

where $D_i N_e$ has the monomials of N_e or fewer and $0 \le R_i, R_i^* \le 1$; and $|c_i| \le 1$. With Step 2(b) there is, for fixed k, a constant C with

$$|H_i| \le C\, e^{-k\varphi}, \qquad |\partial_{y_j} H_i| \le C\, e^{-k\varphi}.$$

(d) If H is C^1 on $\mathbb{R}^2$, tends to 0 at infinity and $\partial_{y_1} H$ is integrable, then $\int \partial_{y_1} H\, dy = 0$: for every y_2, $\int_{-R}^{R} \partial_{y_1} H\, dy_1 = H(R, y_2) - H(-R, y_2) \to 0$, and Fubini applies. By (c) and Step 2, H_1 and H_2 qualify ($e^{-k\varphi} \le e^{-k\|y\|_\infty} \to 0$, and $e^{-k\varphi}$ is integrable), so the right side of (C), times f^k, integrates to 0.

(e) The left side of (C), times f^k, integrates to the left side of (L), all three integrals convergent by Step 2. Hence (L) holds at every real $k > 0$.

Marking. 1 point for (a)–(b), including the mark 3 on its own chart. 2 points for the bounds (c). 1 point for (d). 1 point for the conclusion at every real $k > 0$, with the reason it is needed. Citing Thm. 35.3 at $k = 1, 2, \dots$ and as $k \to 0^+$ without (c): 2 of 5. Any complete extension earns full credit; for instance (C) integrated at large integers by Thm. 35.3 as stated (its hypothesis checked in Step

1), then carried to every k with $\operatorname{Re} k > 0$ by Carlson's theorem: the left side of (L) is holomorphic there, it is bounded by a polynomial in $|k|$ on $\operatorname{Re} k \geq 1$ because $|P(k)| \leq P(\operatorname{Re} k) \leq 6/(\operatorname{Re} k)^2$ (Step 2), and Carlson's theorem applied to $z \mapsto F(N+z)$, with F the left side of (L) and $N \geq 1$ an integer from which on (L) holds at every integer, makes F vanish identically. Integrating $D_3(g_3 f^k)$ by parts "in y_3" on the chart $\alpha_3 = 1$, where y_3 is not a coordinate: -1, unless repaired by Euler's identity or by a certificate with $g_3 = 0$.

Step 6 (2 points). The native values; what positivity can select. The symbol of (L) (Cor. 35.6, p. 308) is

$$\chi(\lambda) = 1 - 10\lambda + 9\lambda^2 = (1-\lambda)(1-9\lambda).$$

The native gauges of the centre ray are the signs η with $\eta_3 = 1$ (Proposition 15.7, Thm. 35.12, p. 310), of native value $K(\eta)^{-2}$: the table's gauge $\eta = (1,1,1) = \alpha^\star$, with $K(\eta) = 3$ and the value $\frac{1}{9} = f(\alpha^\star)$; and $\eta = (-1,1,1), (1,-1,1), (-1,-1,1)$, with $K(\eta) = \pm 1$ and the value 1. They are nondegenerate. By Thm. 35.12(ii) the Hessian of $-\log f$ at η is $2\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e' \neq 3}$, the covariances of the indicators under the signed measure $\mathbb{P}_\eta(\{e\}) = \eta_e / K(\eta)$. Its determinant is $\frac{4}{27}$ at $(1,1,1)$ and -4 at each of the other three signs. So Cor. 35.6, with $g = 1$, makes both native values roots of χ, and since χ has degree 2 they are all its roots. (Cor. 35.6 runs from native values to roots; a root need not be a native value, as the extra root $\frac{131003}{4771109}$ in §18.4 shows.)

In Theorem 18.5 the singular points are $q = 9$, where the slit $[9, \infty)$ begins, and $q = 1$, off the slit. At $q = 1$, a simple root and the value of nondegenerate gauges, positivity imposes $\nu = 1$ condition, and with the corner $1 + \nu = 2 = h$. The condition is independent, and here that is seen directly: by Step 8, $c(K) \geq 3^{1-K}$ and $9c(K) - c(K-1) \geq 2c(K-1) \geq 2 \cdot 3^{2-K}$, so the width of Step 9(d) is at most $1/(K-1)^2$ and the bracket closes on one number. So the law, the corner and positivity determine P (Theorem 18.5(iii)). The rational solutions of (L) have the exponential rate 1 of the signs and P the rate $\frac{1}{9}$ of the gauge, so the bracket shrinks by about $\frac{1}{9}$ per step, $\log_{10} 9 = 0.954$ digits: about seven steps for 10^{-6}, before constants. The constants decide in Step 9.

Marking. 1 point for the native values with their nondegeneracy, matched to the roots of the symbol in the direction Cor. 35.6 gives. "The roots are native values by Cor. 35.6" reverses the corollary and earns 0 for this point unless the roots are matched to gauges directly, as above. 1 point for the count $1 + \nu = h$ with the independence, or any equivalent statement that the corner supplies the condition positivity does not. The step is not used in the proofs of the bounds; a complete solution without it loses these 2 points only.

Step 7 (5 points). The corner. (a) By Step 2(d), $6 \cdot 9^{-k} \leq k^2 P(k) \leq 6$ for real $k > 0$, so

$$\lim_{k \to 0^+} k^2 P(k) = 6.$$

(b) Take (L) at real $k \in (0,1)$ (Step 5) and let $k \to 0^+$. By the continuity of Step 2(d), $P(k+1) \to P(1) = \ell$ and $P(k+2) \to P(2)$, while $10k^2 + 10k + 3 \to 3$ and $9(k+1)^2 \to 9$. So $6 - 3\ell + 9P(2) = 0$:

$$P(2) = \frac{\ell - 2}{3}. \tag{S}$$

(c) This is Corollary 18.2 at $j = 0$,

$$[k^2] c_0 \cdot \mathrm{Cor} + c_1(0) P(1) + c_2(0) P(2) = 0, \qquad \mathrm{Cor} = \sum_\pi \prod_{i=2}^{3} \kappa_{1,U_i}^{-1} = 3! \cdot 1 = 6$$

(Proposition 18.1; for $\binom{\mathcal{E}}{1}$ the corner is $n!$, Proposition 15.3). The six cones of Step 2(c) are the six orders π, each contributing $\prod_i \kappa_{1,U_i}^{-1} = 1$.

Marking. 2 points for (a), 2 for (b), 1 for (c). A correct citation of Proposition 15.3 or 18.1 in place of (a), with the walls of Step 1 and the sum over the 3! orders, earns the 2 points of (a). Errors:

- Reading (L) at $k = 0$ as $0 \cdot P(0) - 3P(1) + 9P(2) = 0$. Here $P(0) = \int dy = \infty$, and the product $0 \cdot \infty$ is the corner, 6, not 0. The reading gives $P(2) = \ell/3$, which is false ($P(2) = 0.1146\ldots$, $\ell/3 = 0.78\ldots$) and contradicts Step 2(d) at once: $P(2) < \frac{1}{9}P(1)$ would say $\ell/3 < \ell/9$. After it every $P(K)$ is ℓ times a positive rational, positivity fixes only ratios, and the bracket is empty. 0 for Step 7, and in Steps 8–9 only the points that do not depend on (S): the signs and their use, 2 points.
- Passing the limit under the integral, $k^2 f^k \to 0$ pointwise, so "$k^2 P(k) \to 0$": 0 for (a). There is no domination, since $\int dy = \infty$.
- The wrong power ($kP(k)$, the corner of a ray on two marks) or Cor = 1 (one order): 0 for (a); (b) is marked as a consequence.
- Citing Corollary 18.2 for (S). Its hypothesis is the law *identically*, as meromorphic functions of k; Step 5 gives it at real $k > 0$, and the continuation of Proposition 18.1 extends it. Citing the corollary with the law proved only at the integers leaves a gap: -1.

Step 8 (4 points). The rational solutions and their signs. For integers $k \geq 1$, $9(k+1)^2 \neq 0$, and (L) gives

$$P(k+2) = \frac{(10k^2 + 10k + 3)\, P(k+1) - k^2 P(k)}{9(k+1)^2}.$$

Let c and r be the solutions of (L) with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. From $P(1) = \ell$, (S) and induction,

$$P(K) = c(K)\,\ell + r(K) \qquad (K \geq 1).$$

Signs, at every K. Put $x'_j = 9^j c(j+1)$. Then (L) at $k = j+1$ reads $(j+2)^2 x'_{j+2} = (10j^2 + 30j + 23)\, x'_{j+1} - 9(j+1)^2 x'_j$, with $x'_0 = 1$, $x'_1 = 3$. If $x'_{j+1} \geq 3x'_j > 0$, then

$$(j+2)^2 x'_{j+2} \geq \big(10j^2 + 30j + 23 - 3(j+1)^2\big) x'_{j+1} = (7j^2 + 24j + 20)\, x'_{j+1} \geq 3(j+2)^2 x'_{j+1},$$

since $7j^2 + 24j + 20 - 3(j+2)^2 = 4(j+1)(j+2) \geq 0$. So $x'_{j+1} \geq 3x'_j > 0$ for every j, hence $x'_j \geq 3^j$, that is,

$$c(K) \geq 3^{1-K} > 0, \qquad 9c(K) \geq 3c(K-1) > c(K-1) \qquad (K \geq 2).$$

The x'_j obey the recurrence of the closings with the same first two terms, so $x'_j = x_j$ and $c(K) = 9^{1-K} x_{K-1}$ wherever that recurrence holds: in the table below by inspection, and for every j by Corollary 18.2 once (L) holds identically. The proof of the signs does not use this.

K	$c(K)$	$r(K)$	$a_K = -r(K)/c(K)$	
1	1	0	0	
2	1/3	−2/3	2	
3	5/27	−23/54	23/10	2.3
4	31/243	−145/486	145/62	2.33870967…
5	71/729	−1331/5832	1331/568	2.34330985…
6	517/6561	−242353/1312200	242353/103400	2.34383945…
7	11723/177147	−5495507/35429400	5495507/2344600	2.34389959…
8	10105/177147	−46422941/347208120	46422941/19805800	2.34390638…
9	26639/531441	−2447622221/20832487200	2447622221/1044248800	2.34390714…
10	5773279/129140163	−530455590781/5062294389600	530455590781/226312536800	2.34390722…

Two checks a grader can run on a student's table: $9^{K-1}c(K)$ is the closing x_{K-1} $(1, 3, 15, 93, \dots)$, and the frame of the two solutions (Prop. 35.8, p. 309: $C_{k+1} = \frac{c_0(k)}{c_2(k)}C_k = \frac{k^2}{9(k+1)^2}C_k$, $C_1 = -\frac{2}{3}$) is

$$c(k)\,r(k+1) - c(k+1)\,r(k) = -\frac{2}{3}\cdot\frac{9^{1-k}}{k^2}.$$

Marking. 2 points for $P(K) = c(K)\ell + r(K)$ with the initial values from (S). 1 point for the signs $c(K) > 0$ and $9c(K) - c(K-1) > 0$, at every K as above or at the K used from the exact table. 1 point for the table, exact, through the K used. A decimal table: 0 for the last point.

Step 9 (8 points). The bracket and the answer. (a) *Below, by positivity.* $P(K) = c(K)\ell + r(K) > 0$, divided by $c(K)$, gives

$$\ell > a_K = -\frac{r(K)}{c(K)}.$$

(b) *Above, by the gauge.* By Step 2(d), $P(K) < \frac{1}{9}P(K-1)$, that is, $c(K)\ell + r(K) < \frac{1}{9}\big(c(K-1)\ell + r(K-1)\big)$, or $\big(9c(K) - c(K-1)\big)\ell < r(K-1) - 9r(K)$, and divided by $9c(K) - c(K-1)$,

$$\ell < b_K = \frac{r(K-1) - 9\,r(K)}{9\,c(K) - c(K-1)}.$$

(c) *The signs.* Both divisions keep the direction of the inequality: $c(K) > 0$ and $9c(K) - c(K-1) > 0$ by Step 8.

At $K = 2$ this is the first bracket: $P(2) = \frac{\ell-2}{3} > 0$ gives $\ell > 2$, and $P(2) < \frac{1}{9}P(1)$ gives $\ell < 3$.

(d) *The width and the answer.* With the frame of Step 8,

$$b_K - a_K = \frac{r(K-1)\,c(K) - r(K)\,c(K-1)}{c(K)\big(9c(K) - c(K-1)\big)} = \frac{2\cdot 9^{2-K}}{3(K-1)^2\,c(K)\big(9c(K) - c(K-1)\big)},$$

and in exact arithmetic:

K	b_K	$b_K - a_K$	width with $P(K) < 9^{1-K}\ell$ instead
2	3	1	1
3	19/8	$7.50\cdot 10^{-2}$	$1.64\cdot 10^{-1}$
4	61/26	$7.44\cdot 10^{-3}$	$2.54\cdot 10^{-2}$
5	3413/1456	$7.84\cdot 10^{-4}$	$3.67\cdot 10^{-3}$
6	104539/44600	$8.43\cdot 10^{-5}$	$5.04\cdot 10^{-4}$
7	149014/63575	$9.17\cdot 10^{-6}$	$6.66\cdot 10^{-5}$
8	909876251/388187800	$1.0036\cdot 10^{-6}$	$8.59\cdot 10^{-6}$
9	6414407843/2736630400	$1.1025\cdot 10^{-7}$	$1.0863\cdot 10^{-6}$
10	232184895407/99058909600	$1.21\cdot 10^{-8}$	$1.35\cdot 10^{-7}$

At $K = 8$ the certified width is $1.0036 \cdot 10^{-6} > 10^{-6}$; at $K = 9$ it is $\frac{6561}{59511099776} < 1.11 \cdot 10^{-7}$. So

$$\frac{2447622221}{1044248800} = a_9 < \ell < b_9 = \frac{6414407843}{2736630400},$$

and $q = a_9$ satisfies $0 < \ell - q < b_9 - a_9 < 1.11 \cdot 10^{-7} < 10^{-6}$.

Marking. (a) 2 points. (b) 2 points: 1 for using $P(K) < \frac{1}{9}P(K-1)$ (its proof is credited in Step 2(d)), 1 for b_K. (c) 1 point for invoking the two signs at the divisions (their proof is credited in Step 8; a sign proved there and used here silently loses nothing). (d) 3 points: 1 for a K at which the width certified by (a)–(c) is below 10^{-6}, 1 for the exact comparison, 1 for a stated q whose inequalities $0 < \ell - q < 10^{-6}$ follow from the bracket.

19.4 Where a solution that looks right is wrong

1. *The answer one step early.* $a_8 = \frac{46422941}{19805800}$ is a correct answer: $\ell - a_8 = 8.58 \cdot 10^{-7}$. But the bracket at $K = 8$ has width $1.0036 \cdot 10^{-6}$, so a solution that certifies a_8 from $K = 8$ alone has an error in it. The upper bound at $K = 9$ certifies it: $\ell - a_8 < b_9 - a_8 = 8.72 \cdot 10^{-7}$. Full credit with that; with the bound claimed at $K = 8$, 5 of the 8 points of Step 9. The true error falls below the tolerance one step before the certified one does.

2. *The observed error.* Comparing a_K with a decimal value of ℓ (from the share side, from a quadrature, or from a closed form recalled elsewhere) proves nothing about $\ell - a_K$: 0 for Step 9(d).

3. *The rate as a bound.* Theorem 18.5(iv) gives the error $O(9^{-K})$ up to powers of K, with no constant: 0 for Step 9(d). Nor is the last step a bound: $a_9 - a_8 = 7.62 \cdot 10^{-7}$ is smaller than $\ell - a_8 = 8.58 \cdot 10^{-7}$, because the a_K increase to ℓ with ratio about $\frac{1}{9}$, so the remaining error is about $\frac{9}{8}$ of the next step.

4. *An unproved upper bound.* "$P(K) \le 9^{-K}\ell$, since $f \le \frac{1}{9}$". The gauge gives $P(K) < \frac{1}{9}P(K-1)$, and iterated $P(K) < 9^{1-K}P(1)$; the factor 9^{-K} does not follow from $f \le \frac{1}{9}$, it would need $\int dy < \infty$. (The inequality is in fact true for $K \ge 8$ and false for $K \le 7$: $9^K P(K)$ decreases by Step 2(d), and $9^7 P(7)/\ell = 1.03$, $9^8 P(8)/\ell = 0.90$. Nothing in such a solution proves it, and a grader who checks it numerically should not be misled.) With it the width at $K = 9$ is $1.21 \cdot 10^{-7}$, while the simple bound actually proved gives $1.09 \cdot 10^{-6}$ there, so the solution certifies nothing below 10^{-6}. The stated $q = a_9$ is right and the proof is not: 1 of 2 in Step 9(b), 0 in 9(d), 4 of the 8 points of Step 9.

5. *The simple bound, done right.* $P(K) < 9^{1-K}P(1) = 9^{1-K}\ell$ gives $\big(c(K) - 9^{1-K}\big)\ell < -r(K)$, and $c(K) - 9^{1-K} = 9^{1-K}(x'_{K-1} - 1) > 0$ for $K \ge 2$ by Step 8, so $\ell < -r(K)/\big(c(K) - 9^{1-K}\big)$: valid and cruder (the last column of the table). It needs $K = 10$, with $q = a_{10} = \frac{530455590781}{226312536800}$. Full credit, with the sign stated.

6. *Decimals.* $q = 2.343907$ is accepted, with the exact comparisons $\frac{2343907}{10^6} < a_9$ and $b_9 - \frac{2343907}{10^6} = 2.53 \cdot 10^{-7}$. A decimal above a_9, such as 2.3439072, is a correct answer ($\ell - 2.3439072 = 3.9 \cdot 10^{-8}$) but needs a new lower bound: $a_{10} = 2.34390722\ldots$ supplies it. A decimal rounded above the lower bound in hand: -1 in Step 9(d), its last point.

7. *The signs.* Signs of $c(K)$ and of $9c(K) - c(K-1)$ neither proved nor checked: -2 in all, the sign point of Step 8 and Step 9(c). They hold here at every K (Step 8), but where a rational solution changes sign the inequality reverses; on the line 1 of $\binom{[5]}{2}$, Remark 17.17 checked the sign only for $m \le 80$.

8. *The corner.* Reading the law at $k = 0$ with $0 \cdot P(0) = 0$, or passing the limit $k \to 0^+$ under the integral: the approximation is lost (Step 7). The first gives $P(2) = \ell/3$, every $P(K)$ is then ℓ times a positive rational, positivity gives only $\ell > 0$, and the gauge bound at $K = 2$, $\ell/3 < \ell/9$, gives $\ell < 0$: the student's own bracket is empty, which the grader can point to.
9. *The theorem at small k.* Thm. 35.3 cited at $k = 1, 2, \dots$ or as $k \to 0^+$ without the decay of Step 2: Step 5.
10. *A certificate checked at sample points only*: Step 4.
11. *The measure $d\alpha_1 d\alpha_2$ on the chart*: Step 1.
12. *A quadrature for ℓ or for $P(2)$ used in the solution*: no credit at the step where it is used. It is a check (§19.5).

19.5 Receipts and the independent check

The script exam_s19.py prints every number of this entry. Exact: the sum of squares; the closings $x_0, \dots, x_{15}$ and their recurrence (14 equations, 9 unknowns, nullity 1); the reflection; the certificate (G) and the identity (C) after clearing $e_1^3 e_2^2$; its three powers of k in the basis of products of e_1, e_2, e_3; the rank 10 of the symmetric ansatz in 18 unknowns; the monomial check of Step 5(c); the spot value $-\frac{2809}{7203}$; the symbol and the signs; the walls and Cor = 6; the table and the brackets to $K = 10$; and to $K = 16$ the signs with $c(K) \geq 3^{1-K}$, the identity $9^{K-1} c(K) = x_{K-1}$, the frame, and the two forms of the width with its bound $1/(K-1)^2$.
Check [computed], not part of the solution, in ball arithmetic, with the integrals cut where their tails are below 10^{-70}. On the chart, the integral over α_2 is elementary, $\int_0^\infty \frac{d\alpha_2}{(\alpha_2+A)(A\alpha_2+B)} = \frac{\log(A^2/B)}{A^2-B}$ with $A = 1 + \alpha_1$, $B = \alpha_1$, so

$$\ell = \int_0^\infty \frac{\log\big((1+x)^2/x\big)}{1+x+x^2}\, dx = 2.343907238689458890601562288872277069095\ldots.$$

On the share side (Proposition 15.2), $\Gamma(k)^2 P(k) = \int_0^\infty y^{k-1} \mathsf{G}_{0;(1,-1)}(1,y)^3\, dy$ gives the same ℓ to 40 digits, $P(2) - \frac{\ell-2}{3} = 0$ to 47 digits, and $P(9) = 4.8152090557463543143 \cdot 10^{-9} = c(9)\ell + r(9)$. With this value,

$$\ell - a_8 = 8.5773 \cdot 10^{-7}, \qquad \ell - a_9 = 9.6062 \cdot 10^{-8}, \qquad b_9 - \ell = 1.4186 \cdot 10^{-8}.$$

The upper side is the sharper. Its slack is $\frac{1}{9}P(K-1) - P(K)$, which is about $P(K)/(K-1)$ because $P(k)$ decays like $9^{-k}/k$ (for three marks the branch of Thm. 35.11, p. 310, is logarithmic), so $b_K - \ell$ is about $\frac{9}{8(K-1)}(\ell - a_K)$.

19.6 What the example shows

The reading needed three inputs, and each was elementary once it was named. The analytic input is one inequality, $e_1 e_2 \geq 9 e_3$, sharpened to $f \asymp e^{-\varphi}$ by the tropical stride φ: it gives the convergence at every real $k > 0$, the corner, and the upper side of the bracket. The algebraic input is one certificate: it gives the law at every real $k > 0$, not only at large k. The arithmetic is exact. Positivity gives the lower side and the gauge the upper side. On every ray through the origin the first two inputs exist [derived]: $f \leq f(\alpha^\star)$ (Thm. 35.10(i)), and the tropical stride, whose volume is Cor (the six cones of Step 2(c) are the six orders of Proposition 18.1). A certificate exists for every law that follows from the relations (Thm. 35.19, p. 313), but Step 5 needs more than existence: bounds on the g_e and their derivatives against the decay of f^k down to $k \to 0^+$. Here they came from the

numerators lying among the monomials of $e_1^2 e_2$, as the exponent condition did in Proposition 17.15; that a certificate with such bounds exists on every ray is [not yet computed]. With $h-1$ masters, positivity of $P(K), \dots, P(K+h-2)$ and the gauge bounds $P(K+i) < f(\alpha^\star) P(K+i-1)$ cut out a polytope in the coordinates of the masters, and Theorem 18.5(iv) is the rate at which it shrinks [not yet computed]. The width of the bracket is a ratio of integers from the other end of the pencil (Remark 18.3):

$$b_K - a_K = \frac{2 \cdot 9^{K-1}}{3(K-1)^2 \, x_{K-1} \, (x_{K-1} - x_{K-2})} \qquad \text{[derived]},$$

from $c(K) = 9^{1-K} x_{K-1}$, which holds at every K by Corollary 18.2: by Step 5 and the continuation of Proposition 18.1, (L) holds identically.
Open.

(i) The same examination for the lean $(2\mathbf{1}; 4, 1)$ of §18.3: its law has no certificate yet (§18.6(i)), and the signs of its rational solutions are unproved. [not yet computed]

(ii) The polytope for the two masters of H (§18.4), with a certificate for its law of order 4. [not yet computed]

(iii) The corner as the boundary term of the certificate at $k = 0$, where $g_e^{(0)}$ does not vanish at infinity, instead of a limit. [not yet computed]

19.7 Corrections

(i) The receipts of Proposition 17.15 say that Thm. 35.3 "applies as stated" to the banana rays. As stated it gives the relations for large k, and Remark 17.17 used the three-mark law from $k = 1$ and its corner from $k \to 0^+$. Step 5 closes that gap for the three-mark law; the four-mark banana ray needs the same bounds, with its own tropical stride.

(ii) In Remark 17.17 the ratios $-r/c$ are lower bounds at every q, since $c(q) > 0$ at every q (Step 8), not only at the q computed.

(iii) Theorem 18.5 said that the roots of the symbol are native values, citing Cor. 35.6. The corollary gives the converse, that the native value of every nondegenerate native gauge is a root, and a root need not be a native value (the extra root $\frac{131003}{4771109}$ of §18.4). The statement is corrected in place: (ii) speaks of roots, and the count of one condition in (iii) is for a simple root that is the value of a nondegenerate native gauge. Step 6 checks that hypothesis here.

Aliases, last: $\ell = 3L(\chi_{-3}, 2)$ [recalled], which agrees with the check to 40 digits; the closings are the sums of squares of trinomial coefficients, the even moments of the planar walk of three unit steps; (L) is an Apéry-like recurrence, and Step 9 is Apéry's limit made two-sided by positivity (Pincherle's minimal solution, read with signs); the certificate is a creative-telescoping certificate for the integral; with $\mathsf{G}_{0;(1,-1)}(1, y) = 2K_0(2\sqrt{y})$ the share side is the Bessel moment, $P(k) = 4^{2-k}\Gamma(k)^{-2} \int_0^\infty x^{2k-1} K_0(x)^3 dx$; φ is the tropicalization of $-\log f$, and $\int e^{-k\varphi} dy = 6/k^2$ its volume.